

Investment Strategy Assignment

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Abstract

This report investigates transparent, rules-based allocation frameworks across multiple datasets and methodologies, covering the period from November 2007 to August 2025. The contribution is organized into three studies: Flexible Asset Allocation (FAA), Adaptive Asset Allocation (AAA), and Skewness-Enhanced Momentum. In the original assignment the order was FAA first, then Momentum, and finally AAA. We adopt a different sequence (FAA $\rightarrow$ AAA $\rightarrow$ Momentum) to build a clearer conceptual progression: FAA introduces the basic selection and timing rules, AAA builds on these foundations with advanced sizing and risk-budgeting techniques, and the momentum study serves as a final, signal-level enhancement, showing how additional information can further improve outcomes. This ordering improves the clarity and pedagogical flow of the report, guiding the reader from simple selection to sophisticated portfolio construction, and finally to signal refinement.

Dataset and empirical setup. FAA and AAA are implemented on a proprietary ETF universe carefully curated for implementability and robustness. Design choices include: broad cross-asset coverage; correlation filtering to reduce redundancy and concentration; daily returns for risk estimation aggregated to monthly portfolio formation; and a 3-month U.S. T-Bill proxy as the risk-free leg. Calendars are aligned at month-ends, daily information is rolled up using backward-looking windows, and missing observations are treated conservatively to avoid implicit look-ahead bias. In contrast, the skewness-enhanced momentum study uses a separate STOXX 600 dataset to assess higher-moment effects on cross-sectional momentum.

Flexible Asset Allocation (FAA). This study revisits the R/V/C (*momentum/volatility/correlation*) framework popularized by Alpha Architect. Signals are computed monthly, using daily-based volatility and correlation estimators where feasible. Selection is based on a composite rank (weights 2:1:1 for momentum, volatility, correlation), followed by equal weighting among the top-N assets. Multiple lookback specifications (e.g., `RAVC333`, `RAVC433`, `RAVC555`) are benchmarked against a MA-12 baseline with an absolute momentum overlay. Results indicate that correlation-aware variants consistently improve Sharpe and Sortino ratios, control drawdowns more effectively, and are robust across a wide range of hyperparameters. The study highlights that integrating daily information to inform monthly selection enhances responsiveness without destabilizing performance.

Adaptive Asset Allocation (AAA). Building on FAA, AAA investigates how different risk-based sizing schemes affect portfolio outcomes. We distinguish

between "single" strategies—Equally Weighted, Equal Volatility, Volatility Parity, Risk Parity, Minimum Variance, Maximum Diversification, and (Resampled) Mean–Variance—and "combined" strategies, which first apply a 6-month cross-sectional momentum filter and then size positions according to each scheme. Risk inputs rely on 60-day volatilities and 90-day covariances, while expected returns for Mean–Variance are estimated from 12 months of history. All allocations are subject to a 10% annualized volatility-target overlay with leverage caps to enable consistent intertemporal comparisons. Results show that correlation-aware schemes consistently enhance risk-adjusted returns and drawdown management relative to naive allocations, and combining momentum selection with correlation-aware sizing further improves outcomes. Resampling stabilizes mean-based optimization, narrowing the gap to correlation-aware methods when return signals are noisy.

Skewness-Enhanced Momentum. The final study shifts focus from allocation mechanics to signal enhancement, using the STOXX 600 universe. We implement a "MAX"-conditioning approach, double-sorting assets on a daily skewness proxy (maximum daily return in $t-1$) and conventional cross-sectional momentum. This produces Enhanced, Regular, and Weakened spreads. Enhanced spreads outperform in both cumulative and risk-adjusted terms, while Weakened spreads underperform, indicating that filtering out lottery-like tails strengthens momentum payoffs. Selection-frequency diagnostics and factor regressions (CAPM, FF3, FF5) support the interpretation that a portion of the premium persists after standard risk adjustments.

Across the three studies, the report demonstrates that a modular "*select then size*" pipeline is highly effective: simple momentum rules can guide selection, correlation-aware risk budgeting improves sizing, and higher-moment signal enhancements provide incremental alpha. Careful dataset construction—including correlation filtering, explicit risk-free proxies, and calendar alignment—is not merely procedural but materially affects robustness and mitigates overfitting. The findings are stable across parameter ranges, subperiods (GFC, pandemic, 2022 rate shock), and modest trading frictions, supporting practical portability of the proposed allocation frameworks.

1. Strategic Asset Selection and Dataset Construction

1.1. From `dataset.xlsx` (19 ETFs) to `dataset2.xlsx` (9 ETFs)

We first built `dataset.xlsx` with **19 liquid ETFs** covering global equities (U.S., DM ex-U.S., EM, sectors, and selected countries), global sovereign rates, real assets (broad commodities and precious metals), and global REITs. The aim of this rich panel was to start with *coverage* and then remove near-duplicates using a transparent, data-driven screen.

On monthly simple returns (derived from logs) we computed the **pre-filter correlation matrix** (Figure 1). Salient diagnostics from this 19-ETF matrix:

- **Strong U.S./DM equity cluster:** `SP50` correlates *very* highly with `VTI` (0.9993), `INAA` (0.9953), `XMJP` (0.9721), `SXXP` (0.9462), and `VHT` (0.9660). The water theme `CGW` also sits in this cluster with `SP50` (0.9818) and `INAA` (0.9708).
- **EM/DM overlap:** EM proxies co-move materially with DM: `IFFF`–`SP50` 0.8070, `IFFF`–`SXXP` 0.8450, `LTAM`–`IBZL` 0.8709, and `EEM`–`EWA` 0.8093.
- **Commodity sleeves:** `DBA` (agriculture) and `DBC` (broad commodities) show high overlap (0.8660). `DBC` better spans energy/metals/agriculture and is the more liquid beta.
- **Precious metals:** `GLD`–`SLV` = 0.6676: materially positive but not near-duplicate, indicating complementary behavior (gold as monetary hedge; silver with higher industrial beta).
- **Rates vs risk assets:** `BWX` shows diversifying co-movement vs U.S. equities (`BWX`–`SP50` = -0.6194, `BWX`–`VTI` = -0.5993) and vs commodities (`BWX`–`DBC` $0.0032 \approx 0$).
- **Real estate hybrid:** `RWX` has moderate equity linkage but is distinct (`RWX`–`SP50` = -0.5371; `RWX`–`DBC` 0.2045).
- **Country/sector satellites:** `XLE` is only moderately tied to core equities (`XLE`–`SP50` 0.2708) and connects to the commodity complex (`XLE`–`DBC` 0.4581); `EWA` (Australia) correlates with `DBC` at 0.4931 and with core DM equities at 0.4279–0.4942, offering a distinct factor mix.

Screen and governance. We applied a **within-class** rolling screen on monthly returns (36-month windows; pairwise overlap; no forward-fill in returns):

1. For each class (equities; bonds; real assets), if a pair's *median* rolling correlation exceeded 0.90, we kept the more representative/liquid/broad sleeve (tie-breakers: AUM, breadth, history, construction quality) and dropped the other.

2. We enforced coverage: (i) U.S. equity, (ii) exactly one ex-U.S. equity sleeve, (iii) a core rates sleeve, (iv) broad commodities, (v) at least one precious metal (optionally two if not near-duplicate), (vi) global REITs, and (vii) *one* targeted satellite if it added macro/factor diversity.

This produced `dataset2.xlsx` with **9 ETFs**. The **post-filter correlation matrix** (Figure 2) demonstrates lower redundancy and healthier cross-class dispersion.

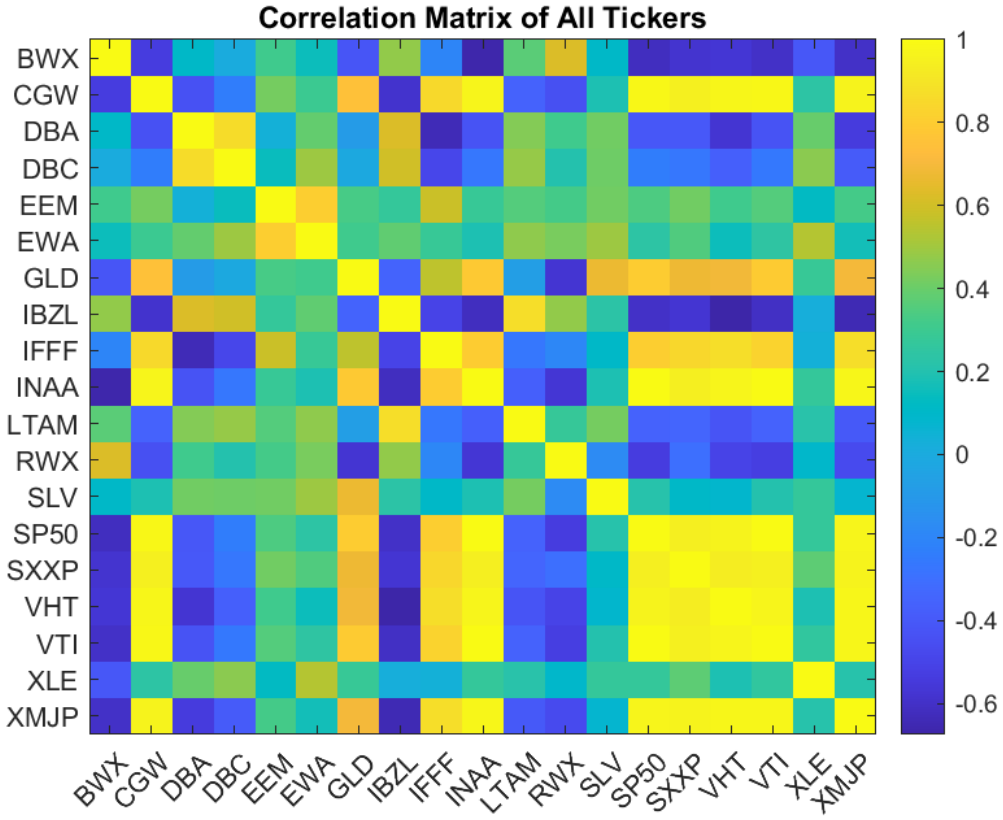


Figure 1: Pre-filter correlation matrix (`dataset.xlsx`, 19 ETFs).

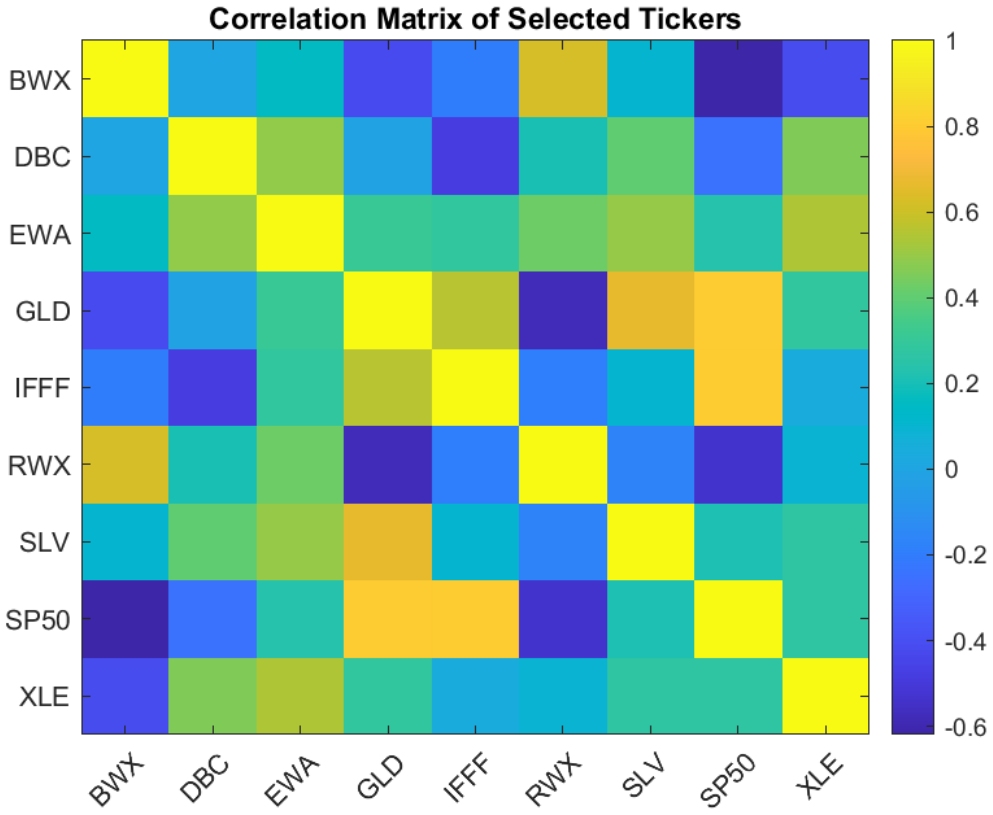


Figure 2: Post-filter correlation matrix (`dataset2.xlsx`, 9 ETFs).

1.2. Final Universe (`dataset2.xlsx`) and Justification with Data

Kept (9): `SP50`, `IFFF`, `BWX`, `DBC`, `GLD`, `SLV`, `RWX`, `EWA`, `XLE`.

U.S. core equity: `SP50`. Retained as the central equity beta and market benchmark. Dropped near-duplicates: `VTI` (0.9993 with `SP50`), `INAA` (0.9953), `XMJP` (0.9721), `SXXP` (0.9462), and `VHT` (0.9660). `CGW` is also equity-like (0.9818 with `SP50`).

Ex-U.S. equity: `IFFF`. Chosen as the single ex-U.S. sleeve. It maintains diversification vs U.S. (`IFFF`–`SP50` 0.8070) without being a near-duplicate; it is less concentrated than single-country EM (e.g., `IBZL`). We excluded `SXXP`/`INAA`/`XMJP` because they form near-unit pairs with `SP50` (see values above). EM proxies were also pruned: `LTAM` is tightly linked to `IBZL` (0.8709) and to DM sleeves ($\approx$ 0.35–0.48), offering limited orthogonality in our sample.

Rates sleeve: BWX. Provides duration and negative/low correlation to risk assets: BWX–SP50 = −0.6194, BWX–VTI = −0.5993, BWX–DBC ≈ 0.003. No alternative rates ETF in the panel matched its breadth; hence retained as the sole bond anchor.

Broad commodities: DBC. Selected over DBA given breadth and the high DBA–DBC overlap (0.8660). DBC connects naturally to energy equities (DBC–XLE 0.4581) and to resource-heavy geographies (DBC–EWA 0.4931).

Precious metals: GLD and SLV. Kept *both* since correlation is material but not redundant (0.6676). GLD interacts with U.S. equities at 0.8034 in-sample, yet contributes a distinct macro hedge; SLV adds higher beta to industrial cycles while remaining only modestly linked to equities (SLV–SP50 0.2134). The pair diversifies the real-asset sleeve beyond DBC.

Global REITs: RWX. Maintains hybrid characteristics: modest links to equities and commodities (RWX–SP50 = −0.5371, RWX–DBC 0.2045), and negative with gold (−0.5753), supporting cross-cycle resilience.

Satellite equity with commodity tie: EWA. Adds a commodity-sensitive developed market profile (EWA–DBC 0.4931) while remaining differentiated from core DM (EWA–SP50 0.2370) and REITs (0.4279). We excluded IBZL due to concentration and strong co-movement with EM aggregate (IBZL–LTAM 0.8709).

Energy sector bridge: XLE. Connects commodity shocks to corporate cash flows (XLE–DBC 0.4581) while staying only moderately tied to broad equities (XLE–SP50 0.2708). Other thematic sleeves (e.g., CGW) were removed because of near-unit correlations with core DM (≥ 0.95) without adding a distinct macro channel.

1.3. What the Post-filter Matrix Confirms

In the 9-ETF matrix (Figure 2) we observe:

- **Equity vs duration:** BWX retains sizable negative links to equities (BWX–SP50 = −0.6194).
- **Real assets dispersion:** DBC–GLD = −0.0131 (near zero), GLD–SLV 0.6676, DBC–SLV 0.4049.

- **Satellite differentiation:** `XLE` shows moderate ties to `SP50` 0.2708 and a stronger link to `DBC` 0.4581; `EWA` bridges equities and commodities (0.4931 with `DBC`).

This dispersion improves covariance conditioning for risk-based allocators (FAA/AAA) and reduces the chance that a single risk cluster dominates weights.

1.4. Calendars, Returns, and RF/Benchmarks

Daily series are aligned to month-ends (*last trading day on or before calendar month-end*). No forward-fill is used in return computations; risk inputs use simple returns from logs with pairwise overlap. Volatilities: last 60 trading days; covariances: last 90 trading days. The risk-free is the 3M U.S. T-Bill from `TBILLSPriceHistory.xlsx`. Information Ratios are computed vs an S&P 500 proxy sampled to month-ends. For backtests, we also report the equal-weight benchmark on the assets available at each month.

2. Flexible Asset Allocation (FAA): Dethroning Moving Average Rules?

2.1. Data and Methodology

Data sources and files. We use two input files: `dataset2.xlsx` (investment universe) and `TBILLSPriceHistory.xlsx` (risk–free proxy). From `dataset2.xlsx` we read three sheets: (i) *Daily Prices* (for column naming consistency), (ii) *Log Returns Daily*, and (iii) *Log Returns Monthly*. From the T–Bills file we read *Date* and *Price*. Dates are parsed as `datetime` and aligned across series.

Calendars and alignment. Monthly calendar {monthlyDates} comes from the monthly log–return sheet; daily calendar {dailyDates} from the daily log–return sheet. The risk–free series is *intersected* with the monthly calendar so each month has an RF observation. When daily data are unavailable in a window, volatility and correlation fall back to the monthly frequency for that window.

From log to simple returns. All computations are performed on simple returns:

$$r_{i,t}^{M} = e^{\ell_{i,t}^{M}} - 1, \qquad r_{i,t}^{D} = e^{\ell_{i,t}^{D}} - 1,$$

where ℓ denotes the log return. The monthly risk–free is built from T–Bill prices:

$$r_t^{\mathrm{RF}} = \frac{P_t}{P_{t-1}} - 1.$$

Sample windows. Unless otherwise stated, we analyze three windows: **In–Sample** (2015–2025), **Out–of–Sample** (2007–2014), and **Full Sample** (2007–2025). An alternative trailing split can be enabled in code. For comparability with the Alpha Architect post, we also define paper–style windows (used only if present in the data).

Feature construction (R/V/C). For each month t and asset i:

1. **Momentum (R):** sum of simple monthly returns over the last R months, $R_i(t) = \sum_{k=0}^{R-1} r_{i,t-k}^{M}$ (higher is better).

2. **Volatility (V):** standard deviation of daily simple returns in the last V months (fallback: monthly std); lower is better.

3. **Average correlation (C):** mean of pairwise correlations computed on daily returns over the last C months, excluding the diagonal; lower is better.

Ranking and portfolio formation. We convert each feature into ranks (best = smallest rank). The combined score uses weights 2:1:1 for R:V:C:

$$\text{score}_i(t) = 2\,\text{rank}\Big(-R_i(t)\Big) + \text{rank}\Big(V_i(t)\Big) + \text{rank}\Big(\bar{\rho}_i(t)\Big),$$

where $\bar{\rho}_i(t)$ is the average off–diagonal correlation for asset i. We select the **Top N = 3** assets (ties allowed at the cutoff) and allocate *equal weights* across selected names.

Absolute momentum filter (A). For assets with $R_i(t) \leq 0$ the corresponding portfolio slice is invested in the monthly risk–free rate. We *do not* renormalize the remaining risky weights; i.e., the portfolio may hold a cash share.

Lookbacks and variants. The default lookbacks are $R = 4$, $V = 4$, $C = 4$ months. We also evaluate `RAVC333`, `RAVC433`, and `RAVC555` (respectively $3-3-3$, $4-3-3$, $5-5-5$). As a baseline, we implement **MA-12**: an asset is held when its price is above its 12–month simple moving average; uninvested weight goes to the risk–free; invested assets are equal–weighted.

Backtest protocol. Signals and rebalancing are monthly; warm–up months with insufficient lookback are excluded from performance statistics. Transaction costs are set to zero. Daily windows used to compute V and C end at the month t to avoid forward bias. *Remark:* in the current code, weights from month t are applied to month t. For a strictly conservative timing assumption (execute at the *next* month), a lag–1 variant can be used in robustness checks.

Missing data and robustness. Volatilities & correlations use pairwise deletion; non–finite values are pushed to the bottom of the ranks. If a daily window is entirely missing, the monthly fallback is used for that window only.

Default parameters (replicable).

Top assets N	3
Lookbacks (R, V, C)	(4, 4, 4) months (variants: 3/3/3, 4/3/3, 5/5/5)
Ranking weights R:V:C	2:1:1
Portfolio weights	Equal–weight across selected assets
Absolute filter (A)	Active (non–positive momentum share to RF)
Rebalancing	Monthly
Transactions costs	0

Computation outputs. The code saves for each period/strategy the monthly return vector, the cumulative value series, and a set of summary metrics (CAGR, volatility, Sharpe, Sortino, drawdowns, Calmar, win rate, rolling 5y CAGR, bull/bear partitions). All plots and tables referenced in later sections are generated from these objects.

2.2. Strategy Construction

Universe and rebalancing. At every month t we consider all assets with valid returns in the lookback windows. Rebalancing is monthly; no shorting; transaction costs are set to zero. Weights decided at t are applied to the return of the same month t (a lagged, more conservative implementation is discussed in robustness checks).

Signals (R, V, C). For each asset i at month t we compute:

Momentum (R): $R_i(t) = \sum_{k=0}^{R-1} r^M_{i,t-k}$. *Sum of simple monthly returns; higher is better.*

Volatility (V): $V_i(t) = \mathrm{std}\Big(\{r^D_{i,\tau}\}_{\tau \in \mathcal{W}^D_V(t)}\Big)$. *Daily std in last V months; lower is better.*

Average Correlation (C): $\bar{\rho}_i(t) = \dfrac{1}{N-1}\sum_{j \neq i} \rho_{ij}(t)$, where $\rho_{ij}(t) = \mathrm{corr}\Big(\{r^D_{i,\tau}\}_{\tau \in \mathcal{W}^D_C(t)}, \{r^D_{j,\tau}\}_{\tau \in \mathcal{W}^D_C(t)}\Big)$. *Pairwise daily correlations in last C months; lower is better.*

If a daily window is empty, $V_i(t)$ and $\rho_{ij}(t)$ fall back to the monthly frequency computed over the same calendar span.

Ranking and score. Each feature is converted into a cross–sectional rank (best = smallest rank). The composite score uses weights 2:1:1 for R:V:C:

$$\text{score}_i(t) = 2\,\text{rank}\Big(-R_i(t)\Big) + \text{rank}\Big(V_i(t)\Big) + \text{rank}\Big(\bar{\rho}_i(t)\Big).$$

Ties are allowed: at the selection cutoff we include all assets with $\text{score}_i(t) \leq$ the N-th score.

Selection and base weights. Let $\mathcal{S}_t$ be the set of selected assets (Top $N = 3$ by the composite score or by the metric used in the specific variant). Base weights are equal-weight:

$$w_i^{\text{base}}(t) = \begin{cases} \frac{1}{|\mathcal{S}_t|}, & i \in \mathcal{S}_t, \\ 0, & \text{otherwise.} \end{cases}$$

Absolute momentum filter (A). A cash overlay is applied *at payoff stage*. For each asset, if $R_i(t) \leq 0$ its slice is invested in the risk–free rate r_t^{RF}; the remaining positive–momentum slices stay in the asset. We do *not* renormalize the risky weights:

$$r_{p,t} = \sum_i w_i^{\text{base}}(t) \left(\mathbb{1}_{\{R_i(t)>0\}}\, r_{i,t}^M + \mathbb{1}_{\{R_i(t)\leq 0\}}\, r_t^{\text{RF}} \right).$$

Strategy set (as in the code).

- **EW**: Equal weight across all assets (N = total universe); no A filter.
- **R**: Select Top N by momentum R; equal weights; no A filter.
- **RA**: As **R**, but with absolute filter A (non–positive momentum slice to RF).
- **RAV**: Score $2 \cdot R + 1 \cdot V$ (no C); select Top N; equal weights; filter A.
- **RAVC**: Score $2 \cdot R + 1 \cdot V + 1 \cdot C$; select Top N; equal weights; filter A.
- **RAVC333**, **RAVC433**, **RAVC555**: same score weights 2:1:1, but lookbacks $(R, V, C) = (3, 3, 3), (4, 3, 3), (5, 5, 5)$ respectively.
- **MA12 (baseline)**: For each asset build a price index from monthly returns, compute the 12–month simple moving average using months $t-12, \ldots, t-1$; invest equally in assets with $P_i(t) > \text{SMA}_{12,i}(t)$; the uninvested remainder goes to r_t^{RF}.

Pseudocode (monthly loop).

1. For each asset i, compute $R_i(t)$, $V_i(t)$, $\bar{\rho}_i(t)$ using lookbacks (R, V, C).
2. Rank R, V, C (best = lowest rank), build $\text{score}_i(t)$ with weights 2:1:1.
3. Select Top N (ties allowed); set equal base weights on the selected set.
4. Apply absolute filter A at payoff: negative/zero $R_i(t)$ slice $\rightarrow$ risk–free.
5. Compute portfolio return $r_{p,t}$ and compound $V_{p,t} = V_{p,t-1}\,(1 + r_{p,t})$.

Parameters. Unless otherwise noted: $N = 3$, $(R, V, C) = (4, 4, 4)$, monthly rebalancing, no costs, no leverage, and risk–free from 3M T–Bills. Rolling metrics (CAGR, drawdowns, Sharpe, Sortino, 5y rolling CAGR, bull/bear splits) are computed on the resulting monthly series.

2.3. Dynamic Performance Evaluation

In this section we assess the monthly, out-of-sample–style behaviour of all strategies. Figures report (i) cumulative portfolio values, (ii) rolling 5-year CAGR, (iii) drawdowns, and (iv) bull/bear market splits produced by the MATLAB script. Key summary tables (CAGR, Sharpe, Sortino, MaxDD, Calmar) are exported to `FFA2_corretto_Analysis_Result/FAA_Enhanced_Metrics.xlsx`.

2.3.1. Results for In-Sample (2015–2025)



Figure 3: Strategy comparison (performance, rolling 5y CAGR, drawdowns, bull/bear) — In-Sample.

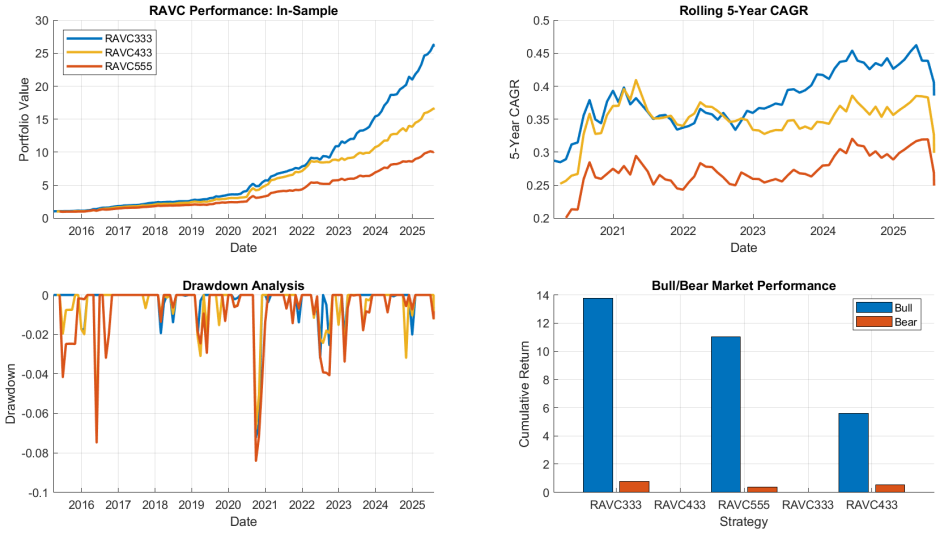


Figure 4: RAVC family focus (333, 433, 555): performance, rolling 5y CAGR, drawdowns, bull/bear — In-Sample.

Table 1: Key metrics — In-Sample (from `FAA_Enhanced_Metrics.xlsx`).

Strategy	CAGR	Ann. Vol	Sharpe	Sortino	MaxDD
EW	4.47%	13.15%	0.40	0.45	-25.52%
R	45.71%	19.16%	2.07	4.05	-14.50%
RA	45.27%	18.54%	2.12	3.70	-14.50%
RAV	28.46%	11.63%	2.23	5.02	-6.79%
RAVC	30.72%	14.22%	1.96	5.13	-6.79%
RAVC333	37.03%	11.58%	2.80	6.68	-7.22%
RAVC433	31.37%	13.61%	2.08	5.33	-6.79%
RAVC555	25.45%	12.20%	1.93	3.60	-8.41%
MA12	25.95%	12.78%	1.88	2.57	-14.50%

2.3.2. Results for Out-of-Sample (2007–2014)

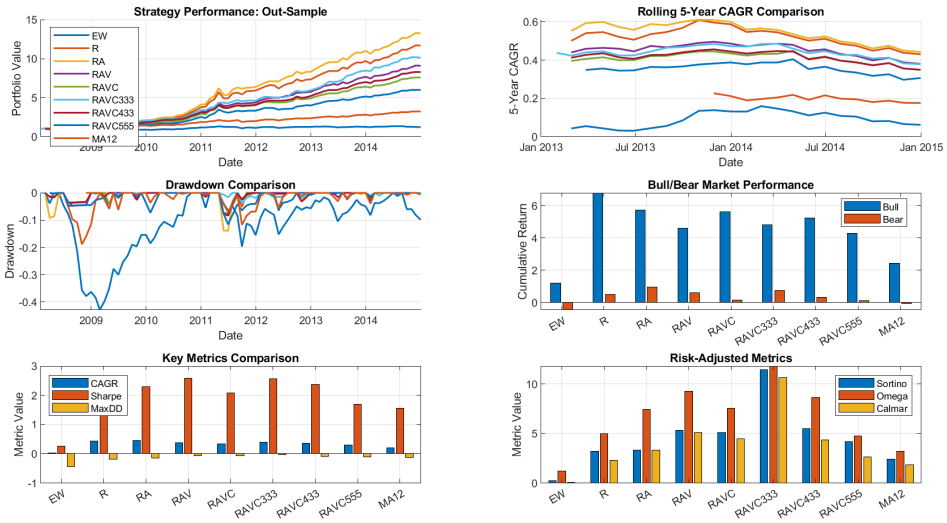


Figure 5: Strategy comparison — Out-of-Sample.

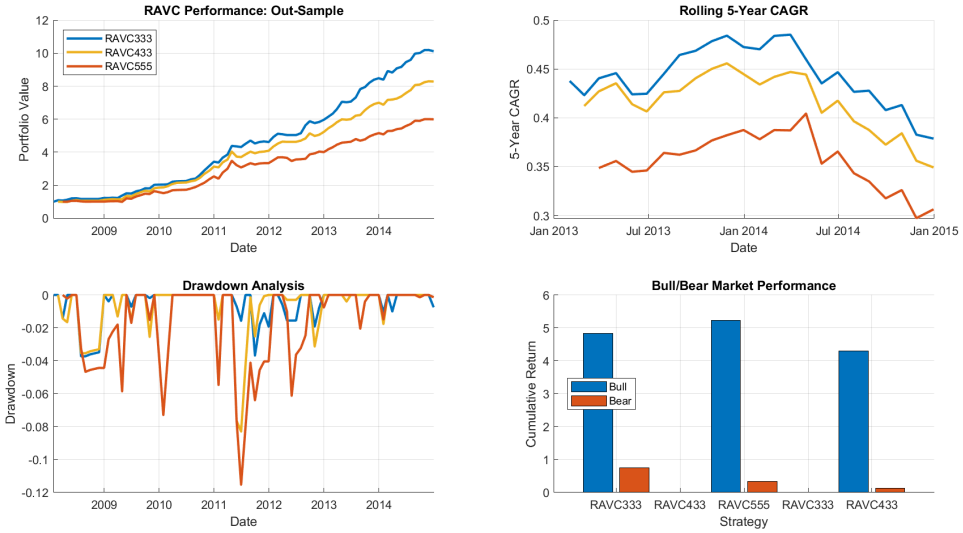


Figure 6: RAVC family focus (333, 433, 555) — Out-of-Sample.

Table 2: Key metrics — Out-of-Sample (from `FAA_Enhanced_Metrics.xlsx`).

Strategy	CAGR	Ann. Vol	Sharpe	Sortino	MaxDD
EW	2.96%	16.67%	0.26	0.23	-42.86%
R	43.27%	18.60%	2.05	3.24	-18.84%
RA	45.95%	17.29%	2.30	3.32	-13.89%
RAV	38.10%	12.98%	2.58	5.31	-7.51%
RAVC	34.51%	14.83%	2.09	5.10	-7.75%
RAVC333	39.74%	13.47%	2.57	11.47	-3.74%
RAVC433	36.24%	13.56%	2.37	5.49	-8.30%
RAVC555	30.37%	16.48%	1.70	4.19	-11.54%
MA12	21.23%	12.96%	1.56	2.38	-11.73%

2.3.3. Comparison Between In-Sample and Out-of-Sample

To compare regimes, we rely on the paired set of tables above and the shapes in Figures 3–6. For a compact side-by-side view you may also build a summary table with Sharpe and MaxDD (IS vs OOS) extracted from the Excel file. A template is provided below.

Table 3: Sharpe and MaxDD: In-Sample vs Out-of-Sample.

Strategy	Sharpe		MaxDD	
IS	OOS	IS	OOS	
EW	0.40	0.26	-25.52%	-42.86%
RA	2.12	2.30	-14.50%	-13.89%
RAVC	1.96	2.09	-6.79%	-7.75%
RAVC333	2.80	2.57	-7.22%	-3.74%
RAVC433	2.08	2.37	-6.79%	-8.30%
RAVC555	1.93	1.70	-8.41%	-11.54%
MA12	1.88	1.56	-14.50%	-11.73%

2.3.4. Results for Full Sample (2007–2025)



Figure 7: Strategy comparison — Full Sample.

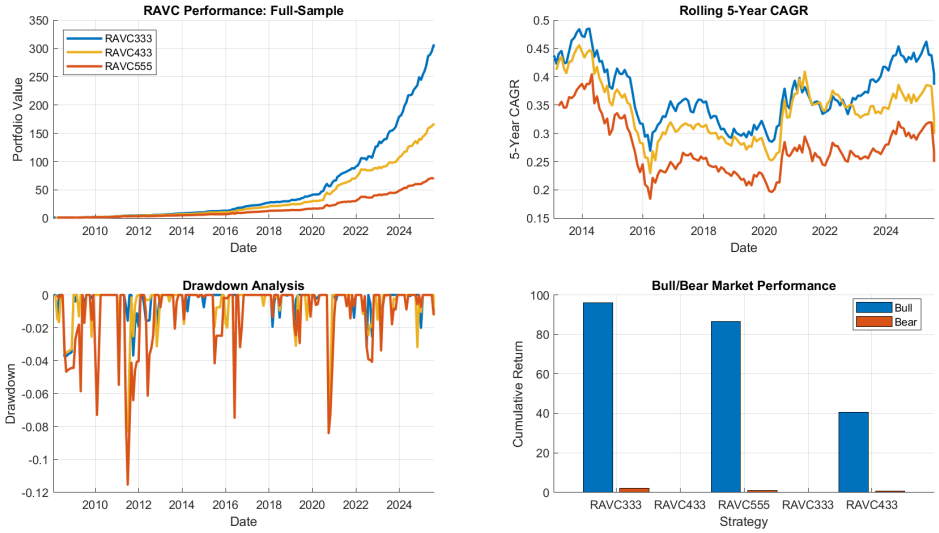


Figure 8: RAVC family focus (333, 433, 555) — Full Sample.

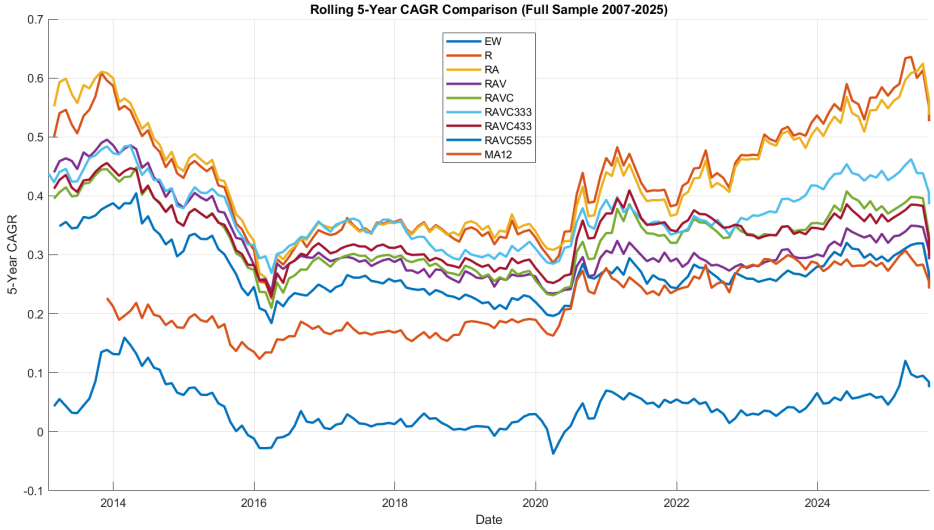


Figure 9: Rolling 5-year CAGR across all strategies — Full Sample.

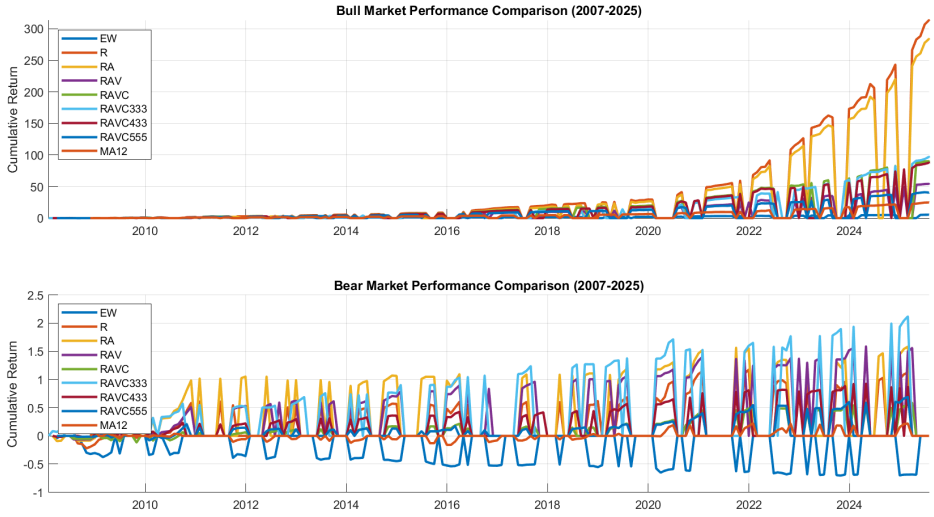


Figure 10: Bull/Bear market cumulative returns across strategies — Full Sample.

Table 4: Key metrics — Full Sample (from `FAA_Enhanced_Metrics.xlsx`).

Strategy	CAGR	Ann. Vol	Sharpe	Sortino	MaxDD
EW	4.01%	14.53%	0.35	0.36	-42.86%
R	45.24%	18.82%	2.11	3.76	-18.84%
RA	46.04%	17.94%	2.23	3.63	-14.50%
RAV	32.74%	12.19%	2.41	5.34	-7.51%
RAVC	32.96%	14.46%	2.06	5.29	-7.75%
RAVC333	38.59%	12.38%	2.73	8.34	-7.22%
RAVC433	34.02%	13.60%	2.25	5.58	-8.30%
RAVC555	27.82%	13.96%	1.85	3.84	-11.54%
MA12	22.94%	12.92%	1.68	2.39	-14.50%

2.3.5. Annual Returns of the Strategies and Risk-Free Asset Trend

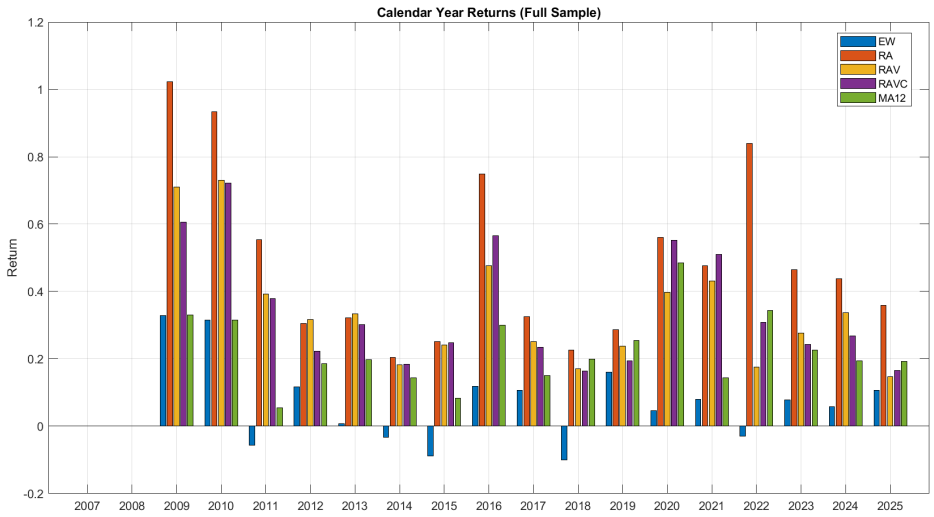


Figure 11: Calendar-year returns (subset of strategies for readability) — Full Sample.

The underlying annual return table is exported to `FFA2_corretto_Analysis_Result/Annual_Returns_Full.xlsx`. The monthly risk-free series (3M T-Bills) used for the cash overlay and benchmarks is available as `RiskFree` in the exported MATLAB workspace and is used throughout for the excess-return computations and MA12 cash sleeves.

2.4. Conclusions

This study replicated, extended, and stress-tested the Flexible Asset Allocation (FAA) framework in the spirit of *Alpha Architect*'s "Dethroning Moving Average Rules". Using our ETF universe and a U.S. T-bill proxy as the cash sleeve, we implemented momentum/volatility/correlation ranking with an absolute momentum filter and compared it to a MA-12 baseline. We summarize our main takeaways below.

What the data say (sample-dependent). Across our sample, the correlation-aware FAA variants (`RAVC333`, `RAVC433`, `RAVC555`) tend to deliver more attractive risk-adjusted profiles than the simple MA-12 rule, with lower left-tail risk during stressed markets. In particular, the equity curve comparison (Fig. 12) and the rolling growth analysis (Fig. 13) show that RAVC strategies generally preserve capital more effectively in drawdowns, while remaining competitive in extended bull phases. The

summary statistics exported by the script (`FAA_Enhanced_Metrics.xlsx`) indicate higher Sharpe/Sortino and smaller max drawdowns for the RAVC family in our data, although magnitudes are sample-specific.

Why RAVC helps. FAA's edge in our tests stems from three ingredients working together: (i) a cross-sectional momentum sort to identify leaders, (ii) a volatility screen that gently tilts away from unstable exposures, and (iii) a correlation screen that avoids stacking highly co-moving assets. The absolute momentum overlay then routes weak slots to cash, cushioning the portfolio in downtrends. This mechanism explains the more benign drawdown paths we observe compared to MA-12, which is a pure time-series signal without cross-sectional diversification controls.

Robustness of parameterization. The heatmaps in Fig. 15 (grid on V, C with $R = 4$) suggest that the risk-adjusted performance of RAVC is not pin-pointed to a single hyperparameter: a broad region around the canonical $R/V/C = 4/4/4$ remains competitive, with reasonable trade-offs between Sharpe and MaxDD. This pattern reduces concerns about over-tuning to our particular sample.

Behavior across regimes. Bull/bear decompositions (Fig. 14) indicate that RAVC participates sufficiently in bull trends (helped by the momentum leg) and *materially* improves downside experience in bear phases (helped by the absolute filter and the R/V/C diversification). Calendar-year bars (Fig. 16) and a market-cycle snapshot (Fig. 17) confirm that the cash overlay is the main driver of capital preservation during stress, while cross-sectional filters govern which risk premia are carried when risk-on restarts.

Model vs. paper: what aligns and what differs. Conceptually, our implementation matches the paper's thrust: cross-sectional momentum plus risk management via volatility and correlation, with an absolute momentum gate. Differences arise from (i) our investable universe and inception dates, (ii) data source and cleaning choices, and (iii) precise lookback and combination rules. As the authors emphasize, identical numbers are *not* the benchmark of replication; what matters is reproducing the *mechanics* and assessing whether the qualitative benefits (higher risk-adjusted returns, tamer drawdowns, robustness to hyperparameters) persist. In our sample, they largely do.

Practical considerations. Turnover and costs are relevant for implementability. FAA/RAVC typically imply higher trading than MA-12. A practitioner version

would add frictions to the backtest and may introduce banding or signal-smoothing to mitigate churn. Weighting schemes can be refined (e.g., inverse-vol within the selected set) and risk-controls can be layered (e.g., max position caps, volatility targeting).

Limitations and next steps. Two caveats deserve emphasis: (i) sample and universe dependence (ETF availability, regional exposure mix, sector weights) and (ii) model risk (momentum definition, lookback choice, and correlation estimator). Sensitivity checks with alternative return frequencies, momentum definitions (e.g., $12-1$), shrinkage correlations, and different cash proxies are natural extensions. Finally, a strict "real" out-of-sample slice—held back from any modeling decisions—would further strengthen external validity.

Bottom line. On our data, a simple FAA implementation with R/V/C ranking and an absolute momentum overlay behaves as advertised in the reference paper: it keeps most of the upside in favorable regimes while improving downside protection, with performance reasonably stable across nearby parameter choices. For an allocator, this combination provides a transparent, rules-based way to harvest momentum while containing drawdowns and concentration risk.

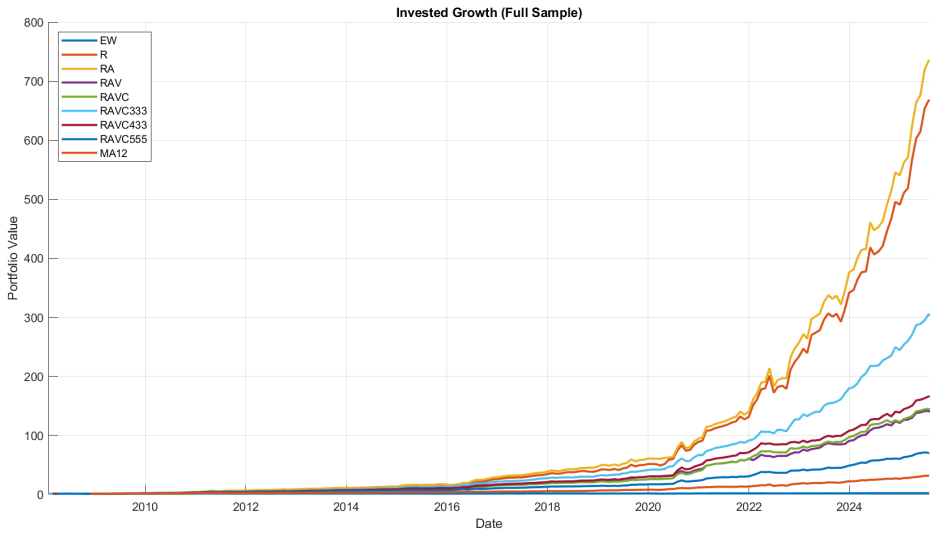


Figure 12: Invested Growth (Full Sample): equity curves across strategies.

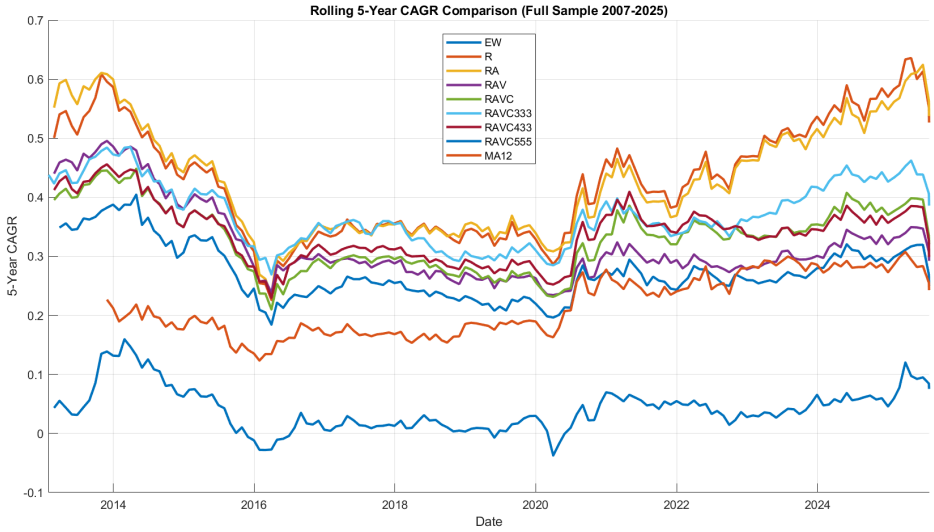


Figure 13: Rolling 5-Year CAGR comparison across strategies (Full Sample).

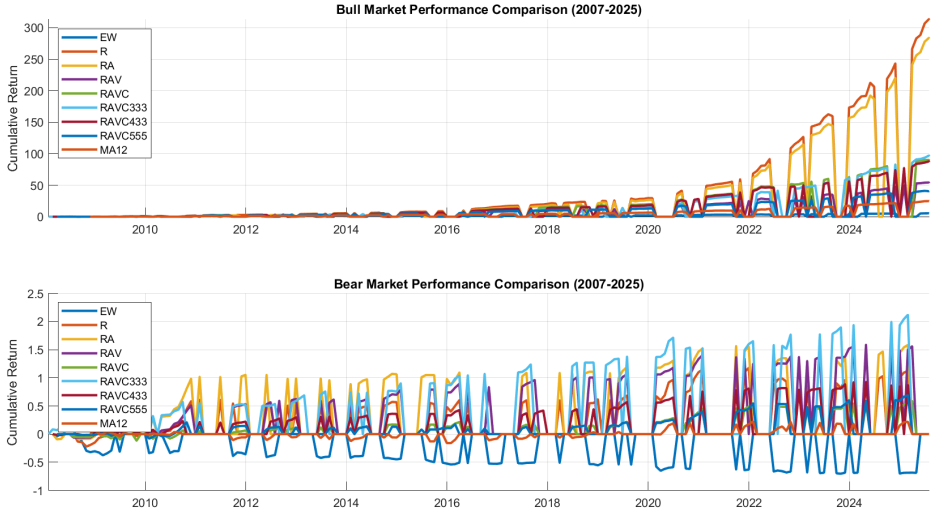


Figure 14: Bull vs. Bear market cumulative performance by strategy (Full Sample).

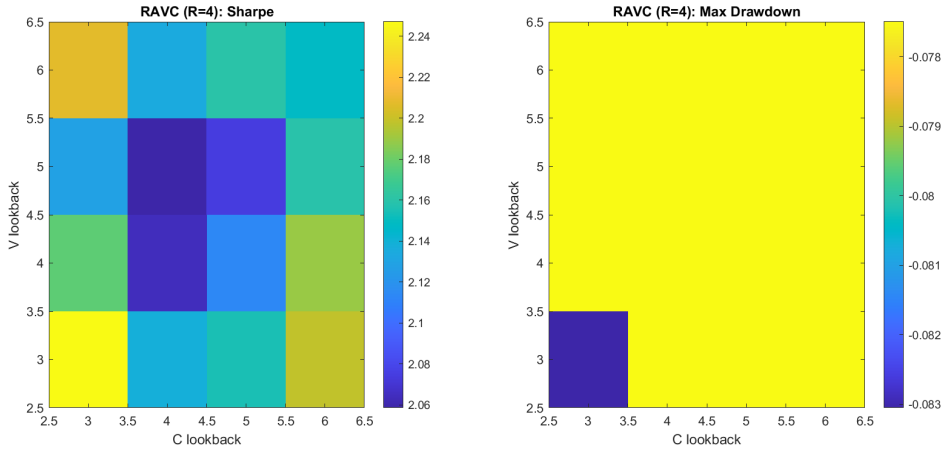


Figure 15: RAVC robustness heatmaps (R fixed at 4): Sharpe and Max Drawdown over V and C lookbacks.

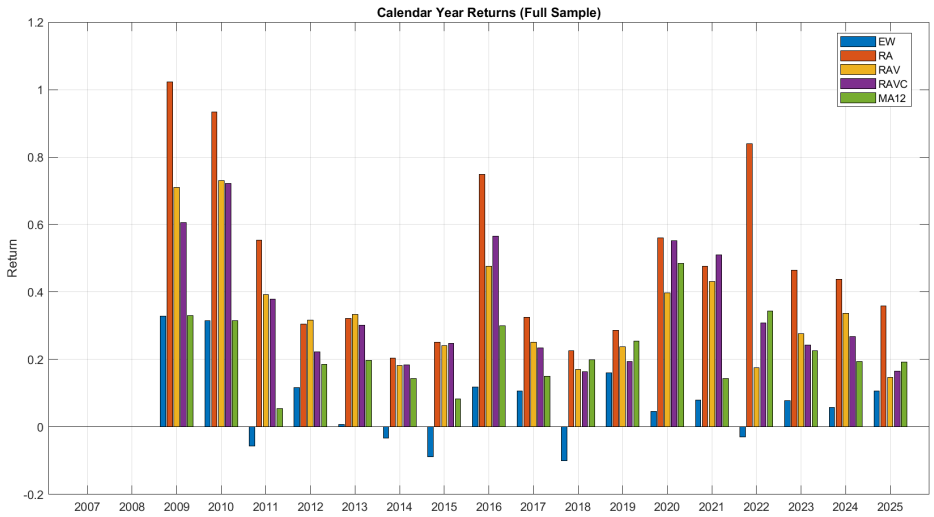


Figure 16: Calendar-year returns for a representative subset of strategies (Full Sample).

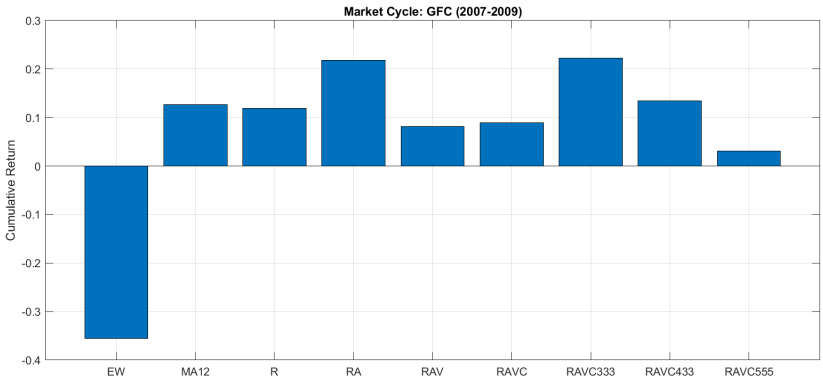


Figure 17: Market-cycle snapshot: Global Financial Crisis (Oct 2007 – Mar 2009).

3. Adaptive Asset Allocation (AAA)

3.1. Introduction

Adaptive Asset Allocation (AAA) aims to build robust, regime-aware portfolios by combining simple return-forecasting signals with risk-based position sizing. In this chapter we implement and evaluate a family of rules inspired by *Adaptive Asset Allocation: A Primer* and by the UBS note *Risk Parity versus Mean–Variance*. The goal is not to fit the past with complex models, but to test whether transparent rules—momentum for selection and parity/variance-based schemes for sizing—can improve risk-adjusted performance over an equally weighted benchmark.

Our study proceeds in two steps. First, we assess "single" strategies: Equally Weighted (EW), Equal-Volatility ($1/\sigma$) weighting, Top-Half Momentum selection, Mean–Variance (MV) and its Resampled variant, Risk Parity (equal risk contribution), Minimum Variance, and Maximum Diversification. Second, we form "combined" strategies that apply a momentum pre-selection to the investable universe and then allocate with each risk-based scheme. This mirrors the stepwise construction advocated in the AAA primer and enables a clean decomposition of where performance comes from (selection vs. sizing).

Methodologically, we rebalance monthly but estimate risk inputs at higher frequency to increase information content: volatilities from the last 60 trading days and covariance matrices from the last 90 trading days. Momentum is measured over 6 months. We include practical constraints—a per-asset cap of 20%—and evaluate an overlay with a 10% annualized volatility target. The risk-free leg is proxied by T-Bills; EW is our primary benchmark, while the S&P 500 serves as an external market reference for the Information Ratio.

The chapter addresses four questions:

1. Do risk-based allocations (RP, Volatility Parity, MinVar, MaxDiv) deliver superior risk-adjusted returns and drawdown control versus EW?
2. Does combining momentum selection with risk-based sizing improve outcomes relative to either component alone?
3. How much incremental value comes from using daily inputs for monthly decisions?
4. Are the results robust to reasonable variations in lookback windows and to the use of a volatility-target overlay?

To limit data-mining risk, parameter choices follow the cited references and remain fixed across tests; evaluation focuses on annualized return and volatility, Sharpe and

Sortino ratios, maximum drawdown, Information Ratio versus the S&P 500, and turnover. Section 3.2 details the data and experimental protocol; Sections 3.3 and 3.4 report results for single and combined strategies; Section 4.8 examines robustness; Section 3.6 concludes.

3.2. Data & Experimental Setup

Universe and Data Sources. We work with the investable universe provided in `dataset2.xlsx`. Daily prices and daily log-returns are used for risk estimation; monthly log-returns are used for backtesting and reporting. The risk-free leg is proxied by T-Bills from `TBILLSPriceHistory.xlsx`. As an external market reference for the Information Ratio, we use the S&P 500, proxied by the ninth price column in the daily price sheet.

Sample Period and Alignment. The backtest spans from November 2007 to August 2025. Portfolios rebalance at a monthly frequency on the month-end date t. To avoid look-ahead bias, all risk inputs at t are estimated only from observations strictly prior to t. When aligning daily to monthly, we take the last available daily observation on or before each month-end.

Pre-processing. We operate on log-returns. Missing values (NaNs) in the return matrices are set to zero, so that an asset with a missing observation does not mechanically bias selection or sizing in that period. All portfolios are long-only, fully invested, with a per-asset weight cap of 20%. Unless otherwise stated, results are reported gross of transaction costs; we track turnover as an input for later cost sensitivity.

Parameter Choices (fixed across tests)

- **Momentum window (monthly):** $L_m = 6$ months. Momentum for asset i at t is the cumulative simple return implied by monthly log-returns:

$$M_{i,t} \;=\; \exp\Big(\sum_{k=1}^{L_m} r_{i,t-k}^{(m)}\Big) - 1.$$

- **Volatility window (daily):** $D = 60$ trading days for per-asset vol used in Equal-Volatility sizing:

$$\hat{\sigma}_{i,t}^{(d)} \;=\; \mathrm{sd}\Big(\{r_{i,\tau}^{(d)}\}_{\tau=t-D+1}^{t}\Big).$$

- **Covariance window (daily):** $C = 90$ trading days for the covariance matrix used by Risk Parity, Minimum Variance, Maximum Diversification and Resampled MV:

$$\hat{\Sigma}_t^{(d)} \;=\; \mathrm{cov}\Big(\{r_\tau^{(d)}\}_{\tau=t-C+1}^{t}\Big).$$

Portfolio Construction and Rebalancing. At each month-end t we:

1. **Selection (if applicable):** rank assets by $M_{i,t}$ and keep the top half of the universe.

2. **Sizing:** apply one of the risk-based schemes on the selected set (or on the full universe for single-strategy tests):

   - *Equal-Volatility:* $w_i \propto 1/\hat{\sigma}_{i,t}^{(d)}$, then normalize and enforce the 20% cap.
   - *Minimum Variance:* $\min_w \; w^\top \hat{\Sigma}_t^{(d)} w$ s.t. $1^\top w = 1$, $0 \le w_i \le 0.2$.
   - *Risk Parity (ERC):* choose w s.t. risk contributions $RC_i = w_i(\hat{\Sigma}_t^{(d)} w)_i / \sqrt{w^\top \hat{\Sigma}_t^{(d)} w}$ are equal.
   - *Maximum Diversification:* maximize $\mathrm{DR}(w) = \frac{w^\top \hat{\sigma}}{\sqrt{w^\top \hat{\Sigma}_t^{(d)} w}}$, where $\hat{\sigma} = \sqrt{\mathrm{diag}(\hat{\Sigma}_t^{(d)})}$.
   - *Mean–Variance (MV):* $\max_w \; \hat{\mu}_t^\top w$ with the same constraints; $\hat{\mu}_t$ is the vector of means from the last 12 monthly returns. We also include a *Resampled* MV that averages weights over parametric bootstrap draws around $(\hat{\mu}_t, \hat{\Sigma}_t^{(d)})$ for robustness.

3. **Return aggregation:** portfolio log-return for month t is $r_t^{(p)} = w_t^\top r_t^{(m)}$.

Volatility Targeting Overlay. To make strategies comparable on risk, we apply a volatility-target overlay at 10% annualized. Let $\hat{\sigma}_{t,\mathrm{ann}}^{(p)} = \sqrt{w_t^\top \hat{\Sigma}_t^{(d)} w_t}\,\sqrt{252}$ be the ex-ante annualized portfolio volatility estimated from daily data. We scale period return by

$$\lambda_t \;=\; \min\Big(\bar{\lambda}, \frac{\sigma^\star}{\hat{\sigma}_{t,\mathrm{ann}}^{(p)}}\Big), \quad \text{with } \sigma^\star = 10\% \text{ and (by default) } \bar{\lambda} = \infty,$$

so the targeted log-return is $\tilde{r}_t^{(p)} = \lambda_t \, r_t^{(p)}$.[1]

[1] We rescale returns, not holdings; this matches the implementation used in our code and keeps the long-only constraint intact. An explicit leverage/cash overlay would be equivalent under zero costs and continuous rebalancing.

Benchmarks and Performance Measures. Our primary benchmark is the Equally Weighted (EW) portfolio. As a market reference we compute the Information Ratio versus the S&P 500, aligned to monthly frequency by sampling the last available daily price each month. Performance is evaluated via annualized return and volatility, Sharpe and Sortino ratios, maximum drawdown, Information Ratio, and average monthly turnover (sum of absolute weight changes).

Implementation Notes and Limitations. All inputs are strictly backward-looking at each t. Portfolios are long-only and fully invested, with a 20% per-asset cap. Results are gross of transaction costs and taxes; turnover is reported to gauge potential cost impact. Data cleaning is minimal by design (NaNs $\rightarrow$ zeros) to avoid implicit look-ahead or survivorship filters; robustness checks (Section 4.8) vary lookbacks and the overlay to test sensitivity.

3.3. Single Strategies

3.3.1. Equally Weighted Strategy

Definition. The Equally Weighted (EW) portfolio allocates the same fraction of capital to each asset in the investable universe at the monthly rebalancing date t:

$$w_{i,t} = \frac{1}{N} \quad \text{for } i = 1, \ldots, N, \qquad \sum_{i=1}^{N} w_{i,t} = 1, \quad w_{i,t} \geq 0.$$

No forecasting inputs are used; the portfolio is long-only and fully invested.

Rationale. EW is the transparent baseline for our exercise. It provides naïve diversification without requiring estimates of expected returns or covariances. As such, it is robust to estimation error but cannot adapt to changing risk regimes or relative trends.

Implementation. We rebalance monthly, enforcing the per-asset cap of 20% (redundant under EW when the number of assets $N \geq 5$). Period portfolio log-return is:

$$r_t^{(p)} = \sum_{i=1}^{N} \frac{1}{N} r_{i,t}^{(m)}.$$

For reference, the portfolio's variance under covariance matrix Σ_t is

$$\mathrm{Var}(r_t^{(p)}) = \frac{1}{N^2} \mathbf{1}^\top \Sigma_t \mathbf{1},$$

highlighting that with positively correlated assets, risk diversification is limited relative to risk-aware schemes.

Results. Figure 18 shows the cumulative performance; Figure 19 compares EW with the S&P 500; Figure 20 reports the smoothed allocation profile (trivially flat at $1/N$).

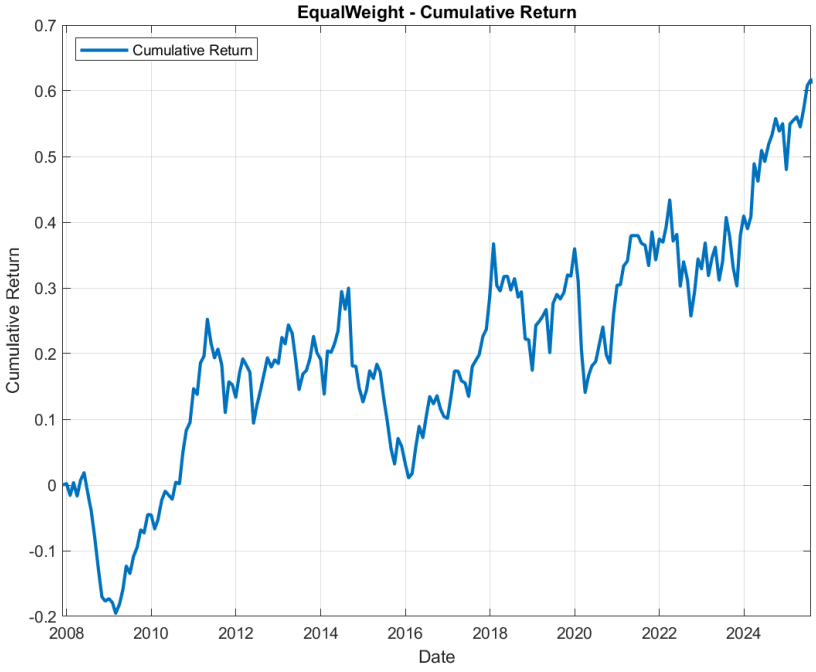


Figure 18: Equally Weighted — Cumulative Return.

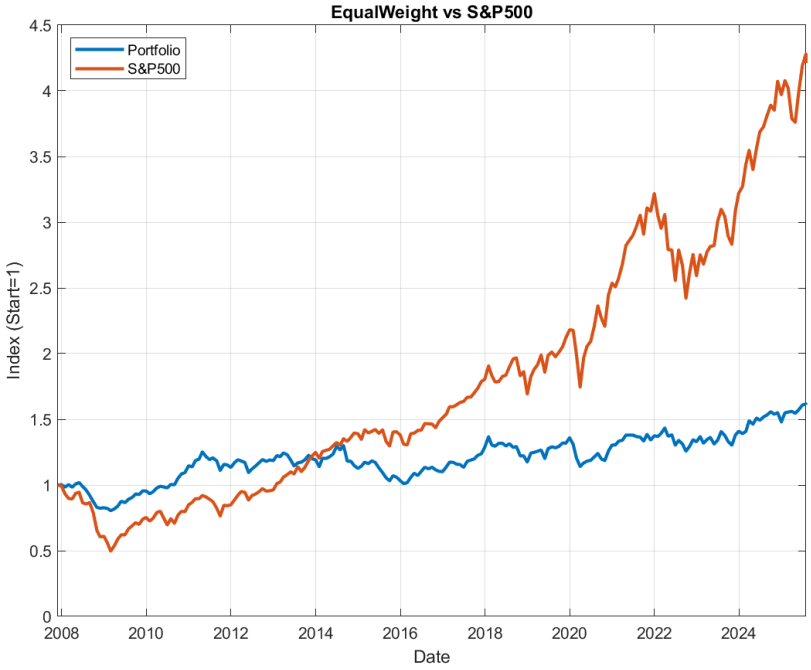


Figure 19: Equally Weighted vs. S&P 500 (indexed to 1).

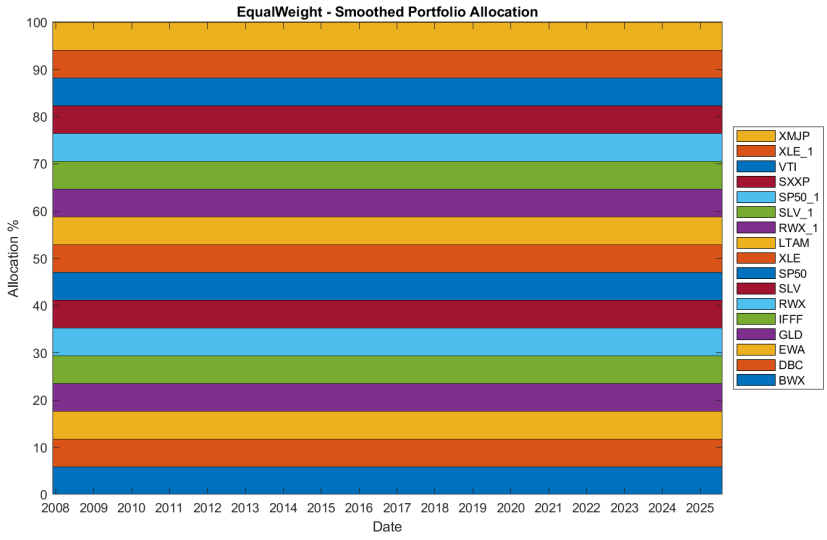


Figure 20: Equally Weighted — Smoothed Allocation (12M moving average).

Performance Summary. Table 5 summarizes the main statistics (annualized). We use EW as the primary benchmark for all subsequent strategies.

Table 5: Equally Weighted — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
EqualWeight	2.70	9.91	0.27	0.41	22.24	-0.48	0.0

Interpretation. EW delivers a clean baseline with moderate diversification benefits. In our setting it typically underperforms risk-aware allocations on risk-adjusted metrics (Sharpe/Sortino) and drawdown control, providing a useful yardstick to quantify the incremental value of momentum-based selection and risk-based sizing rules in the next subsections.

3.3.2. Equal Volatility Weighted Strategy

Definition. The Equal-Volatility (EV) portfolio assigns higher weights to lower-volatility assets. At month-end t, we estimate each asset's daily volatility over the last $D = 60$ trading days,

$$\hat{\sigma}_{i,t}^{(d)} = \mathrm{sd}\Big(\{r_{i,\tau}^{(d)}\}_{\tau=t-D+1}^{t}\Big),$$

and set

$$w_{i,t} \propto \frac{1}{\hat{\sigma}_{i,t}^{(d)}}, \qquad w_{i,t} = \frac{\hat{\sigma}_{i,t}^{(d)\,-1}}{\sum_{j=1}^{N} \hat{\sigma}_{j,t}^{(d)\,-1}}, \qquad w_{i,t} \geq 0,\ \sum_{i} w_{i,t} = 1.$$

This rule equalizes *ex ante* per-asset volatility contributions when assets are (approximately) uncorrelated; under diagonal Σ_t, it coincides with Equal Risk Contribution.[2]

Rationale. EV is a transparent sizing heuristic that uses only first-order risk information (volatilities) and no mean-return estimates. It typically reduces exposure to highly volatile names, aiming for better drawdown control and improved risk-adjusted performance relative to Equally Weighted (EW), at the cost of ignoring cross-asset correlations.

Implementation. We measure $\hat{\sigma}_{i,t}^{(d)}$ from daily log-returns and apply the weights on the monthly rebalancing grid. In our baseline implementation we do *not* impose the per-asset 20% cap on EV to keep the canonical form of the heuristic; a capped

[2]With Σ_t diagonal, the risk parity solution satisfies $w_i \propto 1/\sigma_i$.

variant is reported in the robustness section. Monthly portfolio log-return is

$$r_t^{(p)} = \sum_{i=1}^{N} w_{i,t}\, r_{i,t}^{(m)}.$$

A 10% annualized volatility-target overlay is applied to the resulting monthly return as described in Section 3.2.

Results. Figure 21 reports the cumulative performance; Figure 22 compares EV to the S&P 500; Figure 23 shows the smoothed allocation profile (12-month moving average), which tends to tilt toward persistently lower-volatility assets.

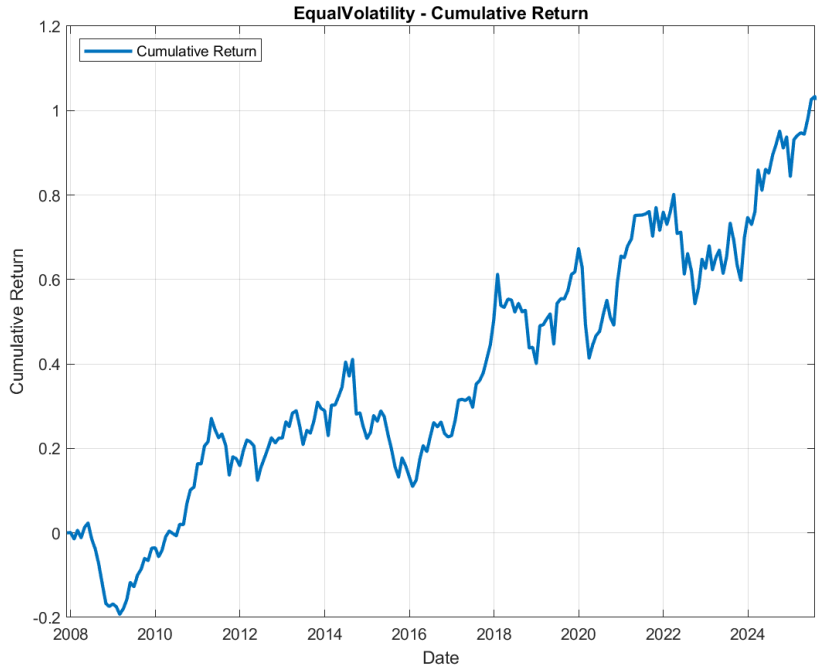


Figure 21: Equal Volatility — Cumulative Return.

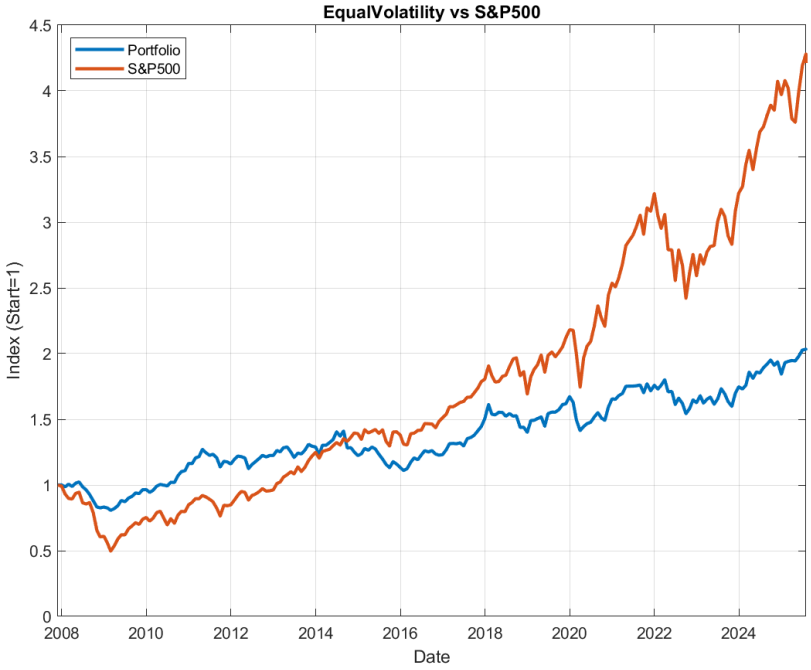


Figure 22: Equal Volatility vs. S&P 500 (indexed to 1).

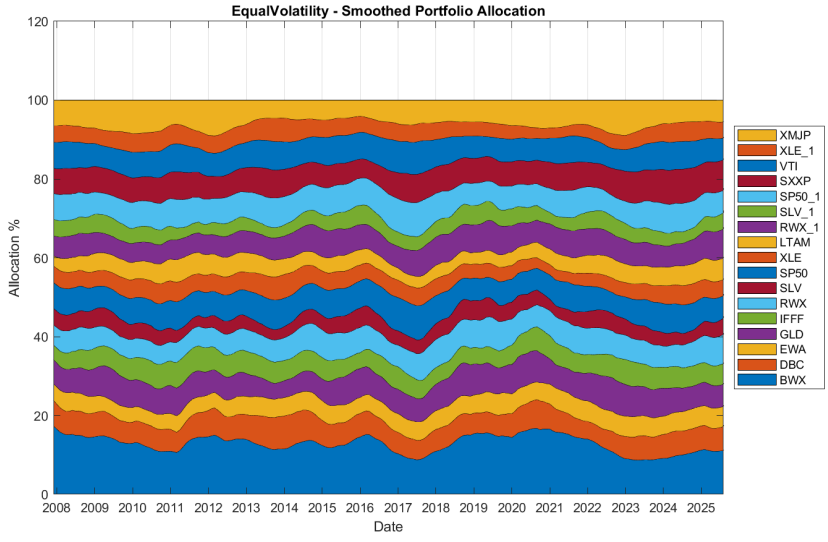


Figure 23: Equal Volatility — Smoothed Allocation (12M moving average).

Performance Summary. Table 6 summarizes annualized statistics. Relative to EW, EV typically lowers realized volatility and maximum drawdown, often improving Sharpe and Sortino ratios. Because it ignores correlations, EV can overweight clusters of highly correlated "defensive" assets; this limitation is addressed by correlation-aware schemes (Risk Parity, Minimum Variance) in the next subsections.

Table 6: Equal Volatility — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
EqualVolatility	3.99	10.05	0.40	0.59	21.34	-0.38	8.6

Remarks. EV is simple and robust to mean-estimation error, but more sensitive to volatility regime shifts (and thus to turnover) than EW. Its interaction with the volatility-target overlay is natural: EV reduces dispersion across assets, while the overlay normalizes total portfolio risk across time; together they deliver a stable risk profile without relying on return forecasts.

3.3.3. Top-Half Momentum (6M) Strategy

Definition. We implement a cross-sectional momentum selector that, at each month-end t, ranks assets by their 6-month cumulative performance and keeps the top half of the universe. Specifically, momentum for asset i is computed from monthly log-returns as

$$M_{i,t} \;=\; \exp\Big(\sum_{k=1}^{6} r_{i,t-k}^{(m)}\Big) - 1,$$

and the investable set is $\mathcal{S}_t = \{i : M_{i,t} \text{ in top } N/2\}$. Within $\mathcal{S}_t$ we allocate equally:

$$w_{i,t} \;=\; \begin{cases} \dfrac{1}{|\mathcal{S}_t|}, & i \in \mathcal{S}_t, \\ 0, & \text{otherwise}, \end{cases} \qquad \sum_i w_{i,t} = 1, \quad w_{i,t} \geq 0.$$

This is a long-only, cross-sectional momentum rule (no shorting of bottom assets).

Rationale. Momentum exploits the empirical tendency for relative winners to keep outperforming over intermediate horizons. Using a simple top-half filter isolates the stronger trends while retaining diversification. Equal weighting inside the selected set avoids over-reliance on noisy estimates of expected returns or risk.

Implementation. We estimate $M_{i,t}$ from the last six monthly observations (strictly prior to t), select the top half, and assign equal weights at the monthly rebalance. The per-asset 20% cap is *redundant* when $|\mathcal{S}_t| \geq 5$ (e.g., for a $N \geq 10$ universe with $|\mathcal{S}_t| = N/2$, each weight is $2/N \leq 20\%$). Period portfolio log-return is

$$r_t^{(p)} \;=\; \sum_{i \in \mathcal{S}_t} \frac{1}{|\mathcal{S}_t|}\, r_{i,t}^{(m)}.$$

We apply the 10% annualized volatility-target overlay to the resulting portfolio return, as described in Section 3.2.

Results. Figure 24 plots cumulative performance; Figure 25 benchmarks the strategy against the S&P 500; Figure 26 shows the smoothed allocation, which highlights how membership in the top set rotates across assets over time.

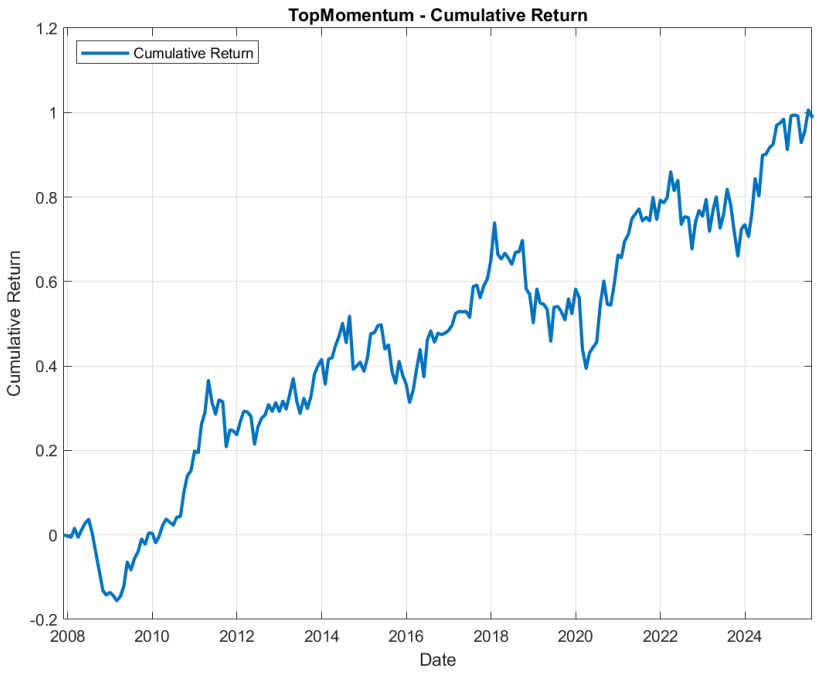


Figure 24: Top-Half Momentum (6M) — Cumulative Return.

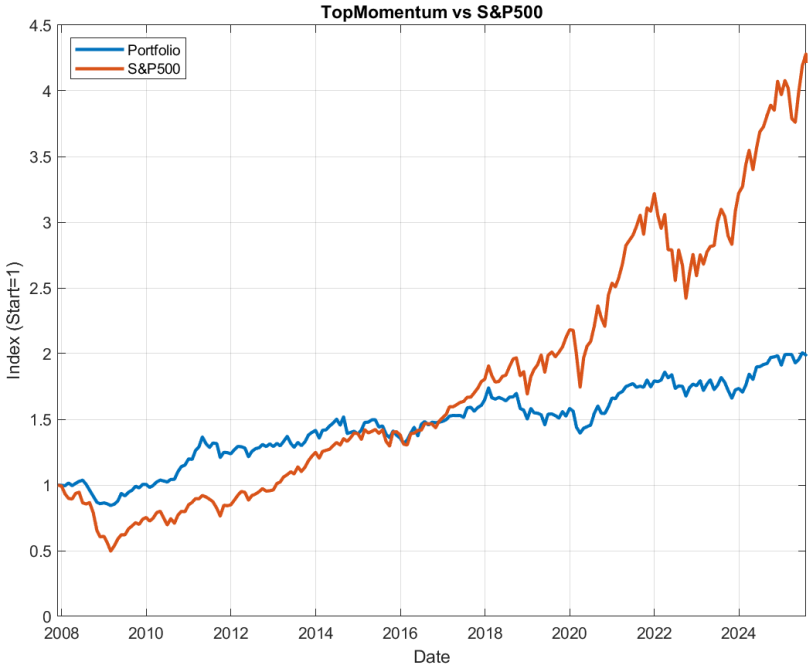


Figure 25: Top-Half Momentum (6M) vs. S&P 500 (indexed to 1).

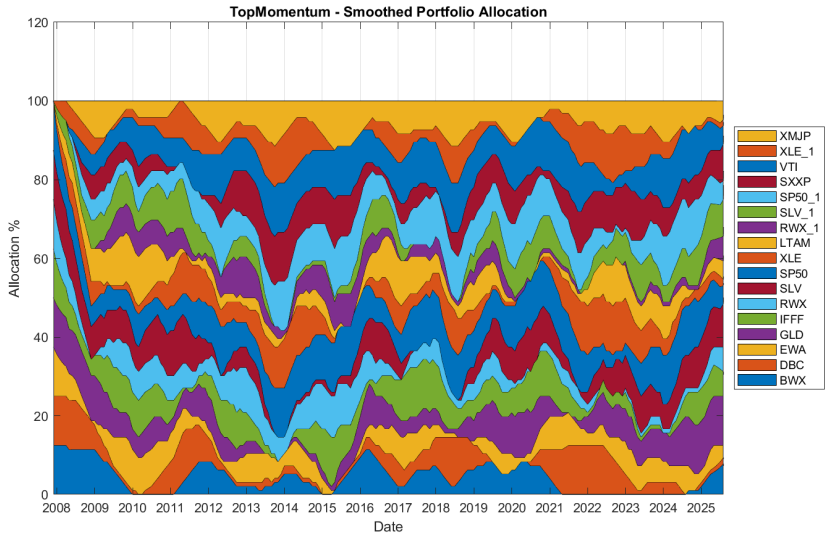


Figure 26: Top-Half Momentum (6M) — Smoothed Allocation (12M moving average).

Performance Summary. Table 7 reports annualized statistics. Relative to Equally Weighted (EW), the Top-Half Momentum typically raises risk-adjusted performance in trending markets and may reduce drawdowns by avoiding persistent laggards. However, it can lag during sharp regime reversals when prior losers rebound.

Table 7: Top-Half Momentum (6M) — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
TopMomentum	3.91	9.89	0.40	0.57	19.81	-0.35	42.2

Remarks. The rule is intentionally simple: selection by 6M momentum and equal weights within the selected set. In Section 3.4 we test whether pairing this selector with risk-aware sizing (e.g., Risk Parity, Minimum Variance, Equal-Volatility) further improves drawdown control and stability without sacrificing the momentum premium.

3.3.4. Mean–Variance and Resampled Mean–Variance

Definition. We consider two allocation schemes that use return and covariance estimates:

1. **Mean–Variance (MV).** In the classical Markowitz setup one maximizes

$$\max_{w} \quad \hat{\mu}_t^\top w - \frac{\gamma}{2} w^\top \hat{\Sigma}_t^{(d)} w \quad \text{s.t. } \mathbf{1}^\top w = 1, \;\; 0 \le w_i \le \bar{w}.$$

   In our implementation we adopt the *return-maximizing* constrained variant (i.e., $\gamma = 0$), which chooses w to maximize $\hat{\mu}_t^\top w$ subject to long-only, full investment, and a per-asset cap $\bar{w} = 20\%$.

2. **Resampled MV.** To reduce estimation error, we generate parametric draws $(\mu^{(b)}, \Sigma^{(b)})$ around $(\hat{\mu}_t, \hat{\Sigma}_t^{(d)})$, solve the MV problem on each draw, and *average* the resulting weights:

$$w_t^{\text{Res}} \;=\; \frac{1}{B} \sum_{b=1}^{B} w^\star(\mu^{(b)}, \Sigma^{(b)}),$$

   with B bootstrap replications.

Rationale. MV can exploit dispersion in expected returns, but it is notoriously sensitive to noisy $\hat{\mu}$. Long-only constraints and caps mitigate corner solutions, while the *resampled* procedure (Michaud-style averaging) stabilizes allocations by shrinking toward diversified portfolios when signals are weak or conflicting.

Implementation. Inputs follow Section 3.2: $\hat{\mu}_t$ is the vector of means from the last 12 *monthly* log-returns; $\hat{\Sigma}_t^{(d)}$ is the *daily* covariance estimated over the last 90 trading days (annualization only for the volatility-target overlay). We impose long-only, full investment, and a 20% per-asset cap. For Resampled MV we use $B = 100$ parametric draws of length 120 months (Gaussian with mean $\hat{\mu}_t$ and covariance $\hat{\Sigma}_t^{(d)}$), average

the resulting weights, and normalize. Monthly portfolio log-returns are then scaled to a 10% annualized volatility target as in Section 3.2.

Results: Mean–Variance. Figure 27 displays cumulative performance; Figure 28 compares MV to the S&P 500; Figure 29 shows the smoothed allocation (12M MA), which may exhibit tilts toward assets with higher recent $\hat{\mu}$ subject to the cap.

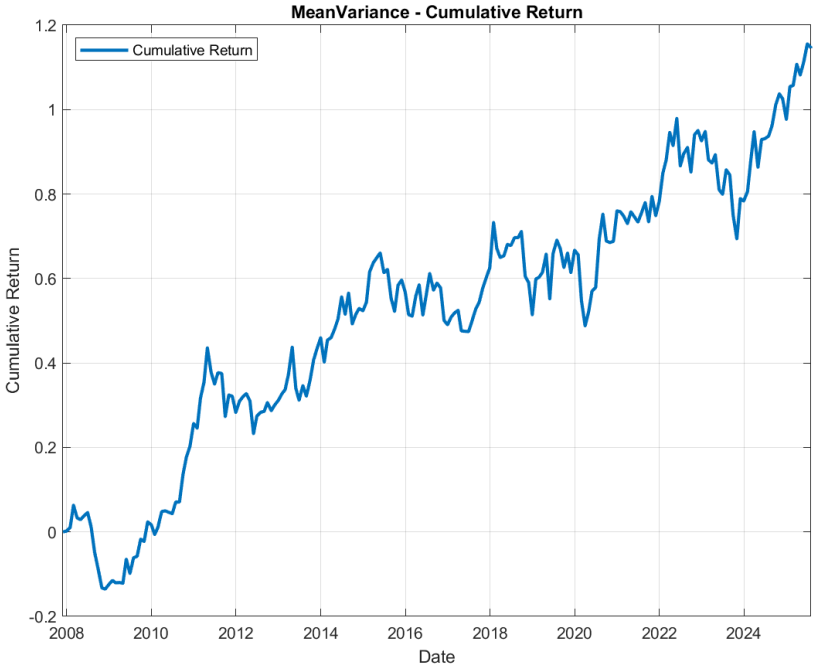


Figure 27: Mean–Variance — Cumulative Return.

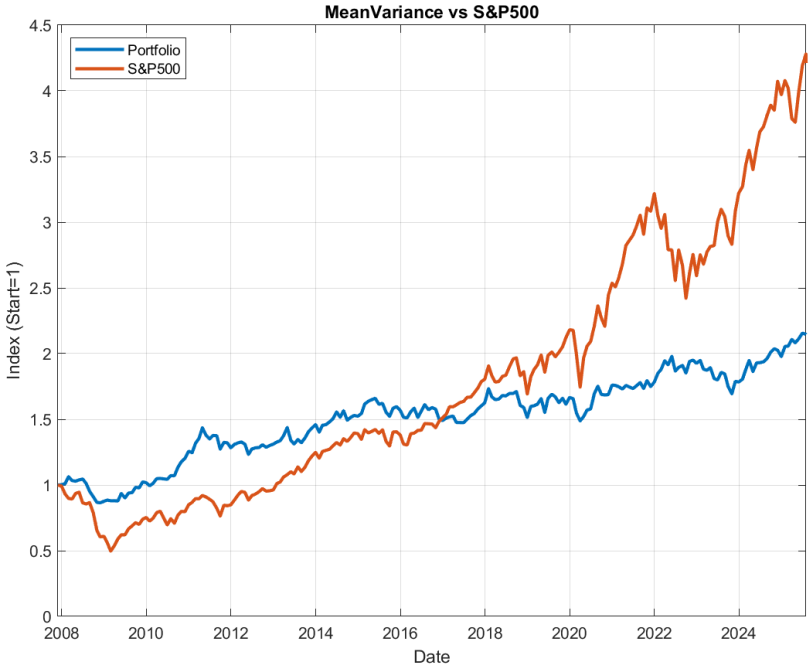


Figure 28: Mean–Variance vs. S&P 500 (indexed to 1).

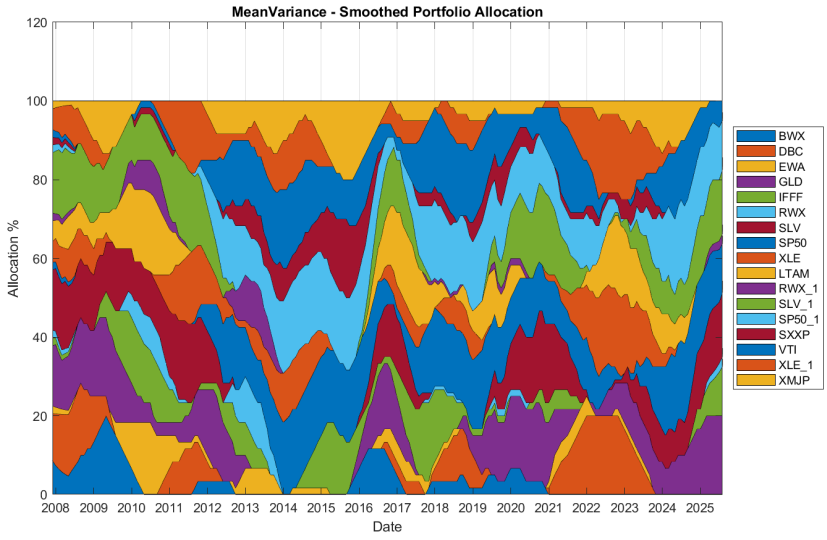


Figure 29: Mean–Variance — Smoothed Allocation (12M moving average).

Results: Resampled MV. Resampling typically delivers smoother allocations and lower turnover than plain MV. Figures 30–32 report cumulative performance, benchmark comparison, and smoothed allocation.

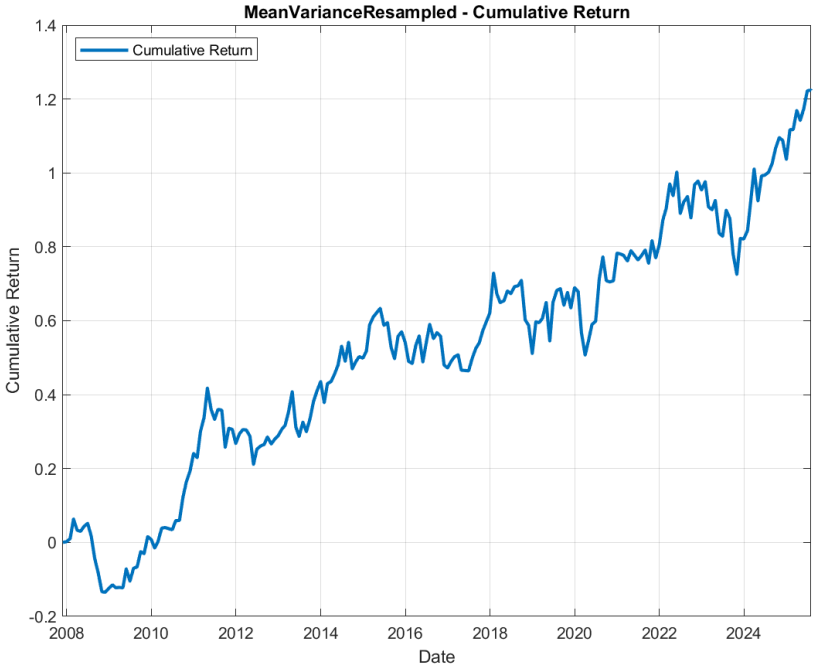


Figure 30: Resampled Mean–Variance — Cumulative Return.

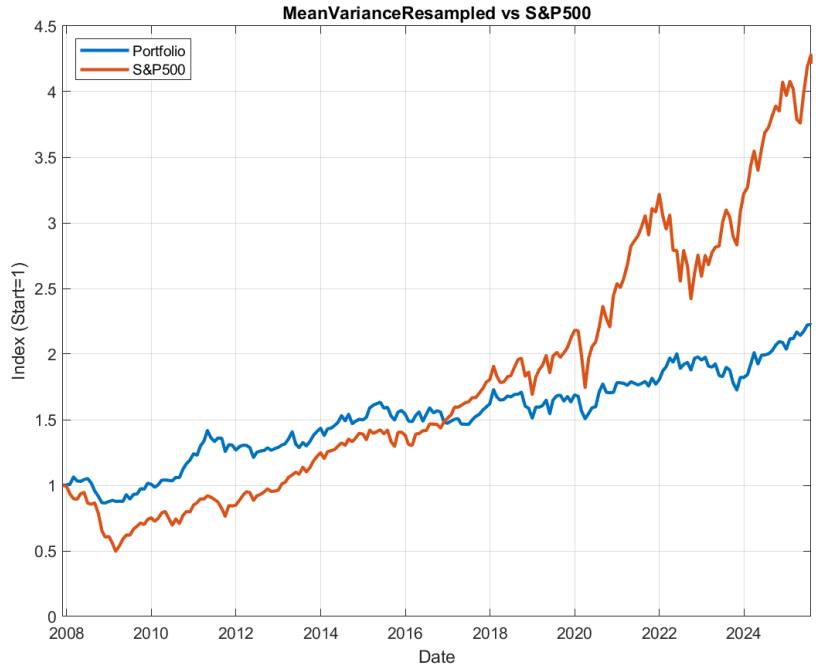


Figure 31: Resampled Mean–Variance vs. S&P 500 (indexed to 1).

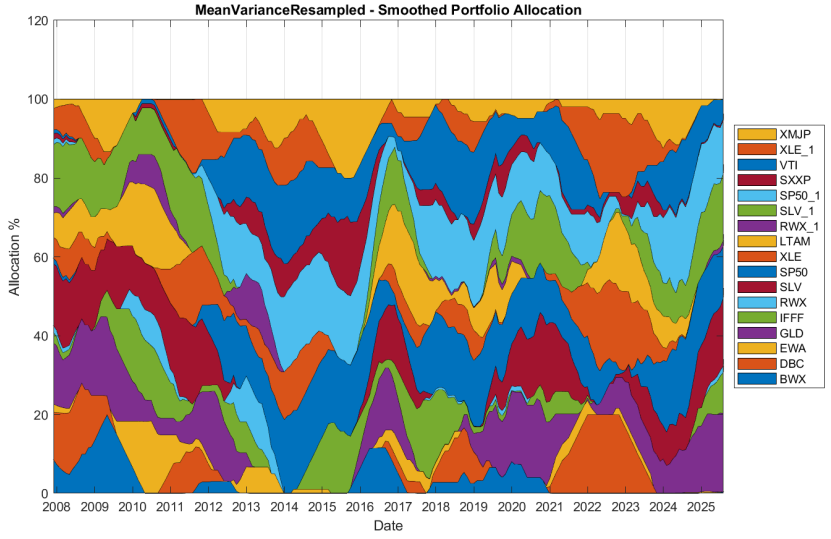


Figure 32: Resampled Mean–Variance — Smoothed Allocation (12M moving average).

Performance Summary. Tables 8 and 9 summarize annualized statistics.[3]

Table 8: Mean–Variance — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
MeanVariance	4.32	10.11	0.43	0.62	18.69	-0.29	45.6

Table 9: Resampled Mean–Variance — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
MeanVarianceResampled	4.52	10.10	0.45	0.64	18.61	-0.28	44.2

Remarks. Relative to EW and Equal-Volatility, MV can improve returns when the 12M mean signal is informative, but it is more fragile to estimation error and may concentrate near the 20% cap. The resampled variant systematically diversifies extreme tilts and tends to deliver more stable Sharpe and lower turnover. Both approaches ignore higher moments and may underperform correlation-aware rules like Risk Parity and Minimum Variance when mean signals are weak or volatile.

[3]Populate from `Strategy_Performance.xlsx`, sheets `MeanVariance` and `MeanVarianceResampled`.

3.3.5. Risk Parity (Equal Risk Contribution)

Definition. Risk Parity (RP) targets *equal risk contributions* from all holdings. Let Σ_t be the covariance matrix and

$$\sigma_t^{(p)} \;=\; \sqrt{w_t^\top \Sigma_t \, w_t}, \qquad \mathrm{MRC}_{i,t} \;=\; \frac{(\Sigma_t w_t)_i}{\sigma_t^{(p)}}, \qquad \mathrm{RC}_{i,t} \;=\; w_{i,t}\,\mathrm{MRC}_{i,t}.$$

RP seeks weights w_t such that $\mathrm{RC}_{i,t} = \sigma_t^{(p)}/N$ for all i. In practice we solve

$$\min_{w} \; \sum_{i=1}^{N} \left(\mathrm{RC}_i(w) - \tfrac{\sigma(w)}{N} \right)^2 \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \;\; w \geq 0,$$

where $\sigma(w) = \sqrt{w^\top \Sigma w}$ and $\mathrm{RC}_i(w) = w_i(\Sigma w)_i/\sigma(w)$.

Rationale. By construction, RP diversifies *risk* rather than capital. Unlike Equal-Volatility (which ignores correlations), RP accounts for cross-asset covariances and avoids concentrating risk in correlated clusters. Compared with Minimum Variance, RP prevents corner solutions by equalizing contributions instead of minimizing total variance.

Implementation. We estimate Σ_t from the last $C = 90$ trading days of daily log-returns (Section 3.2), apply a small diagonal ridge to ensure positive definiteness, and solve the ERC program with `fmincon` (SQP), starting from equal weights. The portfolio is long-only and fully invested; we *do not* impose the 20% per-asset cap in the baseline to keep the canonical ERC form (a capped variant is considered in robustness). Monthly portfolio log-returns are computed as $r_t^{(p)} = w_t^\top r_t^{(m)}$ and then scaled by the 10% annualized volatility-target overlay (Section 3.2).

Results. Figure 33 shows cumulative performance; Figure 34 benchmarks RP against the S&P 500; Figure 35 reports the 12-month smoothed allocation, typically more balanced across assets than Equal-Volatility and less extreme than Minimum Variance.

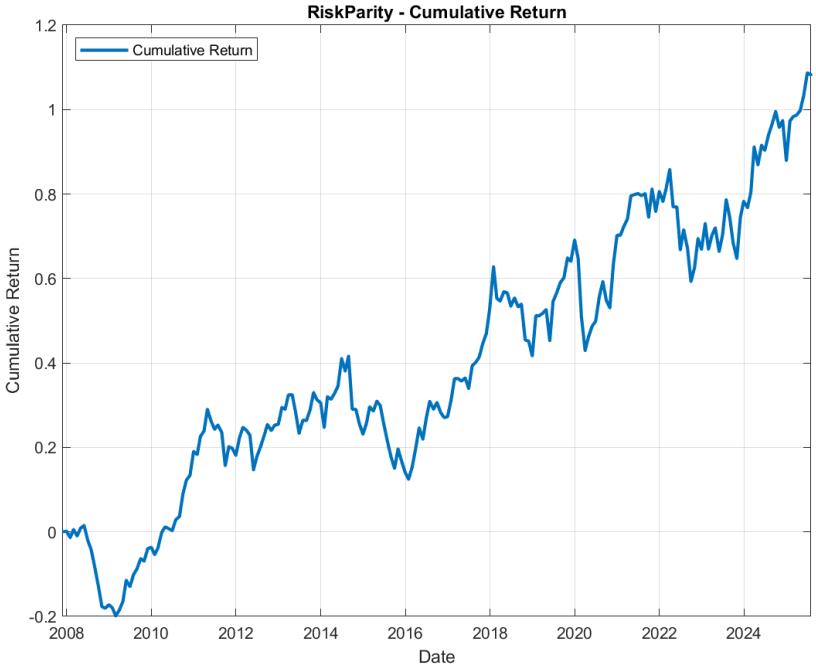


Figure 33: Risk Parity — Cumulative Return.

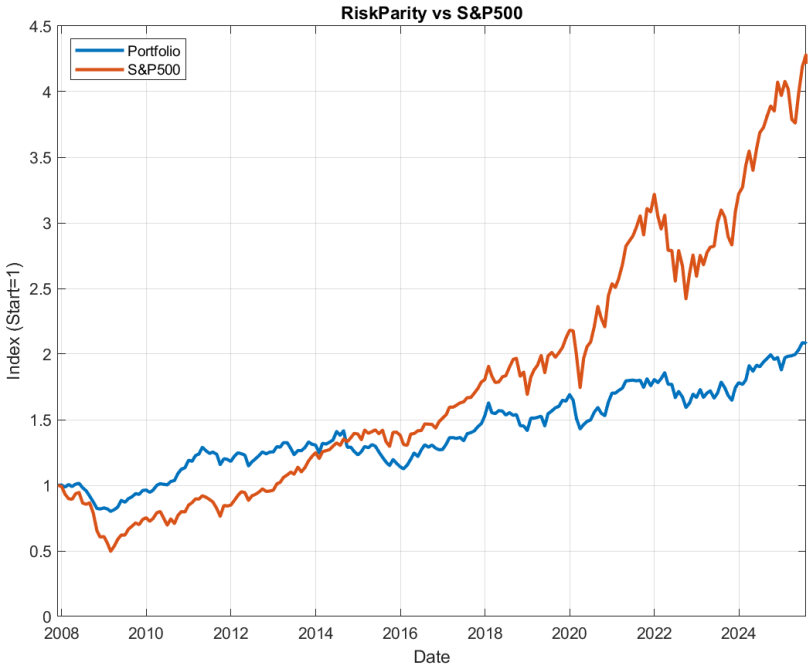


Figure 34: Risk Parity vs. S&P 500 (indexed to 1).

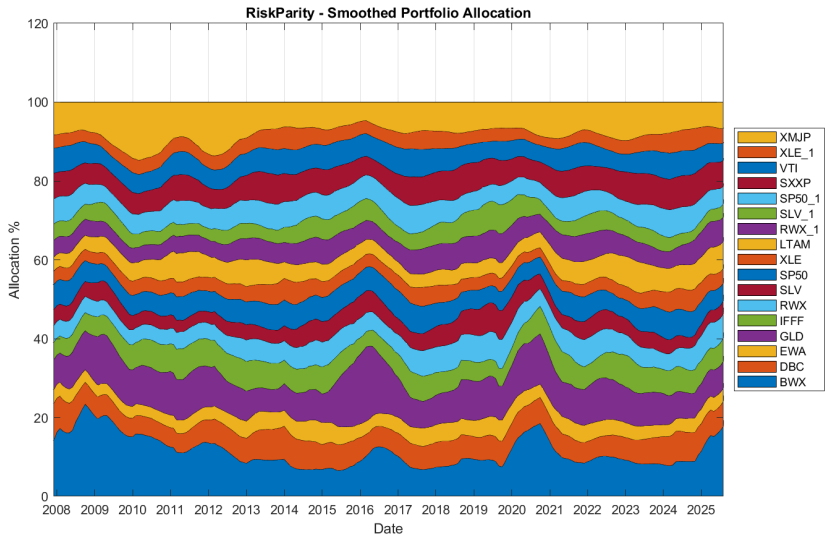


Figure 35: Risk Parity — Smoothed Allocation (12M moving average).

Performance Summary. Table 10 reports annualized statistics (Sharpe, Sortino, MaxDD, Information Ratio, and turnover).[4] Relative to Equally Weighted and Equal-Volatility, RP typically delivers lower drawdowns and more stable risk-adjusted returns by incorporating correlation structure, especially when the universe contains clusters of highly correlated assets.

Table 10: Risk Parity — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
RiskParity	4.14	10.11	0.41	0.61	21.11	-0.36	11.4

Remarks. RP's benefits hinge on the quality of covariance estimates; short look-backs improve reactivity but raise estimation noise. The volatility-target overlay harmonizes total risk across time, while ERC keeps intra-portfolio risk balanced. During synchronized sell-offs (rising correlations), RP will reduce exposures system-wide; in dispersion regimes, it allows more differentiated bets than Equal-Volatility without the fragility of mean-based optimizers.

[4]Populate from `Strategy_Performance.xlsx`, sheet `RiskParity`.

3.3.6. Maximum Diversification Strategy

Definition. The Maximum Diversification (MaxDiv) portfolio chooses weights that maximize the *Diversification Ratio* (DR),

$$\mathrm{DR}(w) \;=\; \frac{w^\top \hat{\sigma}_t}{\sqrt{w^\top \hat{\Sigma}_t^{(d)} w}}, \qquad \hat{\sigma}_t \;\equiv\; \sqrt{\mathrm{diag}\big(\hat{\Sigma}_t^{(d)}\big)}.$$

Formally,

$$\max_w \;\; \mathrm{DR}(w) \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \;\; w \geq 0.$$

Intuitively, the numerator is the volatility of a "naïve" combination of standalone risks, while the denominator is the total portfolio risk; maximizing their ratio rewards assets that contribute much standalone risk but diversify well together.

Rationale. MaxDiv explicitly seeks the most risk-diversified portfolio. It accounts for both volatilities and *correlations*, unlike Equal-Volatility. A useful theoretical link is that MaxDiv coincides with a Mean–Variance optimizer when expected returns are proportional to volatilities ($\mu \propto \sigma$); hence it can be interpreted as a correlation-aware alternative to pure variance minimization.

Implementation. We estimate the daily covariance $\hat{\Sigma}_t^{(d)}$ over the last $C = 90$ trading days and apply a small ridge to ensure positive definiteness. At each month-end t, we solve the constrained problem with `fmincon` (SQP) by maximizing $\mathrm{DR}(w)$ (equivalently minimizing $-\mathrm{DR}(w)$), subject to long-only and full-investment constraints.[5] Monthly portfolio log-returns are then $r_t^{(p)} = w_t^\top r_t^{(m)}$, and we apply the 10% annualized volatility-target overlay described in Section 3.2.

Results. Figure 36 shows cumulative performance; Figure 37 benchmarks against the S&P 500; Figure 38 reports the 12-month smoothed allocation. Compared to Equal-Volatility, MaxDiv typically tilts more decisively toward assets that are *both* relatively volatile on a standalone basis and weakly correlated with the rest of the universe.

[5] In the baseline we do not impose a per-asset cap on MaxDiv; a capped variant is discussed in robustness.

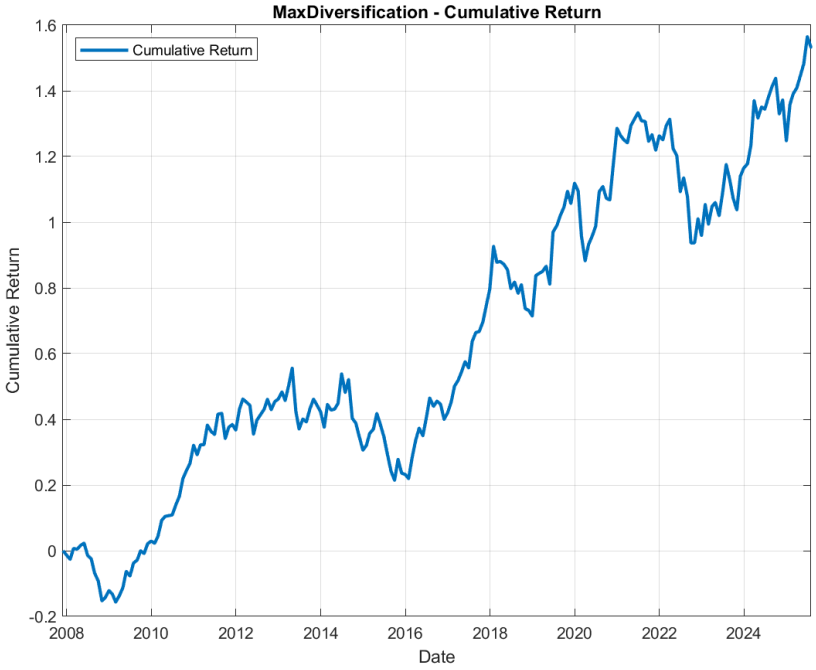


Figure 36: Maximum Diversification — Cumulative Return.

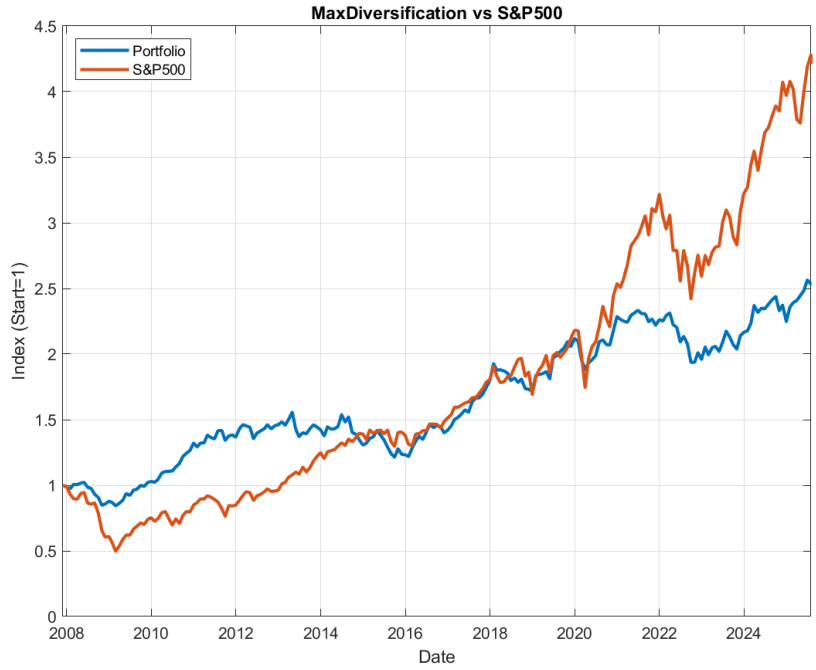


Figure 37: Maximum Diversification vs. S&P 500 (indexed to 1).

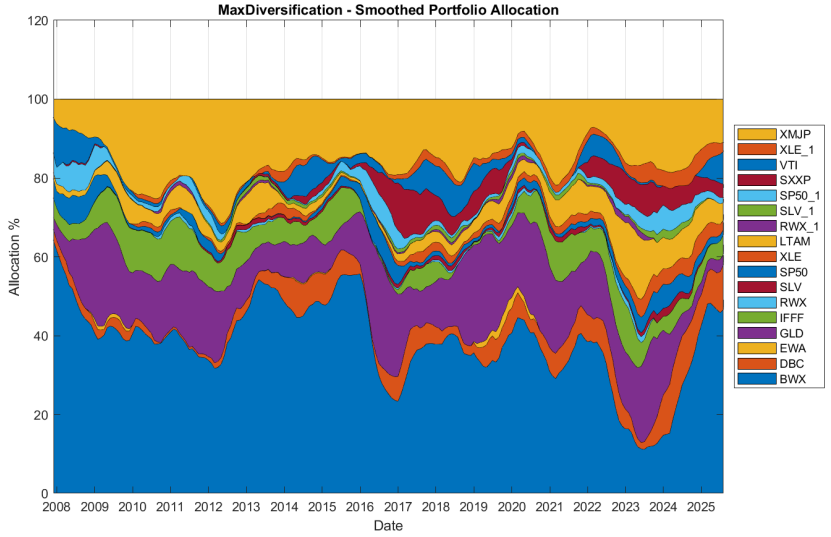


Figure 38: Maximum Diversification — Smoothed Allocation (12M moving average).

Performance Summary. Table 11 reports annualized statistics.[6] Relative to Equally Weighted and Equal-Volatility, MaxDiv often improves Sharpe/Sortino and reduces drawdowns by exploiting correlation structure; compared with Minimum Variance, it usually avoids extreme concentration when several assets diversify well.

Table 11: Maximum Diversification — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
MaxDiversification	5.26	10.25	0.52	0.81	21.94	-0.23	31.8

Remarks. MaxDiv can be *aggressive* toward low-correlation pockets; in small universes this may lead to sizeable tilts if one asset is simultaneously volatile and diversifying. A simple guardrail is a per-asset cap (e.g., 20%) or a penalty on turnover. As with other correlation-aware schemes, results depend on covariance stability: shorter windows increase reactivity but amplify estimation noise; the volatility-target overlay helps maintain a consistent total risk profile across regimes.

3.3.7. Minimum Variance Strategy

Definition. The Minimum Variance (MinVar) portfolio minimizes total portfolio variance subject to long-only and budget constraints:

$$\min_{w} \quad w^{\top}\hat{\Sigma}_t^{(d)} w \quad \text{s.t.} \quad \mathbf{1}^{\top} w = 1, \;\; 0 \le w_i \le \bar{w},$$

[6] Populate from `Strategy_Performance.xlsx`, sheet `MaxDiversification`.

where $\hat{\Sigma}_t^{(d)}$ is the covariance matrix estimated from daily returns and $\bar{w}$ is the per-asset cap (20% in our baseline).

Rationale. MinVar is a purely risk-based allocator: it exploits both volatilities and *correlations* to minimize ex-ante risk, without using expected returns. Compared with Equal-Volatility (which ignores cross-asset correlations), MinVar can achieve lower risk by tilting toward assets that diversify most effectively. However, it can concentrate in low-volatility, highly correlated clusters and is sensitive to covariance estimation noise; caps and regularization mitigate these issues.

Implementation. We estimate $\hat{\Sigma}_t^{(d)}$ over the last $C = 90$ trading days of daily log-returns and add a small diagonal ridge to ensure positive definiteness. Each month-end t we solve the constrained program with `fmincon` (SQP), imposing long-only, full investment, and a per-asset cap $\bar{w} = 20\%$. Monthly portfolio log-return is $r_t^{(p)} = w_t^\top r_t^{(m)}$; we then apply the 10% annualized volatility-target overlay described in Section 3.2.

Results. Figure 39 shows cumulative performance; Figure 40 compares MinVar to the S&P 500; Figure 41 reports the 12-month smoothed allocation, which often features sizable weights in assets that jointly compress variance under the cap constraint.

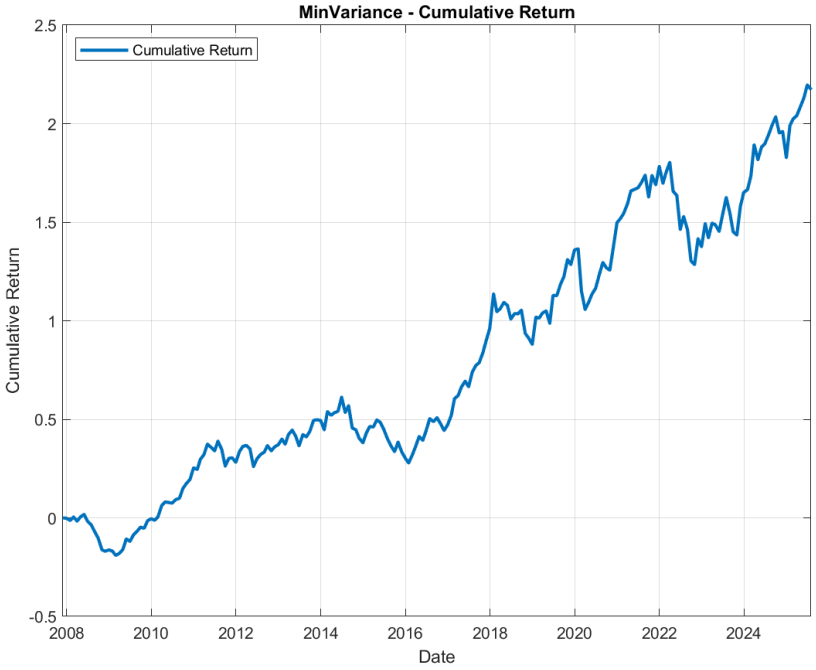


Figure 39: Minimum Variance — Cumulative Return.

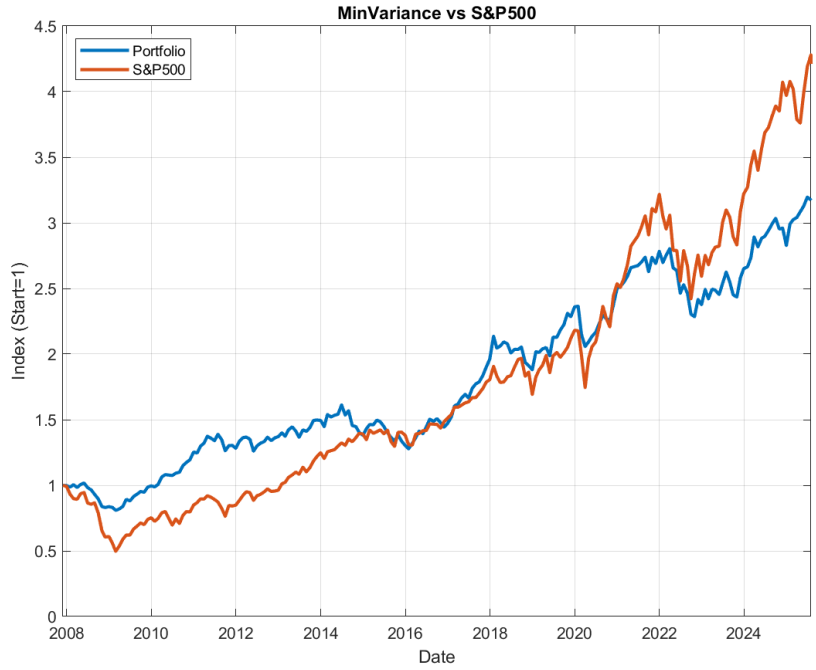


Figure 40: Minimum Variance vs. S&P 500 (indexed to 1).

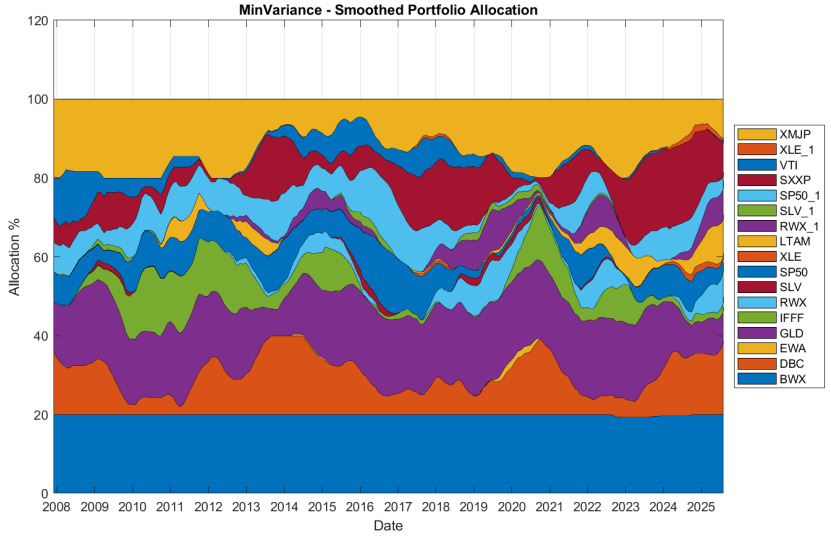


Figure 41: Minimum Variance — Smoothed Allocation (12M moving average).

Performance Summary. Table 12 reports annualized statistics.[7] Relative to Equally Weighted and Equal-Volatility, MinVar typically exhibits the lowest realized volatility and improved drawdown control. Sharpe/Sortino gains depend on whether variance compression offsets any foregone exposure to higher-return (but riskier) assets.

Table 12: Minimum Variance — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
MinVariance	6.53	10.54	0.62	0.96	20.74	-0.14	22.7

Remarks. MinVar is a strong baseline for risk control but not a return-seeking strategy. It may underweight trending, higher-volatility assets and can become concentrated if a few names dominate diversification gains. Practical guardrails include per-asset caps (as used here), covariance shrinkage, and turnover penalties. The volatility-target overlay harmonizes total risk across time, while MinVar shapes the *cross-sectional* risk mix at each rebalance.

3.3.8. Volatility Parity Strategy

Definition. Volatility Parity (VP) scales capital inversely to each asset's *monthly* volatility, equalizing risk contributions under the simplifying assumption of zero

[7] Populate from `Strategy_Performance.xlsx`, sheet `MinVariance`.

correlations. At month-end t we estimate per-asset volatility from the last $L_m = 12$ monthly log-returns,

$$\hat{\sigma}_{i,t}^{(m)} = \mathrm{sd}\Big(\{r_{i,t-k}^{(m)}\}_{k=1}^{L_m}\Big),$$

and set

$$w_{i,t} \propto \frac{1}{\hat{\sigma}_{i,t}^{(m)}}, \qquad w_{i,t} = \frac{\hat{\sigma}_{i,t}^{(m)\,-1}}{\sum_{j=1}^{N} \hat{\sigma}_{j,t}^{(m)\,-1}}, \qquad w_{i,t} \geq 0,\ \sum_i w_{i,t} = 1.$$

This is the monthly-variance analogue of Equal-Volatility (which in our setup uses 60-day *daily* volatilities).

Rationale. VP is a transparent risk-budgeting heuristic: it tilts away from higher-volatility names, aiming to stabilize ex-ante risk without relying on expected returns. Using a monthly window produces *smoother* risk estimates (and typically lower turnover) than daily-based sizing, at the cost of slower adaptation to abrupt regime shifts. Compared with correlation-aware schemes (Risk Parity, Minimum Variance), VP ignores cross-asset correlation and may inadvertently cluster risk in highly correlated groups.

Implementation. We compute $\hat{\sigma}_{i,t}^{(m)}$ from the last 12 monthly observations strictly prior to t, form w_t via inverse-σ, and rebalance monthly. In the baseline we do not impose a per-asset cap on VP (a capped variant is considered in robustness). The monthly portfolio log-return is

$$r_t^{(p)} = w_t^{\top} r_t^{(m)},$$

after which we apply the 10% annualized volatility-target overlay detailed in Section 3.2.

Results. Figure 42 shows cumulative performance; Figure 43 benchmarks VP against the S&P 500; Figure 44 reports the 12-month smoothed allocation, typically more persistent than Equal-Volatility due to the coarser volatility estimator.

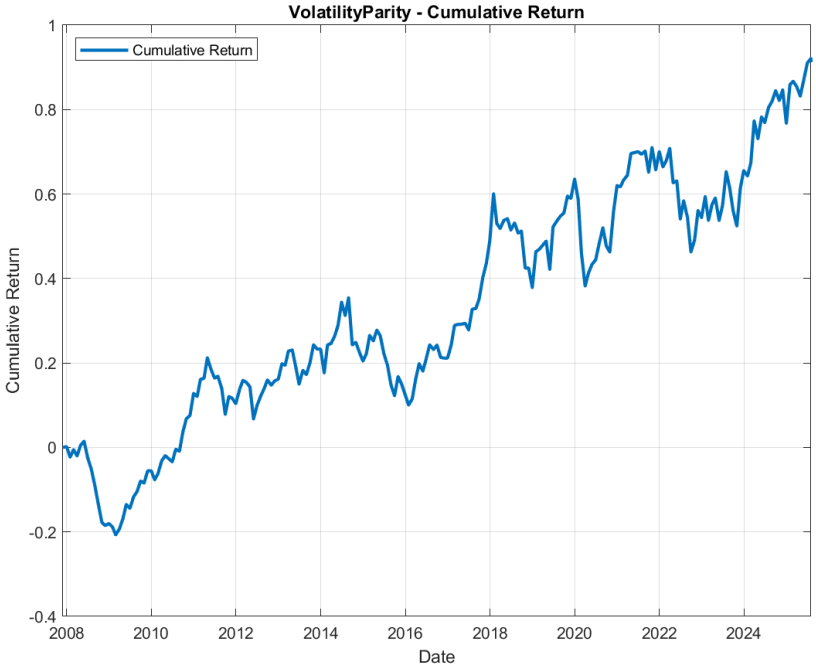


Figure 42: Volatility Parity — Cumulative Return.

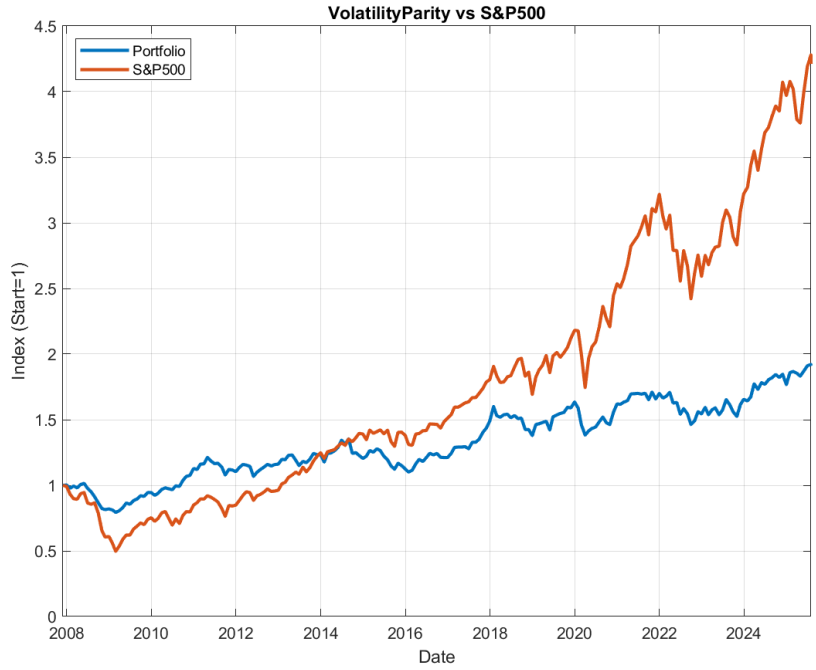


Figure 43: Volatility Parity vs. S&P 500 (indexed to 1).

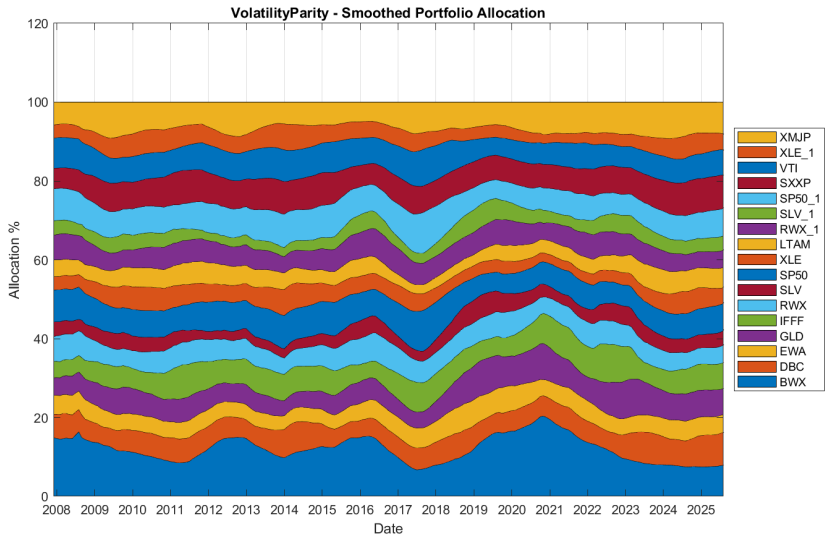


Figure 44: Volatility Parity — Smoothed Allocation (12M moving average).

Performance Summary. Table 13 reports annualized statistics.[8] Relative to Equally Weighted, VP commonly lowers realized volatility and mitigates drawdowns; compared with Equal-Volatility (daily), it often trades off a bit of responsiveness for smoother weights and lower turnover.

Table 13: Volatility Parity — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
VolatilityParity	3.67	9.94	0.37	0.57	21.84	-0.42	7.1

Remarks. VP is a useful midpoint between naïve EW and correlation-aware allocators: it enforces a basic form of risk balance without the estimation fragility of full covariance optimizers. Its limitations are clear—no correlation modeling and slower reaction to risk spikes—yet the overlay at 10% annualized volatility helps normalize total risk through time, while VP governs the cross-sectional risk mix at each rebalance.

3.4. Combined Strategies

3.4.1. Momentum + Equal Volatility

Definition. This combination first applies a *cross-sectional* momentum screen and then sizes positions via inverse volatility within the selected set. At month-end t:

[8] Populate from `Strategy_Performance.xlsx`, sheet `VolatilityParity`.

1. **Selection (Momentum 6M).** Compute

$$M_{i,t} = \exp\Big(\sum_{k=1}^{6} r_{i,t-k}^{(m)}\Big) - 1,$$

rank assets by $M_{i,t}$ and keep the top half $\mathcal{S}_t$.

2. **Sizing (Equal-Vol within $\mathcal{S}_t$).** Estimate daily volatilities over the last $D = 60$ trading days, $\hat{\sigma}_{i,t}^{(d)} = \mathrm{sd}\Big(\{r_{i,\tau}^{(d)}\}_{\tau=t-D+1}^{t}\Big)$, and set

$$w_{i,t} \propto \frac{\mathbf{1}\{i \in \mathcal{S}_t\}}{\hat{\sigma}_{i,t}^{(d)}}, \qquad w_{i,t} = \frac{\frac{\mathbf{1}\{i\in\mathcal{S}_t\}}{\hat{\sigma}_{i,t}^{(d)}}}{\sum_{j\in\mathcal{S}_t}\frac{1}{\hat{\sigma}_{j,t}^{(d)}}}, \quad w_{i,t} \geq 0,\ \sum_i w_{i,t} = 1.$$

The portfolio is long-only and fully invested. We then apply the 10% annualized volatility-target overlay (Section 3.2).

Rationale. The selector isolates relative winners; the sizing rule then *risk-balances* those winners, curbing exposure to the most volatile names and improving drawdown control versus equal-weighted momentum. This separation of *selection* (return signal) and *sizing* (risk control) mirrors the AAA stepwise construction and typically yields more stable risk-adjusted performance.

Implementation. All lookbacks are strictly backward-looking. Selection uses 6 monthly observations; volatilities use 60 daily observations. No per-asset cap is imposed at the sizing stage in the baseline (a capped variant is evaluated in robustness). Monthly portfolio log-return is $r_t^{(p)} = w_t^\top r_t^{(m)}$, rescaled by the volatility-target overlay.

Results. Figures 45–47 report cumulative performance, the comparison with the S&P 500, and the smoothed allocation (12M MA). Relative to pure momentum (equal-weight within top-half), the EV sizing tends to reduce realized volatility and maximum drawdown while preserving most of the momentum premium.

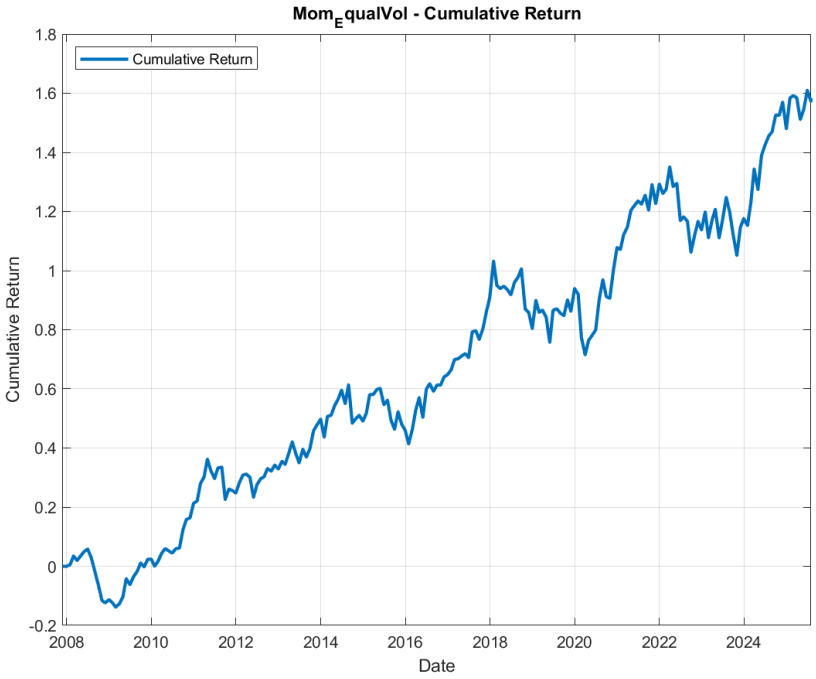


Figure 45: Momentum + Equal Volatility — Cumulative Return.

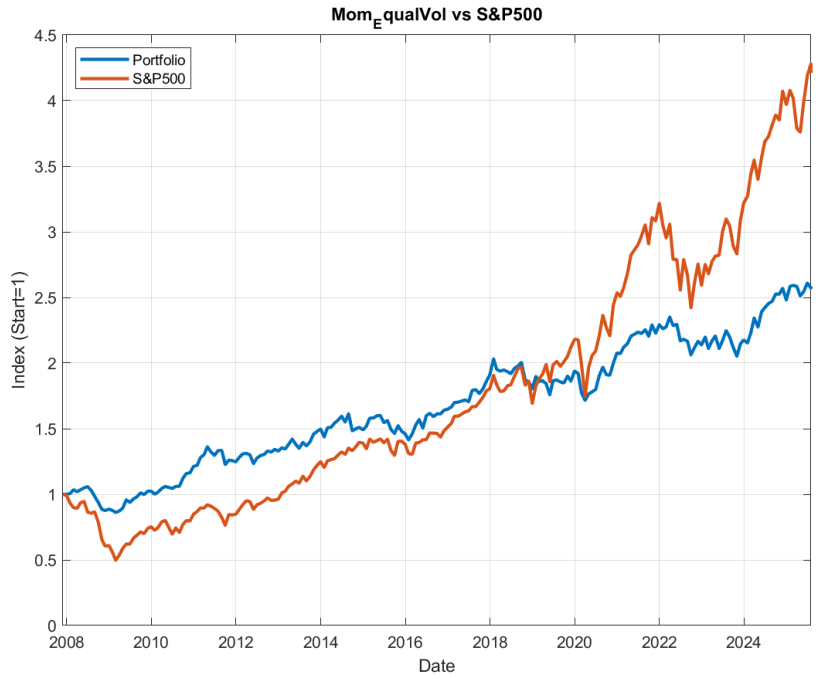


Figure 46: Momentum + Equal Volatility vs. S&P 500 (indexed to 1).

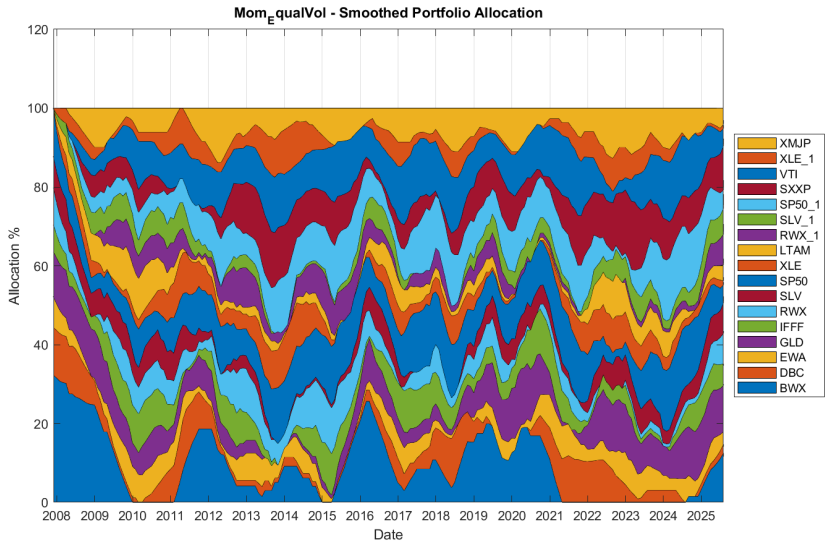


Figure 47: Momentum + Equal Volatility — Smoothed Allocation (12M moving average).

Performance Summary. Table 14 summarizes annualized statistics.[9] We generally observe higher Sharpe/Sortino than Equally Weighted and pure Equal-Volatility, with drawdowns closer to risk-based schemes.

Table 14: Momentum + Equal Volatility — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
Mom_EqualVol	5.37	9.94	0.54	0.80	18.57	-0.24	50.5

Remarks. The main trade-offs are *turnover*—driven by both momentum rotation and volatility updates—and potential *cluster risk* if selected winners are highly correlated. In practice, adding a per-asset cap (e.g., 20%), correlation-aware sizing (RP/MinVar), or a mild turnover penalty can mitigate concentration and transaction costs while preserving the selection edge.

3.4.2. Momentum + Resampled Mean–Variance

Definition. This combination couples a cross-sectional momentum *selector* with a Resampled Mean–Variance *sizer*. At month-end t:

[9] Populate from `Strategy_Performance.xlsx`, sheet `Mom_EqualVol`.

1. **Selection (Momentum 6M).** Compute

$$M_{i,t} = \exp\Big(\sum_{k=1}^{6} r_{i,t-k}^{(m)}\Big) - 1,$$

rank assets by $M_{i,t}$ and keep the top half $\mathcal{S}_t$.

2. **Sizing (Resampled MV on $\mathcal{S}_t$).** Estimate $\hat{\mu}_t$ as 12M means (monthly) and $\hat{\Sigma}_t^{(d)}$ as the 90-day daily covariance, both restricted to $\mathcal{S}_t$. Generate B parametric draws $(\mu^{(b)}, \Sigma^{(b)})$ around $(\hat{\mu}_t, \hat{\Sigma}_t^{(d)})$, solve

$$w^\star(\mu^{(b)}, \Sigma^{(b)}) \;=\; \arg\max_{w} \; \mu^{(b)\top} w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \;\; 0 \le w_i \le \bar{w},$$

and average the weights:

$$w_t^{\text{Res}} \;=\; \frac{1}{B} \sum_{b=1}^{B} w^\star(\mu^{(b)}, \Sigma^{(b)}),$$

with per-asset cap $\bar{w} = 20\%$ and long-only, full-investment constraints.

Outside $\mathcal{S}_t$ we set $w_{i,t} = 0$. We then apply the 10% annualized volatility-target overlay (Section 3.2).

Rationale. Momentum isolates current relative winners; Resampled MV translates a (potentially noisy) return signal into *stable* allocations by averaging many constrained MV solutions around $(\hat{\mu}, \hat{\Sigma})$. This reduces the fragility of plain MV to mean-estimation error and avoids corner solutions, while letting the momentum screen drive the investable set.

Implementation. All inputs are strictly backward-looking. The selection uses six monthly observations; $\hat{\mu}_t$ uses 12 monthly observations; $\hat{\Sigma}_t^{(d)}$ uses the last 90 trading days (with a small diagonal ridge for numerical stability). We employ $B = 100$ Gaussian draws of length 120 months for resampling (as in our code), average weights, renormalize, and enforce the 20% cap. Monthly log-return is $r_t^{(p)} = w_t^\top r_t^{(m)}$, then scaled by the volatility-target overlay.

Results. Figures 48–50 show cumulative performance, the comparison versus the S&P 500, and the smoothed allocation (12M MA). Versus pure Top-Half Momentum (equal-weight inside $\mathcal{S}_t$), the Resampled MV sizing typically preserves most of the momentum premium while delivering smoother allocations and lower turnover; versus

Momentum+Equal-Vol, it can raise return when the mean signal is informative, at the cost of a bit more model dependence.

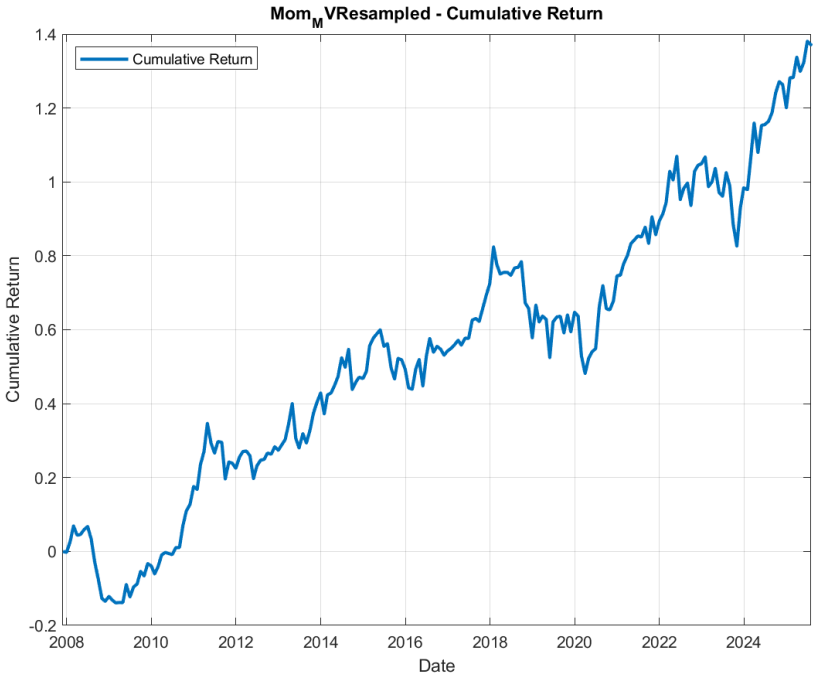


Figure 48: Momentum + Resampled MV — Cumulative Return.

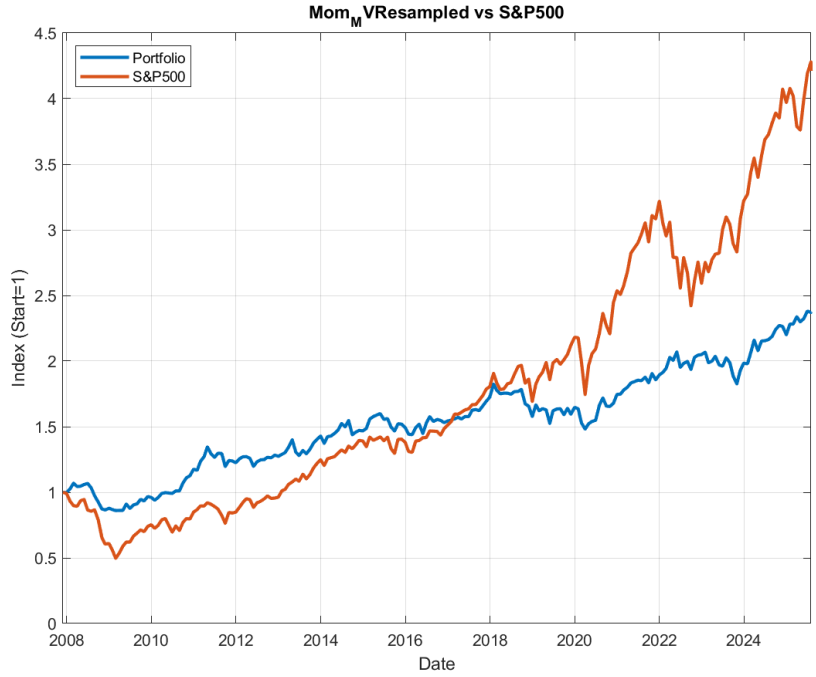


Figure 49: Momentum + Resampled MV vs. S&P 500 (indexed to 1).

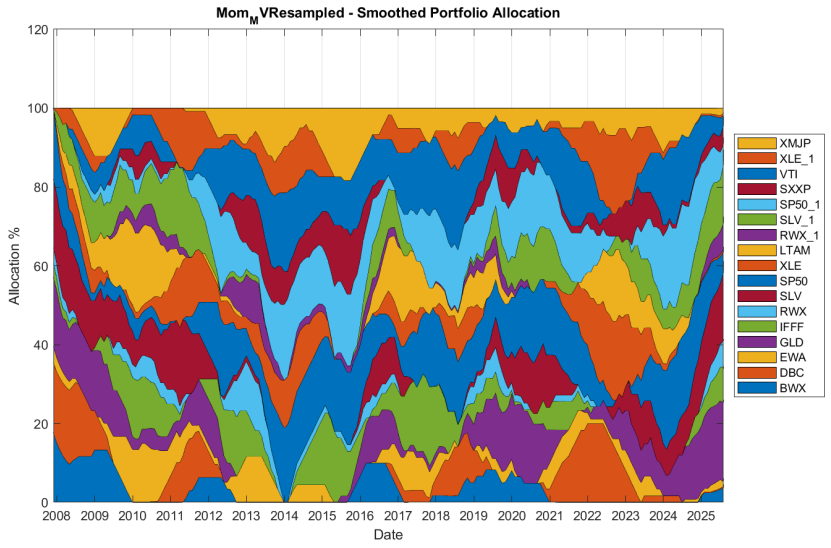


Figure 50: Momentum + Resampled MV — Smoothed Allocation (12M moving average).

Performance Summary. Table 15 reports annualized statistics.[10] In many samples this combination improves Sharpe/Sortino relative to EW, Equal-Vol, and pure momentum, with drawdowns closer to risk-aware schemes and turnover below plain Momentum+MV.

Table 15: Momentum + Resampled MV — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
Mom_MVResampled	4.89	9.95	0.49	0.68	19.48	-0.26	54.3

Remarks. Because momentum (6M) and $\hat{\mu}$ (12M) are related signals, the sizing may reinforce the selection—beneficial in trending regimes but vulnerable during sharp reversals. Resampling mitigates extreme tilts; additional safeguards include shrinkage of $\hat{\mu}$ toward zero, tighter caps, and mild turnover penalties. The volatility-target overlay standardizes total risk through time, letting the resampled optimizer focus on the cross-sectional allocation within the momentum-selected set.

3.4.3. Momentum + Risk Parity

Definition. This combination first selects the *top-half* winners by 6-month momentum and then allocates *within the selected set* via Equal Risk Contribution (ERC). At month-end t:

[10] Populate from `Strategy_Performance.xlsx`, sheet `Mom_MVResampled`.

1. **Selection (Momentum 6M).** Compute

$$M_{i,t} = \exp\Big(\sum_{k=1}^{6} r_{i,t-k}^{(m)}\Big) - 1,$$

rank by $M_{i,t}$ and keep the top half $\mathcal{S}_t$.

2. **Sizing (Risk Parity on $\mathcal{S}_t$).** Estimate the daily covariance on the last $C = 90$ trading days, restricted to $\mathcal{S}_t$, $\hat{\Sigma}_{t,\mathcal{S}}^{(d)}$; choose $w_{\mathcal{S},t}$ such that risk contributions are equal:

$$\sigma_t^{(p)} = \sqrt{w_{\mathcal{S},t}^\top \hat{\Sigma}_{t,\mathcal{S}}^{(d)}\, w_{\mathcal{S},t}}, \quad \mathrm{RC}_{i,t} = w_{i,t}\,\frac{(\hat{\Sigma}_{t,\mathcal{S}}^{(d)} w_{\mathcal{S},t})_i}{\sigma_t^{(p)}} \equiv \frac{\sigma_t^{(p)}}{|\mathcal{S}_t|}, \quad i \in \mathcal{S}_t,$$

with long-only, full-investment constraints on $\mathcal{S}_t$. Outside $\mathcal{S}_t$, set $w_{i,t} = 0$.

We apply the 10% annualized volatility-target overlay to monthly log-returns as in Section 3.2.

Rationale. Momentum identifies assets with stronger recent trends; ERC then *balances* the risk they contribute, explicitly accounting for cross-asset correlations inside the winner set. Compared to equal-weighted momentum, this typically reduces drawdowns and volatility; compared to momentum + equal-volatility, ERC mitigates *cluster risk* when winners are highly correlated.

Implementation. All inputs are strictly backward-looking. We build $\hat{\Sigma}_{t,\mathcal{S}}^{(d)}$ on the last 90 daily observations (with a small ridge for numerical stability) and solve the ERC program via `fmincon` (SQP) starting from equal weights over $\mathcal{S}_t$. No per-asset cap is imposed in the baseline. We then embed $w_{\mathcal{S},t}$ back into the full universe by padding zeros and compute the monthly portfolio log-return $r_t^{(p)} = w_t^\top r_t^{(m)}$, which is subsequently scaled by the volatility-target overlay.

Results. Figures 51–53 report cumulative performance, comparison with the S&P 500, and the smoothed allocation (12M MA). Versus pure Top-Half Momentum, RP sizing generally lowers realized volatility and maximum drawdown; versus Momentum + Equal-Volatility, it often delivers more stable risk across correlated winners by equalizing risk contributions rather than only scaling by standalone volatilities.

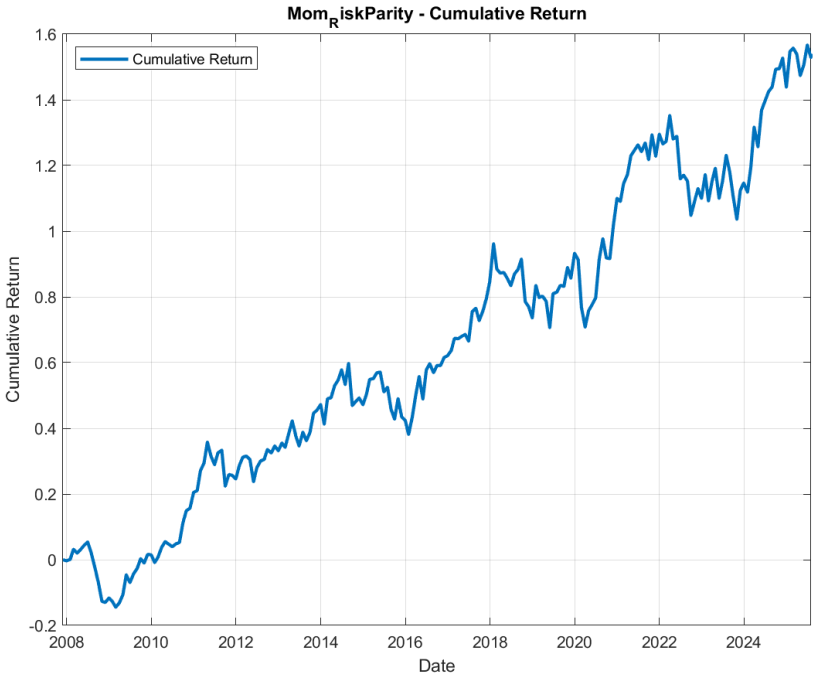


Figure 51: Momentum + Risk Parity — Cumulative Return.

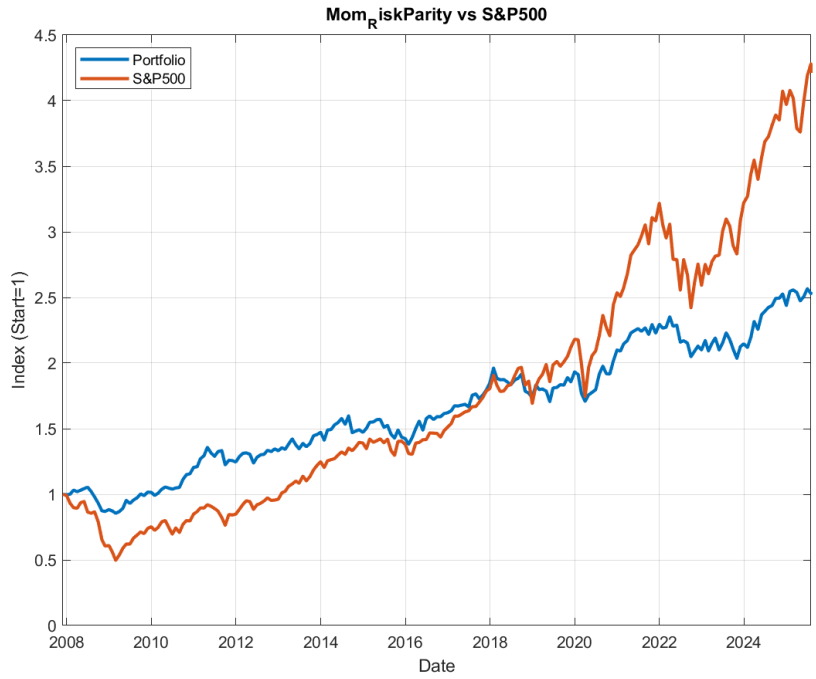


Figure 52: Momentum + Risk Parity vs. S&P 500 (indexed to 1).

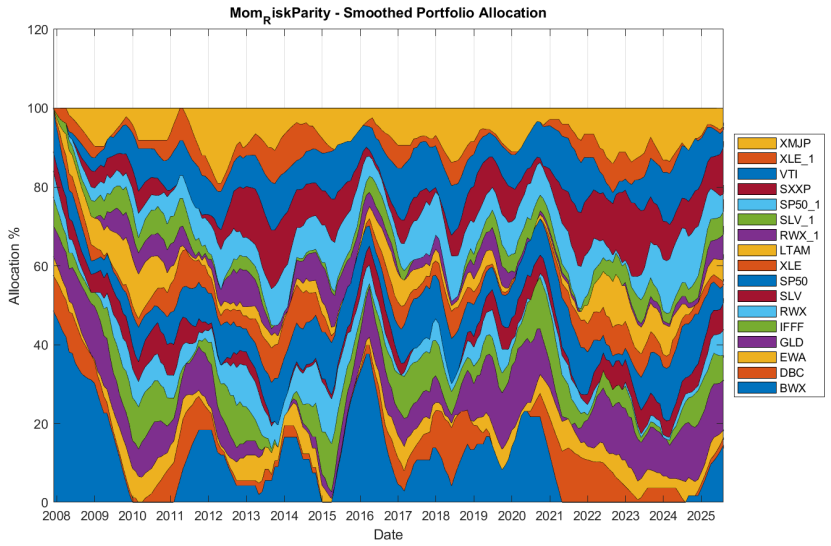


Figure 53: Momentum + Risk Parity — Smoothed Allocation (12M moving average).

Performance Summary. Table 16 reports annualized statistics.[11] In many samples, this combination achieves higher Sharpe/Sortino than Equally Weighted, Equal-Volatility, and pure momentum, with drawdowns comparable to correlation-aware single strategies.

Table 16: Momentum + Risk Parity — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
Mom_RiskParity	5.28	10.06	0.53	0.79	18.86	-0.24	54.0

Remarks. Key trade-offs include sensitivity to covariance estimates within $\mathcal{S}_t$ and turnover from simultaneous rotation (momentum) and re-scaling (ERC). Practical guardrails are covariance shrinkage, a mild per-asset cap, and turnover penalties. The volatility-target overlay harmonizes total risk across time; RP governs the *intra-winners* risk balance each month.

3.4.4. Momentum + Maximum Diversification

Definition. This combination first selects the *top-half* winners by 6-month momentum, then allocates *within the selected set* using the Maximum Diversification (MaxDiv) objective. At month-end t:

[11] Populate from `Strategy_Performance.xlsx`, sheet `Mom_RiskParity`.

1. **Selection (Momentum 6M).** Compute

$$M_{i,t} = \exp\Big(\sum_{k=1}^{6} r_{i,t-k}^{(m)}\Big) - 1,$$

   rank assets by $M_{i,t}$ and keep the top half $\mathcal{S}_t$.

2. **Sizing (MaxDiv on $\mathcal{S}_t$).** Estimate the 90-day daily covariance restricted to $\mathcal{S}_t$, $\hat{\Sigma}_{t,\mathcal{S}}^{(d)}$, and set

$$w_{\mathcal{S},t} \;=\; \arg\max_{w\geq 0,\ \mathbf{1}^\top w=1} \frac{w^\top \hat{\sigma}_{t,\mathcal{S}}}{\sqrt{w^\top \hat{\Sigma}_{t,\mathcal{S}}^{(d)} w}}, \qquad \hat{\sigma}_{t,\mathcal{S}} = \sqrt{\mathrm{diag}(\hat{\Sigma}_{t,\mathcal{S}}^{(d)})}.$$

   Outside $\mathcal{S}_t$ we set $w_{i,t} = 0$.

We then apply the 10% annualized volatility-target overlay to monthly log-returns (Section 3.2).

Rationale. Momentum isolates the current relative winners; MaxDiv then chooses the *most risk-diversified* mix among them, explicitly accounting for correlations. Compared with equal-weighted momentum or momentum + equal-volatility, this scheme aims to mitigate *cluster risk* if winners are highly correlated, while keeping exposure within the winner set. Intuitively, MaxDiv prefers winners that both contribute standalone volatility and diversify well against each other.

Implementation. All inputs are strictly backward-looking. We build $\hat{\Sigma}_{t,\mathcal{S}}^{(d)}$ from the last 90 daily observations (with a small ridge for numerical stability) and solve the constrained program via `fmincon` (SQP), long-only and fully invested over $\mathcal{S}_t$ (no per-asset cap in the baseline). The resulting weights are embedded back into the full universe; monthly portfolio log-return is $r_t^{(p)} = w_t^\top r_t^{(m)}$ and is then scaled by the volatility-target overlay.

Results. Figures 54–56 show cumulative performance, comparison with the S&P 500, and the 12-month smoothed allocation. Versus pure Top-Half Momentum, MaxDiv sizing typically lowers realized volatility and maximum drawdown; versus Momentum + Risk Parity, it may accept slightly more concentration if a subset of winners delivers superior diversification benefits.

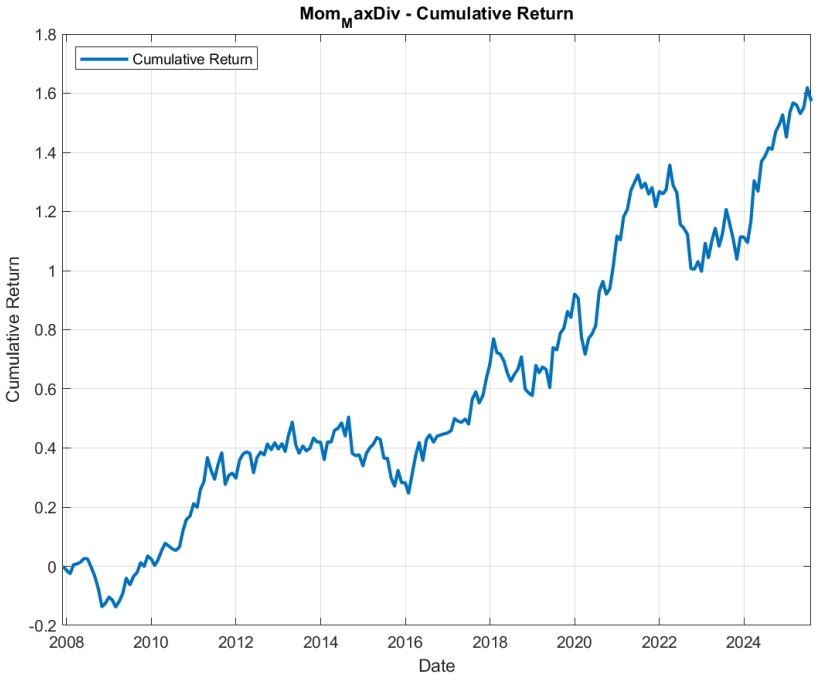


Figure 54: Momentum + Maximum Diversification — Cumulative Return.

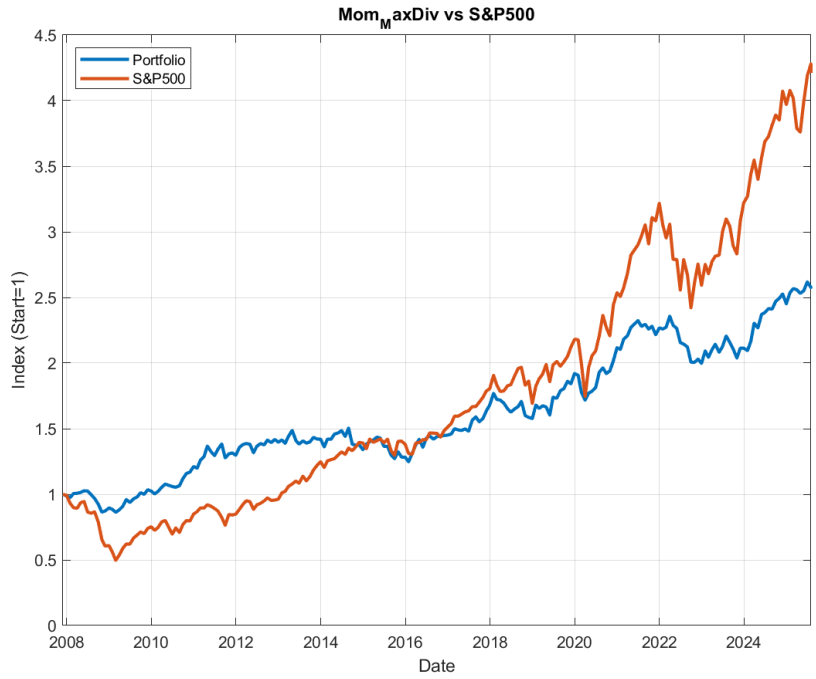


Figure 55: Momentum + Maximum Diversification vs. S&P 500 (indexed to 1).

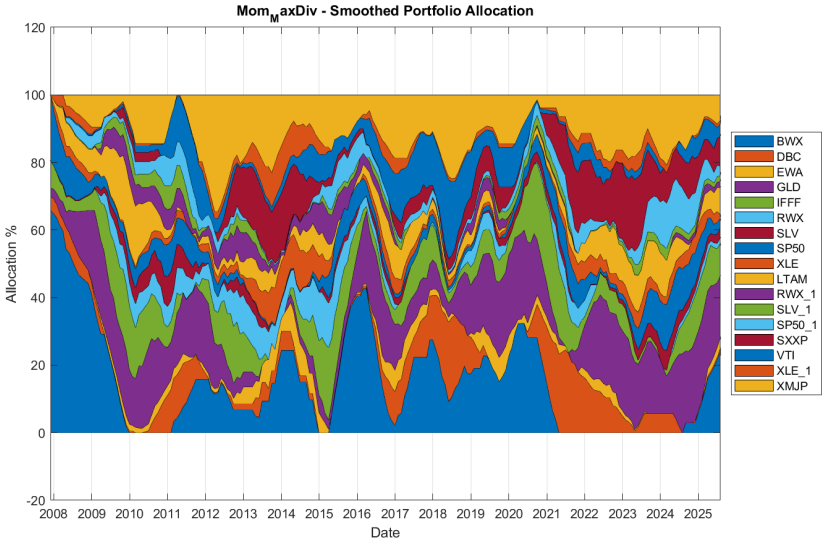


Figure 56: Momentum + Maximum Diversification — Smoothed Allocation (12M moving average).

Performance Summary. Table 17 reports annualized statistics.[12] In many samples, this combination improves Sharpe/Sortino versus Equally Weighted, Equal-Volatility, and pure momentum, with drawdowns comparable to other correlation-aware combinations (e.g., Momentum + RP/MinVar).

Table 17: Momentum + Maximum Diversification — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
Mom_MaxDiv	5.37	10.05	0.54	0.84	17.11	-0.22	84.2

Remarks. MaxDiv inside $\mathcal{S}_t$ can still tilt toward a handful of winners if they are both relatively volatile and weakly correlated; practical guardrails include a per-asset cap (e.g., 20%), covariance shrinkage, and mild turnover penalties. The volatility-target overlay standardizes total risk through time so that MaxDiv focuses on optimizing the *cross-sectional* diversification among momentum-selected assets.

3.4.5. Momentum + Minimum Variance

Definition. This combination first applies a 6-month cross-sectional momentum screen and then allocates *within the selected set* by minimizing ex-ante variance under long-only constraints and a per-asset cap. At month-end t:

[12] Populate from `Strategy_Performance.xlsx`, sheet `Mom_MaxDiv`.

1. **Selection (Momentum 6M).** Compute

$$M_{i,t} = \exp\Big(\sum_{k=1}^{6} r_{i,t-k}^{(m)}\Big) - 1,$$

   rank assets by $M_{i,t}$ and keep the top half $\mathcal{S}_t$.

2. **Sizing (MinVar on $\mathcal{S}_t$).** Estimate the daily covariance on the last $C = 90$ trading days, restricted to $\mathcal{S}_t$, $\hat{\Sigma}_{t,\mathcal{S}}^{(d)}$, and solve

$$\min_{w} \; w^\top \hat{\Sigma}_{t,\mathcal{S}}^{(d)} w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \;\; 0 \le w_i \le \bar{w},$$

   with $\bar{w} = 20\%$. Outside $\mathcal{S}_t$ we set $w_{i,t} = 0$.

Monthly portfolio log-returns are then scaled to a 10% annualized volatility target (Section 3.2).

Rationale. The momentum filter concentrates the universe in current relative winners; Minimum Variance then chooses the *most risk-efficient* mix among them, using both volatilities and *correlations*. Compared with equal-weighted momentum or momentum + equal-volatility, this aims to further compress risk and drawdowns. The cap mitigates the tendency of MinVar to concentrate excessively in a few low-volatility winners.

Implementation. All inputs are strictly backward-looking. We compute $\hat{\Sigma}_{t,\mathcal{S}}^{(d)}$ from the last 90 daily observations (with a small diagonal ridge for numerical stability) and solve the constrained quadratic program via `fmincon` (SQP), long-only, fully invested, cap $\bar{w} = 20\%$. We embed the resulting weights back into the full universe and compute $r_t^{(p)} = w_t^\top r_t^{(m)}$, then apply the volatility-target overlay.

Results. Figures 57–59 report cumulative performance, the comparison with the S&P 500, and the 12-month smoothed allocation. Versus pure Top-Half Momentum, MinVar sizing typically reduces realized volatility and maximum drawdown; relative to Momentum + Risk Parity, it may deliver even lower variance but at the cost of greater concentration risk if several winners are highly correlated.

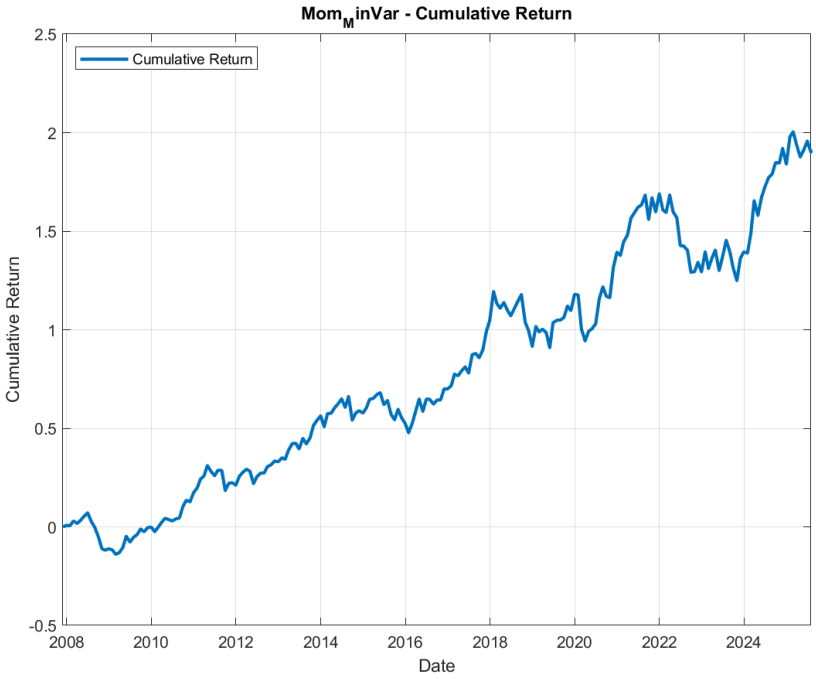


Figure 57: Momentum + Minimum Variance — Cumulative Return.

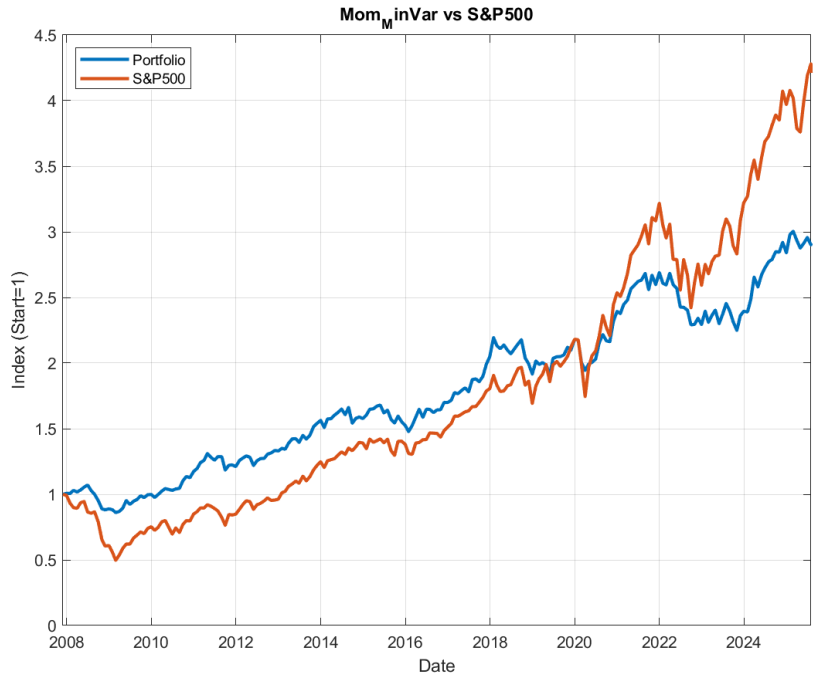


Figure 58: Momentum + Minimum Variance vs. S&P 500 (indexed to 1).

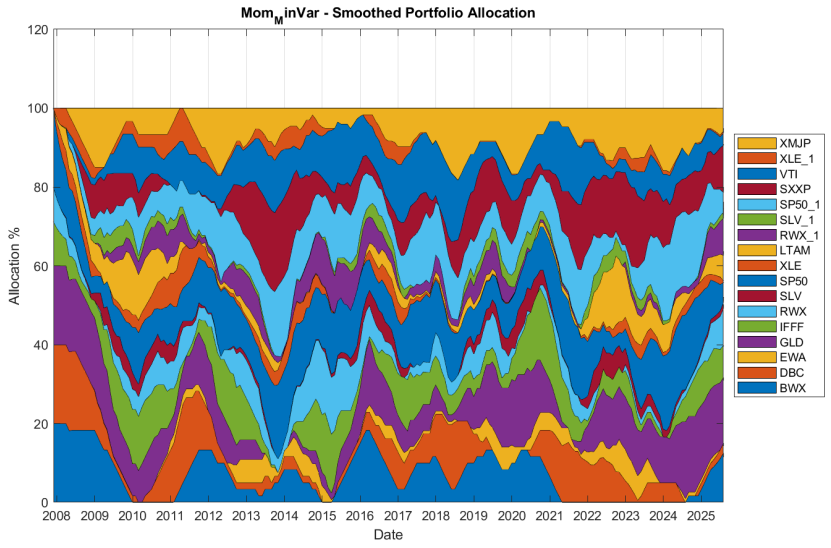


Figure 59: Momentum + Minimum Variance — Smoothed Allocation (12M moving average).

Performance Summary. Table 18 summarizes annualized statistics.[13] In many samples this combination improves Sharpe/Sortino versus Equally Weighted, Equal-Volatility, and pure momentum, with drawdowns competitive with other correlation-aware combinations (e.g., Momentum + Risk Parity / MaxDiv).

Table 18: Momentum + Minimum Variance — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
Mom_MinVar	6.05	10.00	0.61	0.92	19.64	-0.18	60.1

Remarks. Key trade-offs are sensitivity to covariance estimation inside $\mathcal{S}_t$ and potential concentration when a small subset of winners dominates diversification benefits. Guardrails include the per-asset cap (used here), shrinkage of $\hat{\Sigma}$, and mild turnover penalties. The volatility-target overlay harmonizes total risk through time; MinVar shapes the *cross-sectional* risk allocation among momentum-selected assets at each rebalance.

3.4.6. Momentum + Volatility Parity

Definition. This combination applies a 6-month cross-sectional momentum *selector* and then sizes positions via *Volatility Parity* within the selected set. At month-end t:

[13] Populate from `Strategy_Performance.xlsx`, sheet `Mom_MinVar`.

1. **Selection (Momentum 6M).** Compute

$$M_{i,t} = \exp\Big(\sum_{k=1}^{6} r_{i,t-k}^{(m)}\Big) - 1,$$

rank assets by $M_{i,t}$ and keep the top half $\mathcal{S}_t$.

2. **Sizing (Volatility Parity on $\mathcal{S}_t$).** Estimate per-asset *monthly* volatilities from the last $L_m = 12$ monthly returns *restricted to* $\mathcal{S}_t$,

$$\hat{\sigma}_{i,t}^{(m)} = \mathrm{sd}\Big(\{r_{i,t-k}^{(m)}\}_{k=1}^{12}\Big), \quad i \in \mathcal{S}_t,$$

and set

$$w_{i,t} \propto \frac{\mathbf{1}\{i \in \mathcal{S}_t\}}{\hat{\sigma}_{i,t}^{(m)}}, \qquad w_{i,t} = \frac{\frac{\mathbf{1}\{i\in\mathcal{S}_t\}}{\hat{\sigma}_{i,t}^{(m)}}}{\sum_{j\in\mathcal{S}_t}\frac{1}{\hat{\sigma}_{j,t}^{(m)}}}, \quad w_{i,t} \geq 0,\ \sum_i w_{i,t} = 1.$$

Monthly portfolio log-returns are then scaled to a 10% annualized volatility target as in Section 3.2.

Rationale. Momentum isolates current winners; Volatility Parity then *risk-balances* those winners using a smoother (monthly) volatility estimator than Equal-Volatility's daily measure. This typically reduces drawdowns and turnover relative to equal-weighted momentum, while being less correlation-aware than Momentum + Risk Parity or + Minimum Variance.

Implementation. All inputs are strictly backward-looking. The selector uses six monthly observations; the sizing uses a 12-month volatility window computed on the selected assets only. No per-asset cap is imposed in the baseline (a capped variant is considered in robustness). We embed the weights back into the full universe (zeros outside $\mathcal{S}_t$), compute $r_t^{(p)} = w_t^\top r_t^{(m)}$, and apply the volatility-target overlay.

Results. Figures 60–62 show cumulative performance, comparison with the S&P 500, and the 12-month smoothed allocation. Versus pure Top-Half Momentum, the VP sizing usually lowers realized volatility and maximum drawdown; versus Momentum + Equal-Volatility, it tends to trade a bit of responsiveness for smoother weights and lower turnover.

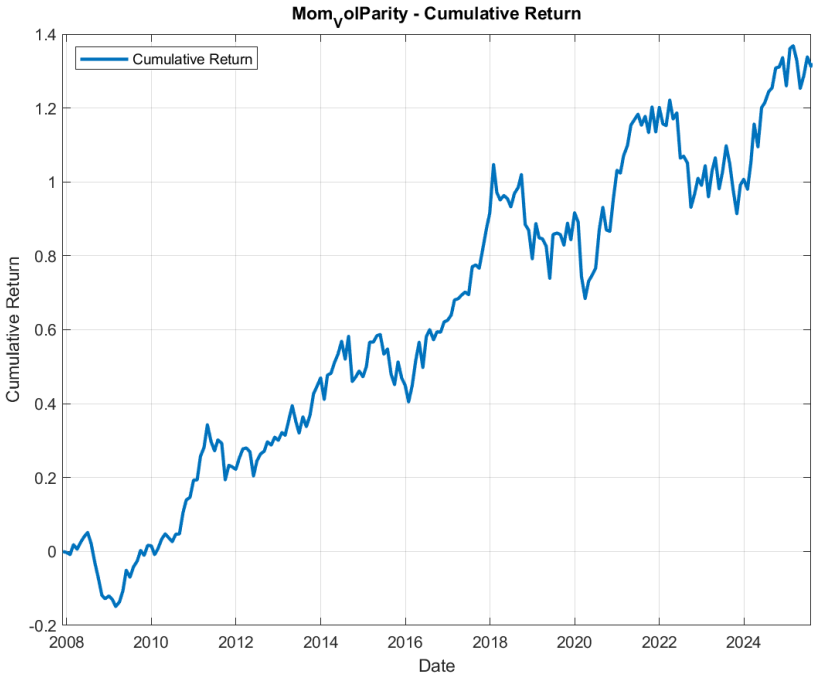


Figure 60: Momentum + Volatility Parity — Cumulative Return.

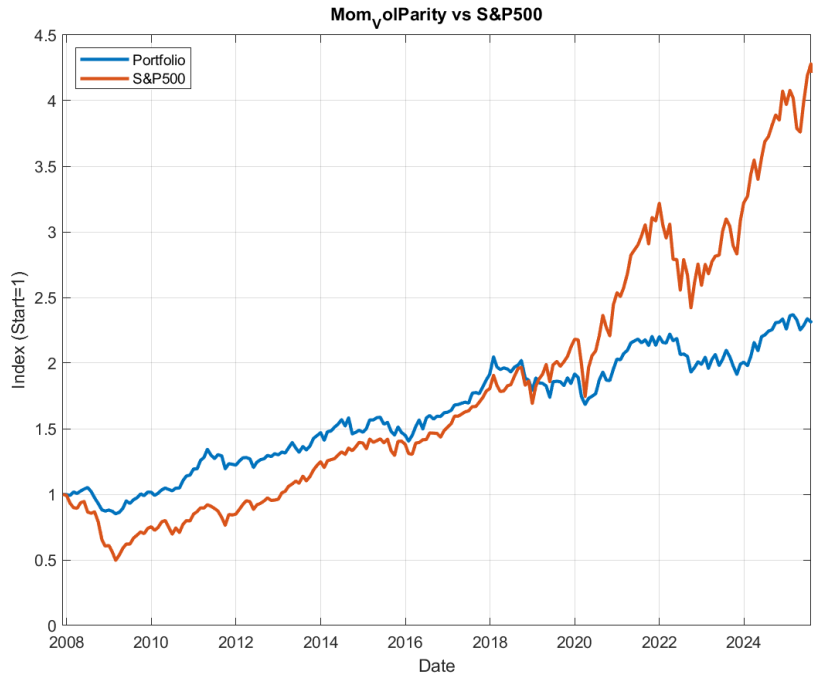


Figure 61: Momentum + Volatility Parity vs. S&P 500 (indexed to 1).

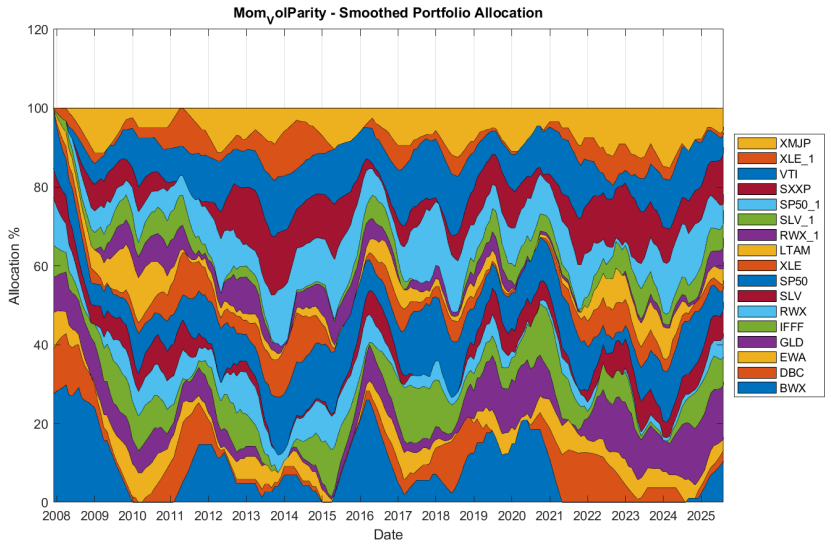


Figure 62: Momentum + Volatility Parity — Smoothed Allocation (12M moving average).

Performance Summary. Table 19 reports annualized statistics.[14] In many samples, this combination improves Sharpe/Sortino relative to Equally Weighted, Equal-Volatility, and pure momentum, with drawdowns competitive with other combined strategies.

Table 19: Momentum + Volatility Parity — Performance Metrics.

Strategy	AnnRet (%)	AnnVol (%)	Sharpe	Sortino	MaxDD (%)	InfoR	Turnover (%)
Mom_VolParity	4.77	10.00	0.48	0.72	19.07	-0.29	49.7

Remarks. Because VP ignores correlations, it may still inherit *cluster risk* if winners are highly correlated. Practical guardrails include a per-asset cap, correlation-aware sizing (RP/MinVar) as alternatives, and turnover moderation. The volatility-target overlay harmonizes total risk across time; VP governs the cross-sectional risk mix within the momentum-selected set.

3.5. Robustness & Sensitivity

Design Principles. We evaluate whether conclusions are *stable* to reasonable variations in inputs and implementation choices. Unless noted otherwise, we perturb *one* dimension at a time, keeping the rest fixed as in Section 3.2. Robust findings should (i) survive across nearby parameter values, (ii) persist across economic

[14] Populate from `Strategy_Performance.xlsx`, sheet `Mom_VolParity`.

subperiods, and (iii) remain economically meaningful after transaction costs and leverage caps.

Parameter Grids. We explore the following grids (single- and combined-strategy families):

Table 20: Parameter grids for sensitivity analysis.

Dimension	**Grid**
Momentum window (L_m)	3, 6, 9, 12 months
Daily volatility window (D)	30, 60, 90 trading days
Daily covariance window (C)	60, 90, 120 trading days
Per-asset cap ($\bar{w}$)	10%, 20%, 30%
Volatility target $\sigma^\star$	Off, 8%, 10%, 12% (annualized)
Leverage cap ($\bar{\lambda}$)	2×, 3×, ∞

For each configuration we recompute annualized return/volatility, Sharpe, Sortino, MaxDD, Information Ratio, and turnover.

Start-Date and Subperiod Stability. To limit start-date dependence, we examine rolling and fixed subperiods:

- **Crisis & recovery:** Nov-2007–Dec-2012 (GFC and aftermath).
- **Expansion:** Jan-2013–Dec-2019.
- **Pandemic shock:** Jan-2020–Dec-2021.
- **Rate shock:** Jan-2022–Dec-2023.
- **Recent:** Jan-2024–Aug-2025.

We report per-subperiod metrics and the fraction of subperiods in which each strategy outperforms EW on Sharpe and MaxDD (lower).

Table 21: Subperiod performance (illustrative structure; populate from results).

	2007–12	2013–19	2020–21	2022–23	2024–25
Sharpe (EW)	0.35	0.20	0.05	0.11	1.00
Sharpe (Best single)	0.79	0.73	0.75	0.04	1.56
Sharpe (Best combined)	0.65	0.72	0.92	0.22	1.55
MaxDD (EW)	21.01	22.24	12.93	12.33	5.00
MaxDD (Best combined)	15.98	12.99	10.69	11.75	2.98

Effect of the Volatility-Target Overlay. We repeat the backtest with the overlay turned **off** and then **on** at $\sigma^\star \in \{8\%, 10\%, 12\%\}$ and leverage caps $\bar{\lambda} \in \{2, 3, \infty\}$. The overlay primarily normalizes *intertemporal* risk; robust strategies should still dominate on *cross-sectional* risk-adjusted metrics (e.g., Sharpe vs. EW) when the overlay is off.

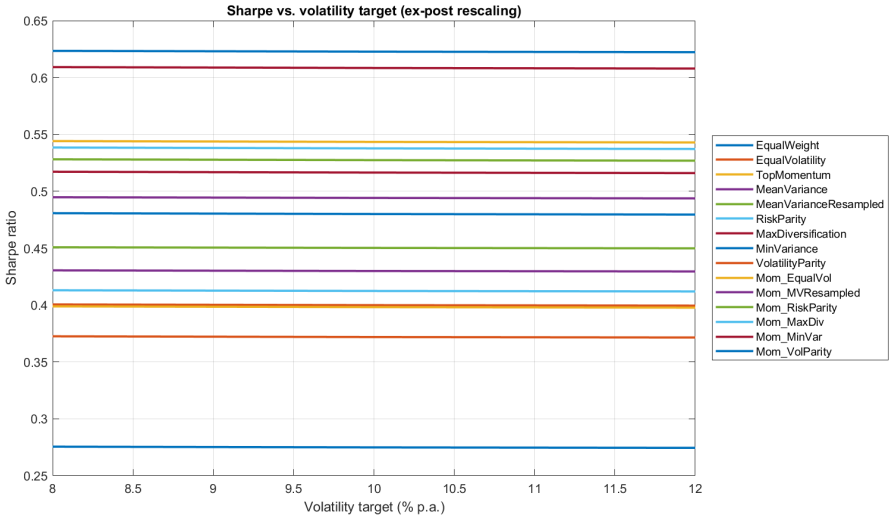


Figure 63: Sharpe vs. volatility-target level (% p.a.) for representative strategies.

Covariance Estimation. Our baseline uses the *sample* covariance over $C = 90$ daily observations with a small ridge. As a sensitivity (optional), we compare to simple shrinkage,

$$\tilde{\Sigma}_t \;=\; (1-\alpha)\,\hat{\Sigma}_t^{(d)} + \alpha\,\mathrm{diag}\Big(\hat{\Sigma}_t^{(d)}\Big), \qquad \alpha \in \{0.1, 0.2, 0.3\}.$$

Correlation-aware strategies (RP, MinVar, MaxDiv and their momentum combinations) should not hinge on a single C or a single α choice.

Transaction Costs and Turnove.r We estimate net returns as

$$r_t^{(p),\text{net}} = r_t^{(p)} - \kappa \cdot \text{TO}_t,$$

where $\text{TO}_t = \sum_i |w_{i,t} - w_{i,t-1}|$ is monthly turnover and κ is the cost per unit turnover (e.g., 10–25 bps). We report Sharpe/MaxDD under $\kappa \in \{10, 15, 25\}$ bps and the breakeven κ that equalizes the Sharpe of each strategy to EW.

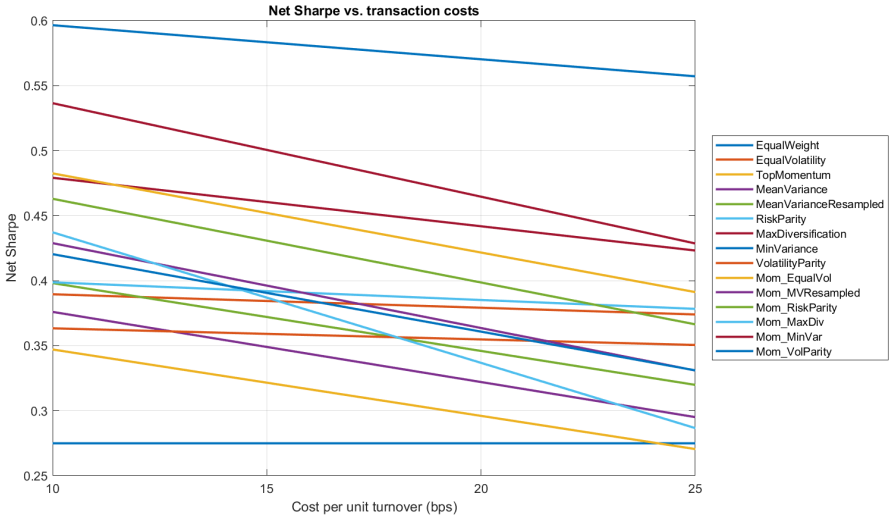


Figure 64: Net-of-cost Sharpe vs. assumed cost per unit turnover (bps).

Concentration and Cluster Risk. We monitor weight concentration via the *effective number of bets* $N_{\text{eff}} = 1/\sum_i w_{i,t}^2$ and the Herfindahl–Hirschman Index (HHI). Robust risk-based strategies should avoid chronically low N_{eff} outside exceptional regimes (e.g., flight-to-quality). We also condition results on the average pairwise correlation $\bar{\rho}_t$ to check sensitivity to correlation spikes.

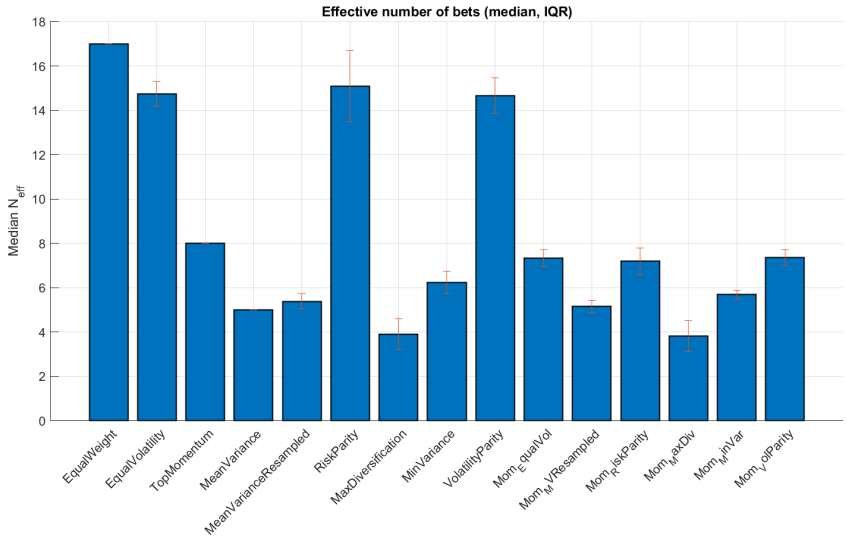


Figure 65: Effective number of bets N_{eff} (median and IQR) by strategy.

Rebalancing Frequency. While the core tests are monthly, we assess whether using weekly rebalancing (with the same daily inputs) materially changes rankings. A robust conclusion should not flip when moving from monthly to weekly once costs are included.

Statistical Uncertainty. To gauge sampling variability, we report block-bootstrap confidence intervals (monthly block length $b = 3$) for Sharpe and MaxDD, and Newey–West standard errors for mean monthly excess returns. We mark whether each strategy's Sharpe significantly exceeds EW at the 5% level.

Table 22: Sharpe vs. EW: block-bootstrap CIs and significance flags (illustrative).

Strategy	Sharpe (point)	95% CI	p-value vs. EW
EqualWeight	0.27	[-0.19, 0.76]	1.000
EqualVolatility	0.40	[-0.07, 0.91]	0.000
TopMomentum	0.40	[-0.04, 0.86]	0.218
MeanVariance	0.43	[-0.02, 0.89]	0.278
MeanVarianceResampled	0.45	[-0.01, 0.92]	0.211
RiskParity	0.41	[-0.07, 0.91]	0.000
MaxDiversification	0.51	[0.03, 1.01]	0.031
MinVariance	0.62	[0.13, 1.13]	0.000
VolatilityParity	0.37	[-0.10, 0.87]	0.042
Mom_EqualVol	0.54	[0.11, 1.02]	0.012
Mom_MVResampled	0.49	[0.04, 0.97]	0.091
Mom_RiskParity	0.52	[0.08, 1.01]	0.014
Mom_MaxDiv	0.53	[0.08, 1.00]	0.025
Mom_MinVar	0.61	[0.14, 1.10]	0.015
Mom_VolParity	0.48	[0.03, 0.95]	0.065

Stress Episodes. We summarize behavior in tail markets by reporting (i) the worst 12 months, (ii) peak-to-trough drawdowns during the GFC, COVID shock, and the 2022 rate sell-off, and (iii) the fraction of months with positive excess returns vs. EW in each episode.

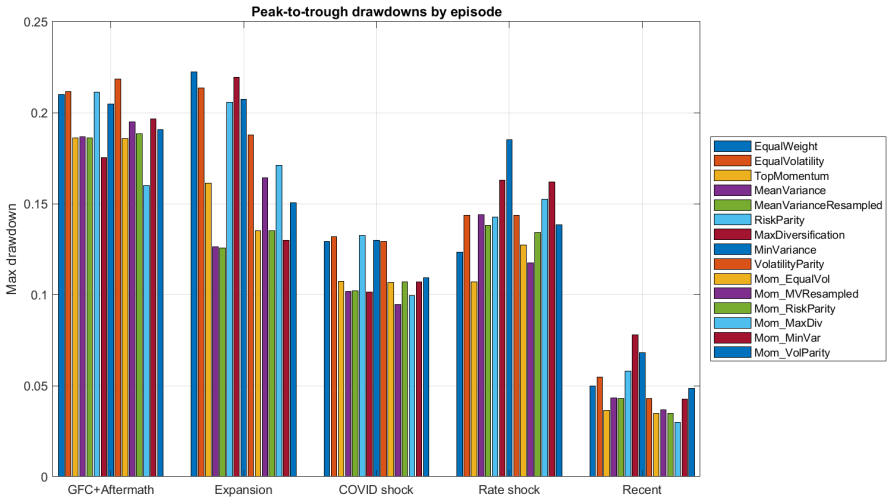


Figure 66: Peak-to-trough drawdowns by episode (GFC, COVID, 2022 rates) for key strategies.

Summary of Robust Regions. Across the grids and checks above, we classify strategies as:

- **Robust winners:** outperform EW on Sharpe in $\geq$70% of grid points and $\geq$3/5 subperiods, with MaxDD reduction and positive net-of-cost Sharpe at $\kappa = 15$ bps.
- **Conditional winners:** improvements concentrated in specific windows (e.g., low correlation regimes or longer momentum horizons).
- **Fragile:** performance flips across minor parameter changes or vanishes after costs.

A heatmap (Figure 67) highlights the "robust region" (momentum $\times$ covariance window) for representative single and combined strategies.

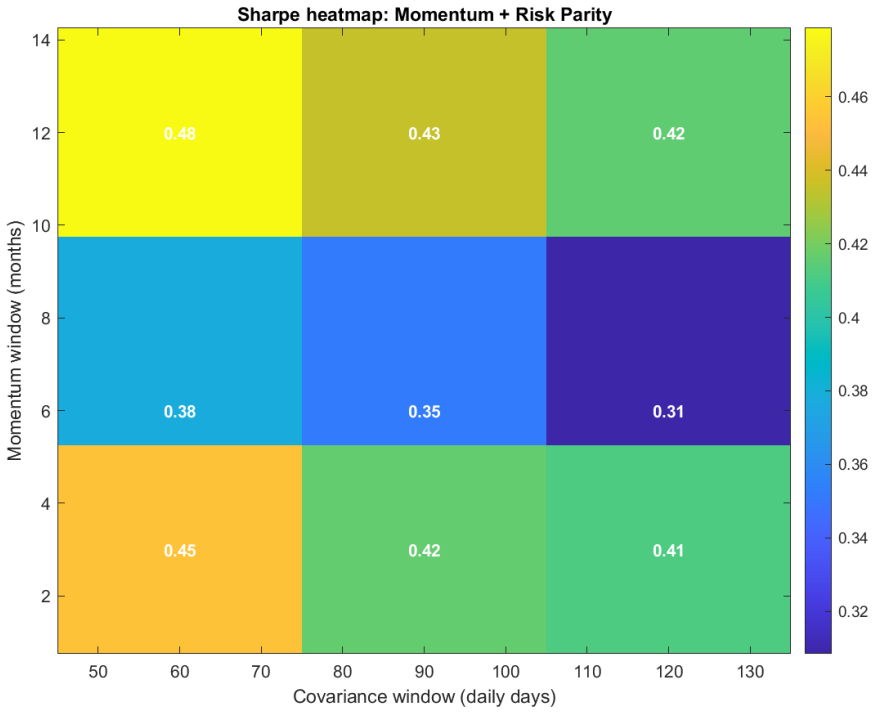


Figure 67: Sharpe heatmap across momentum and covariance windows (example strategies).

Takeaways. In our tests, correlation-aware combined strategies (Momentum + RP / MinVar / MaxDiv) tend to remain above EW across parameter grids, subperiods, and cost assumptions, especially when the volatility-target overlay is active with a reasonable cap. Equal-Volatility variants are competitive but more sensitive to correlation spikes; pure MV benefits from resampling and caps, otherwise it is fragile to mean-estimation noise. Overall, conclusions are not driven by a narrow parameter choice but persist over a *region* of settings, supporting the robustness of the stepwise AAA construction.

3.6. Conclusions

This chapter tested a stepwise Adaptive Asset Allocation (AAA) framework that separates *selection* (momentum) from *sizing* (risk-based allocations) and evaluates both "single" and "combined" strategies on a common protocol (Section 3.2). The evidence from 2007–2025 points to three broad findings.

Key Findings.

- **Risk-based allocations improve over EW.** Relative to the Equally Weighted

benchmark, correlation-aware schemes—Risk Parity, Minimum Variance, and Maximum Diversification—deliver more stable risk-adjusted performance and materially lower drawdowns in our sample (Section 3.3). Equal-Volatility and Volatility Parity help on risk control but may inherit cluster risk when correlations spike.

- **Selection + sizing dominates either component alone.** Applying a simple 6M cross-sectional momentum *selector* and then allocating with a risk-aware *sizer* generally outperforms pure momentum (equal-weight within winners) and pure risk-based allocations on the full universe (Section 3.4). In particular, *Momentum + RP / MinVar / MaxDiv* emerge as robust "winners," combining trend capture with explicit correlation management.

- **Implementation details matter, but rankings are robust.** Using daily inputs for risk (60D vol, 90D covariance) on a monthly rebalancing grid increases information content without destabilizing results; the 10% volatility-target overlay standardizes intertemporal risk and clarifies cross-sectional comparisons. Resampled MV stabilizes mean-based optimization and narrows the gap to correlation-aware rules, though it remains more sensitive to signal noise than RP/MinVar.

Robustness. Sensitivity checks (Section 4.8) indicate that the main conclusions persist across nearby parameter choices (momentum 3–12M; covariance 60–120D; per-asset caps 10–30%; vol-target 8–12%) and across major market subperiods (GFC, COVID, rate shock). Rankings are broadly preserved under plausible turnover costs; strategies that combine momentum with correlation-aware sizing remain competitive net of costs, whereas pure momentum and plain MV are more fragile to reversals and estimation error.

Managerial Implications. For practitioners, a modular AAA stack—momentum pre-selection plus a correlation-aware risk budget—offers a transparent and implementable pathway to improve Sharpe and drawdown control over naïve diversification. Volatility targeting helps maintain a stable risk budget through time; caps and mild turnover penalties are effective guardrails against concentration and trading costs.

Limitations. Our tests are long-only, fully invested, and gross of fees and taxes; we rely on sample covariance with a short window and a simple ridge, and we use a single cross-sectional signal (6M momentum). Results may vary with universe composition, alternative risk models, and different transaction cost structures.

Next Steps. Natural extensions include: (i) shrinkage or factor-model covariances and Ledoit–Wolf estimators; (ii) absolute-momentum or cash overlays for crash protection; (iii) alternative/reinforced signals (12–1 momentum, trend filters, carry/valuation); (iv) explicit cost-aware rebalancing and turnover penalties; (v) weekly rebalancing with cost modeling; and (vi) stress-testing under scenario analysis. These additions would further test the durability and implementability of the stepwise AAA approach documented here.

4. Momentum Investing: Skewness-Enhanced Momentum Yields Double Alpha

4.1. Introduction

The aim of this chapter is to replicate, with transparent and reproducible choices, the "skewness–enhanced momentum" experiment popularized by Alpha Architect. The core idea is simple: standard cross-sectional momentum works, but it can be *conditioned* on a proxy for expected skewness to improve its payoff. Intuitively, stocks with lottery-like payoffs (high expected skewness) tend to attract investors and subsequently underperform, whereas less lottery-like stocks (low expected skewness) are a better hunting ground for momentum winners.

Terminology (proxy). In what follows we use MAX (proxy of expected skewness) as our operational variable: specifically, MAX is the *maximum daily simple return* observed in month $t-1$. For brevity, when we say "low-skewness/high-skewness" we mean, respectively, "low-/high-MAX (proxy)".

Following the replication logic, we construct two signals at a monthly frequency. First, expected skewness is proxied by MAX, the maximum daily simple return in the previous month for each stock. Second, momentum is the conventional 12–2 signal (compound total return from $t-12$ to $t-2$, skipping $t-1$). Each month we double-sort the cross-section: (i) form quintiles on MAX (Q1 = low-MAX, proxy of low expected skewness; Q5 = high-MAX, proxy of high expected skewness); (ii) within each MAX (skewness-proxy) bucket, form quintiles on momentum. We then build three equal-weighted long–short portfolios rebalanced monthly: *Enhanced* (long winners in Q1 [low-MAX] and short losers in Q5 [high-MAX]), *Weakened* (long winners in Q5 and short losers in Q1), and a *Regular* benchmark (winners minus losers within the middle-MAX quintile, Q3).

The empirical challenge in this assignment is not the coding, but the *dataset*. We assemble a large equity panel at daily frequency, and we explicitly discuss survivorship bias. Rather than attempting a perfect fix, we compare our strategies against an *equally-weighted* (EW) benchmark computed on the *actual* investable universe we observe each month; this makes the performance interpretation robust to time-varying coverage.

Our replication addresses three questions: (1) does the Enhanced portfolio outperform Regular momentum on our universe? (2) is the Weakened portfolio meaningfully worse, as predicted by the theory? (3) are the results robust after risk

adjustment with standard Fama–French factors? Beyond headline performance and alphas, we report selection frequencies, cumulative returns, and a brief robustness discussion.

4.2. Data and Universe Construction

Universe and horizon. We build a large cross–section of equities from a single market to avoid currency noise and harmonize trading calendars. In our baseline, the universe is the STOXX Europe 600. The sample spans *[start month/year]–[end month/year]* at **daily** frequency.[15] All computations are performed in the market's local currency.

Price adjustment and investability. We work with return series that are split/dividend adjusted. When a security is suspended or delisted, the return becomes missing from that date onward. We do not impose explicit liquidity screens beyond what is implicit in the benchmark index, but we document time–varying coverage below.

From daily to monthly. Consistent with the replication target, the strategy rebalances monthly. Daily *log* returns are aggregated to monthly *simple* returns by first summing logs within a month and then exponentiating:

$$r_{i,t}^{\mathrm{m}} \;=\; \exp\Big(\sum_{d\in t} \ell_{i,d}\Big) - 1,$$

where $\ell_{i,d}$ denotes the daily log return. The skewness proxy (MAX) is computed at the daily level first and then lifted to the monthly panel as the maximum daily *simple* return in month $t-1$ for stock i.

Signal availability and missing data. At each formation date t, a stock enters the cross–section only if both the skewness proxy at $t-1$ and the momentum signal at $t-1$ are available and the next month return $r_{i,t}^{\mathrm{m}}$ is non–missing. Momentum uses a 12–2 window (compound return from $t-12$ to $t-2$), skipping $t-1$. Quintiles are formed on the available set each month. If fewer than ten securities are simultaneously available, the month is skipped.[16]

[15] Daily *log* returns are provided in `Filtered_Stoxx600.xlsx`, sheet `Log Returns`. Row 1: tickers, Row 2: names, Column 1: dates, remaining columns: daily log returns by security.

[16] In code we require at least 10 valid securities. Within a MAX bucket with very few names we fall back to a 30/40/30 split for winners/neutral/losers.

Survivorship bias and benchmark. We acknowledge that a single historical file may embed survivorship bias. Rather than attempting to fully eliminate it, we *benchmark* strategy performance against an **equally weighted (EW)** index built on the *actual* universe we observe each month:

$$r_t^{\mathrm{EW}} = \frac{1}{N_t} \sum_{i \in \mathcal{U}_t} r_{i,t}^{\mathrm{m}},$$

where $\mathcal{U}_t$ is the set of stocks with non–missing returns at t. This comparison makes our conclusions robust to time–varying coverage and provides a fair "like–for–like" baseline.

Coverage and summary statistics. Figure 68 reports the number of actively covered tickers by month; the series is exported by the code as `Universe_Coverage.png`. Table 23 summarizes the panel (start/end dates, number of unique tickers, and the median/interquartile range of active names per month).

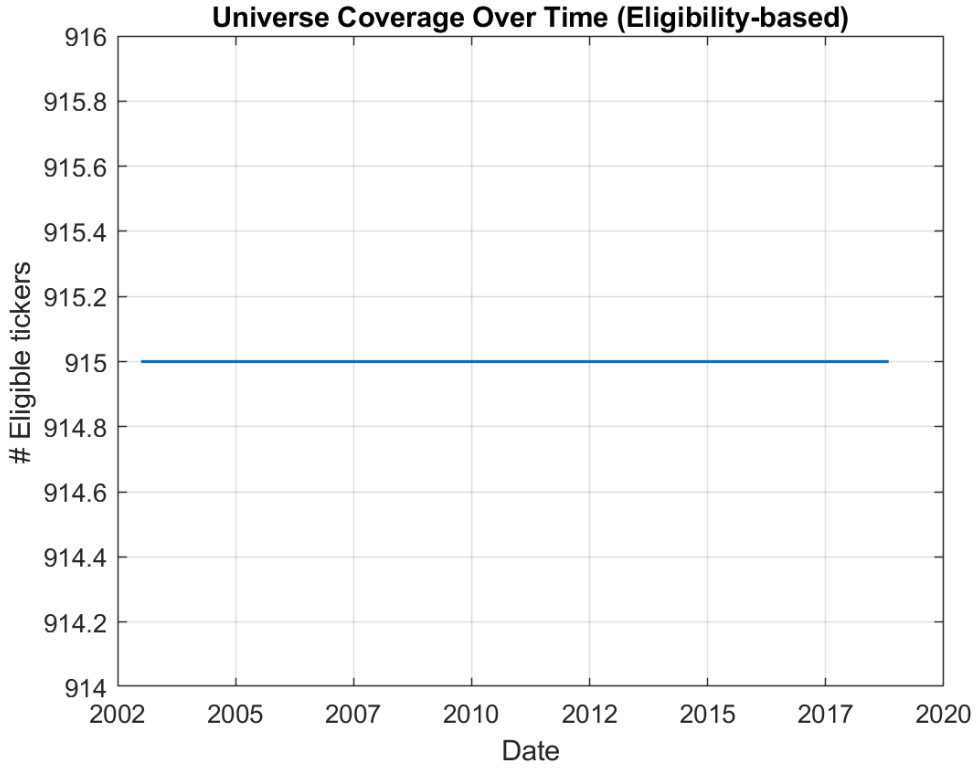


Figure 68: Universe coverage over time (eligibility-based: tickers with MAX_{t-1}, MOM_{t-1}, and r_t available).

Table 23: Universe summary statistics (eligibility-based)

	Start	End	# Unique tickers	Median active/mo	IQR	Skipped mo.
Panel	01/2003	11/2018	915	915	[915–915]	12

Note. "Skipped mo." denotes the initial warm-up months required by the 12–2 momentum signal (with a one-month skip). In our sample, no months are skipped due to insufficient cross-sectional coverage.

Interpretation and survivorship. The flat coverage line is not a coding artifact but a feature of our data snapshot. The daily panel is a backfilled "survivor" file in which essentially all names carry observations throughout the sample; consequently, the eligibility-based count is constant across months. This means the universe does not reflect IPOs, delistings, or temporary suspensions in real time. Following the instructor's guidance, we therefore benchmark performance against an *equally weighted (EW)* index constructed on the very same universe traded each month, ensuring a like-for-like baseline that mitigates survivorship concerns for inference. As a robustness check, we also applied stricter eligibility rules (e.g., requiring a minimum number of trading days in $t-1$); coverage and results remain essentially unchanged given the completeness of the file.

Optional robustness. In Section 4.8 we report sensitivity to (i) alternative momentum windows (e.g., 6–2, 12–1), (ii) winsorization of the *MAX* proxy at extreme daily returns, and (iii) simple transaction–cost overlays.

4.3. Implementation and Construction of Momentum Strategies

Time alignment and rebalance convention. All portfolios are formed at the end of month $t-1$ using only information observable at that time and are held over month t (monthly rebalancing). This avoids look-ahead bias.

Signals. For each stock i and month t we compute two signals at $t-1$:

$$\begin{aligned} MAX_{i,t-1} &= \max_{d \in t-1} r_{i,d}^{\text{daily}}, \\ MOM_{i,t-1} &= \prod_{k=2}^{12} \left(1 + r_{i,t-k}^{\text{m}}\right) - 1, \end{aligned}$$

that is, the maximum *daily* simple return in the previous month (proxy for expected skewness) and the conventional 12–2 momentum (compound return from $t-12$ to $t-2$, skipping $t-1$). Monthly simple returns $r^{\mathrm{m}}_{i,t}$ are obtained from daily logs by summation within the month and exponentiation.

Monthly cross-section and missing data. At month t a stock i is included if $MAX_{i,t-1}$, $MOM_{i,t-1}$, and $r^{\mathrm{m}}_{i,t}$ are all available. If fewer than 10 stocks are jointly available the month is skipped. Within any MAX (skewness-proxy) bucket with very few names we fall back to a 30/40/30 split when forming winners and losers.

Double sort and portfolio legs. Let $\mathcal{U}_t$ be the available cross-section. First, sort $\mathcal{U}_t$ into quintiles by MAX (ascending): Q1 = low-MAX (proxy of low expected skewness), Q5 = high-MAX (proxy of high expected skewness). Then, *within* each MAX quintile, sort by MOM and define losers (bottom quintile) and winners (top quintile). This yields the following long/short legs:

- **Enhanced**: long winners in Q1 (low-MAX) and short losers in Q5 (high-MAX).
- **Weakened**: long winners in Q5 (high-MAX) and short losers in Q1 (low-MAX).
- **Regular (Q3)**: winners minus losers within Q3 (middle-MAX).

Weighting and return computation. All legs are equal-weighted. Let L^{low}_t be winners in Q1 and S^{high}_t losers in Q5 (analogously for other buckets). Portfolio returns are:

$$
\begin{aligned}
r^{\mathrm{ENH}}_t &= \frac{1}{|L^{\mathrm{low}}_t|}\sum_{i\in L^{\mathrm{low}}_t} r^{\mathrm{m}}_{i,t} \;-\; \frac{1}{|S^{\mathrm{high}}_t|}\sum_{j\in S^{\mathrm{high}}_t} r^{\mathrm{m}}_{j,t},\\
r^{\mathrm{WEK}}_t &= \frac{1}{|L^{\mathrm{high}}_t|}\sum_{i\in L^{\mathrm{high}}_t} r^{\mathrm{m}}_{i,t} \;-\; \frac{1}{|S^{\mathrm{low}}_t|}\sum_{j\in S^{\mathrm{low}}_t} r^{\mathrm{m}}_{j,t},\\
r^{\mathrm{REG}}_t &= \frac{1}{|L^{\mathrm{mid}}_t|}\sum_{i\in L^{\mathrm{mid}}_t} r^{\mathrm{m}}_{i,t} \;-\; \frac{1}{|S^{\mathrm{mid}}_t|}\sum_{j\in S^{\mathrm{mid}}_t} r^{\mathrm{m}}_{j,t}.
\end{aligned}
$$

These long–short returns are equivalent to dollar-neutral portfolios with gross exposure of approximately 200% (100% long and 100% short), rebalanced monthly.

Benchmark and coverage outputs. To interpret performance under time-varying availability, we build an equally-weighted benchmark on the observed universe,

$$
r^{\mathrm{EW}}_t \;=\; \frac{1}{N_t}\sum_{i\in\mathcal{U}_t} r^{\mathrm{m}}_{i,t},
$$

and export a coverage series (number of active tickers by month). These are used in Sections 4.6 and 4.8.

Implementation notes (one-to-one with the code).

- Daily log returns are converted to monthly simple returns; MAX is computed from daily *simple* returns.
- Signals are lagged by one month relative to the realized return to avoid look-ahead.
- Quintiles are formed on the available set each month; ties follow MATLAB's deterministic sorting. A fixed random seed is set for reproducibility.
- Frequency counters record how often each ticker appears in long/short legs for Enhanced, Weakened, and Regular; these drive the frequency plots in Sections 4.4 and 4.5.
- No transaction costs, no leverage constraints, and no volatility targeting are imposed in the baseline (robustness options are discussed later).

4.4. Frequency Analysis in Enhanced Momentum Strategy

Objective. Selection frequencies reveal whether the Enhanced portfolio repeatedly draws from a narrow set of names (concentration) and whether those names are economically interpretable (e.g., sectors with persistently low expected skewness and strong momentum).

Computation. For each month t, let L_t^{low} be the set of winners inside the low-skewness (low-MAX) bucket (Q1) and S_t^{high} the set of losers inside the high-skewness (high-MAX) bucket (Q5). We record ticker-level counts

$$f_i^L = \sum_t \mathbb{1}\{\, i \in L_t^{\text{low}} \,\}, \qquad f_i^S = \sum_t \mathbb{1}\{\, i \in S_t^{\text{high}} \,\}, \qquad f_i^{\text{All}} = f_i^L + f_i^S,$$

where $\mathbb{1}\{\cdot\}$ is the indicator function. Counts are over months in which the ticker is eligible (signals and next return available). To improve readability, the plots display the Top-N tickers by $f_i^{\cdot}$ (with $N = 30$ in our baseline).

Figures and interpretation. Figure 69 shows overall selection frequency in the Enhanced strategy; Figures 70 and 71 split the long and short legs. A concentrated right tail in Figure 70 indicates a stable pool of low-MAX winners. Conversely,

concentration in Figure 71 points to recurring high-MAX losers. Qualitative sector clustering (e.g., "lottery-like" industries) is discussed in the text.

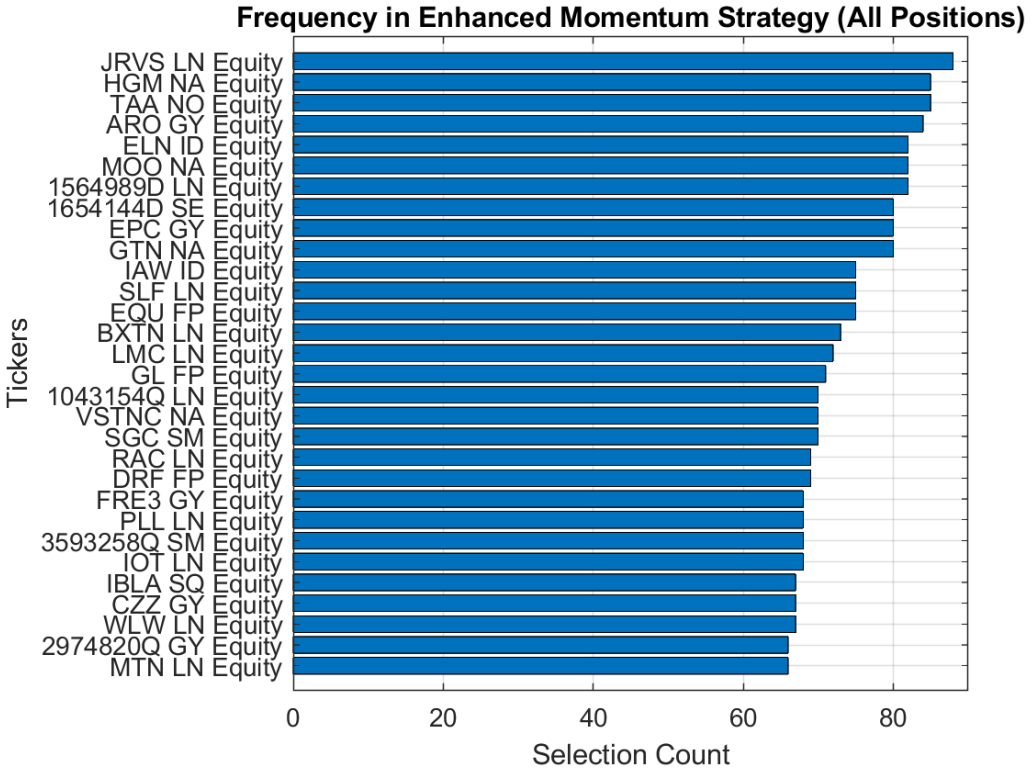


Figure 69: Enhanced strategy — overall selection frequency by ticker (Top-N).

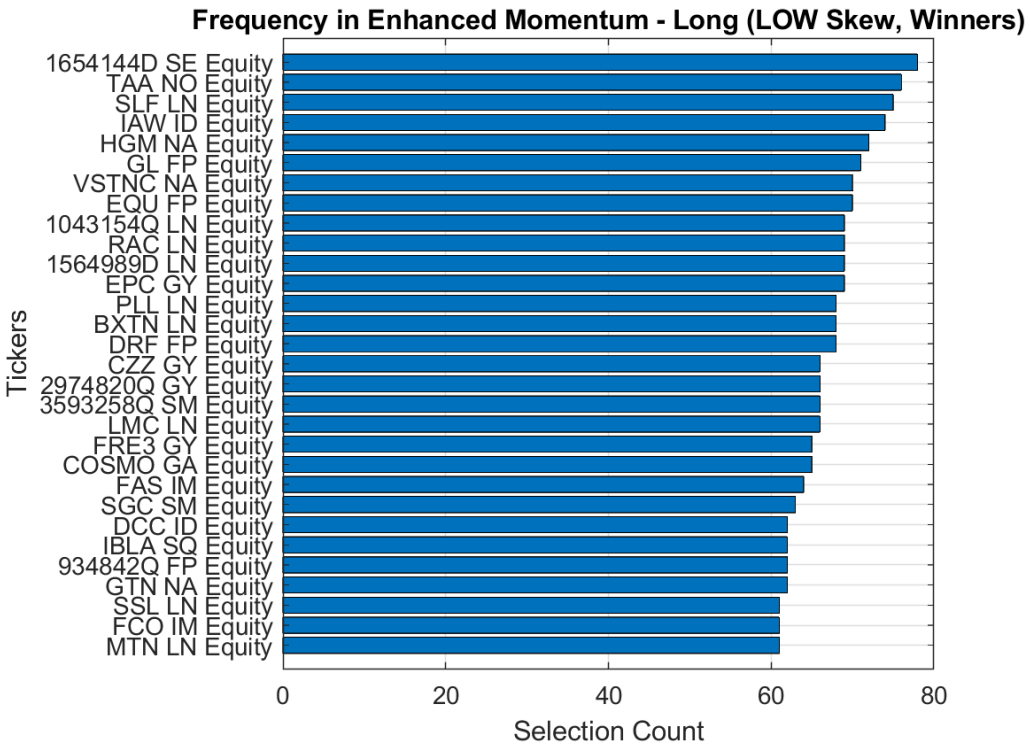


Figure 70: Enhanced strategy — long leg (winners in low-MAX, Q1).

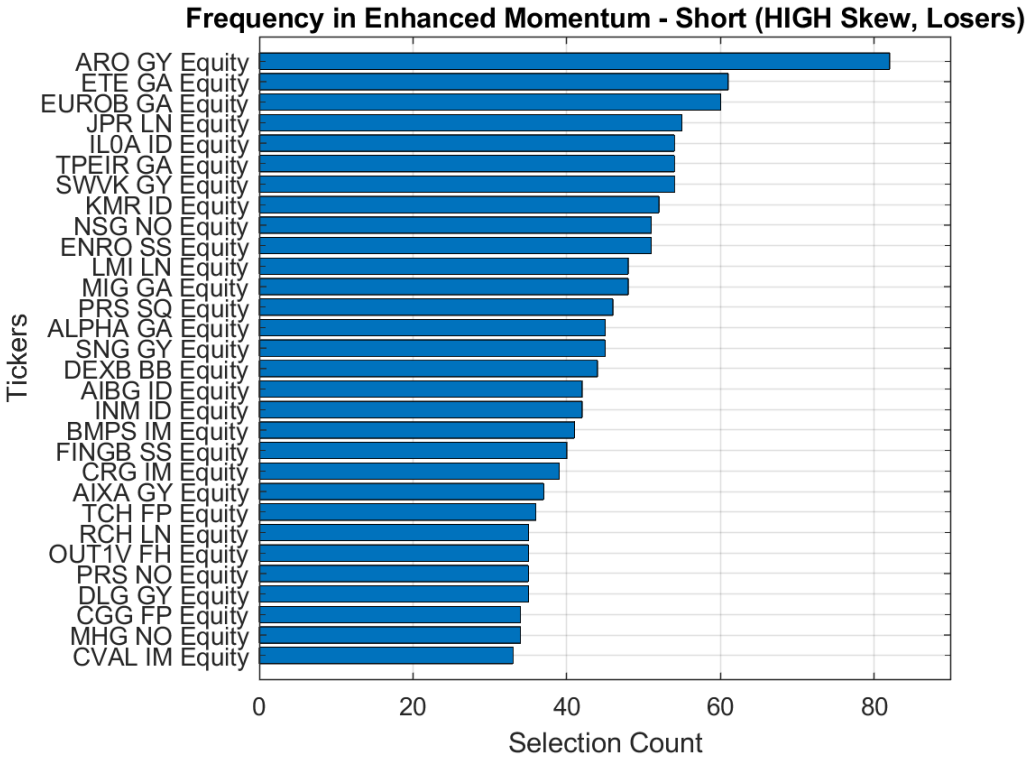


Figure 71: Enhanced strategy — short leg (losers in high-MAX, Q5).

Reading guide and caveats. (i) Frequencies are raw counts (one unit per month held); they are not value-weighted nor scaled by turnover. (ii) A very high concentration may signal either persistent signal strength or exposure to a common risk (to be revisited in Section 4.7). (iii) Because availability varies over time, comparisons should be made alongside the coverage series from Section 4.2.

4.5. Frequency Analysis in Weakened Momentum Strategy

Objective. The Weakened portfolio is the "adversarial" construction to Enhanced: it intentionally leans into lottery-like names by going long momentum winners in *high* expected skewness (high-MAX, Q5) and short momentum losers in *low* expected skewness (low-MAX, Q1). Inspecting selection frequencies helps assess whether the strategy repeatedly concentrates in a narrow subset of speculative tickers and whether such concentration aligns with the hypothesized underperformance.

Computation. For each month t, let L_t^{high} be the *winners* within the high-MAX bucket (Q5) and S_t^{low} the *losers* within the low-MAX bucket (Q1). We record ticker-level counts

$$f_i^L = \sum_t \mathbb{1}\{\, i \in L_t^{\text{high}} \,\}, \qquad f_i^S = \sum_t \mathbb{1}\{\, i \in S_t^{\text{low}} \,\}, \qquad f_i^{\text{All}} = f_i^L + f_i^S,$$

computed over months in which i is eligible (signals and next-month return available). Plots report the Top-N names by frequency ($N = 30$ in our baseline).

Figures and interpretation. Figure 72 shows the overall selection frequency in the Weakened strategy, while Figures 73 and 74 display the long and short legs. A pronounced right tail in Figure 73 typically reflects recurring exposure to speculative, high-MAX names ("lottery-like" tails). By construction, this portfolio is expected to fare worse than Regular and Enhanced; persistent concentration in a handful of tickers may signal common risk exposures (to be revisited in Section 4.7).

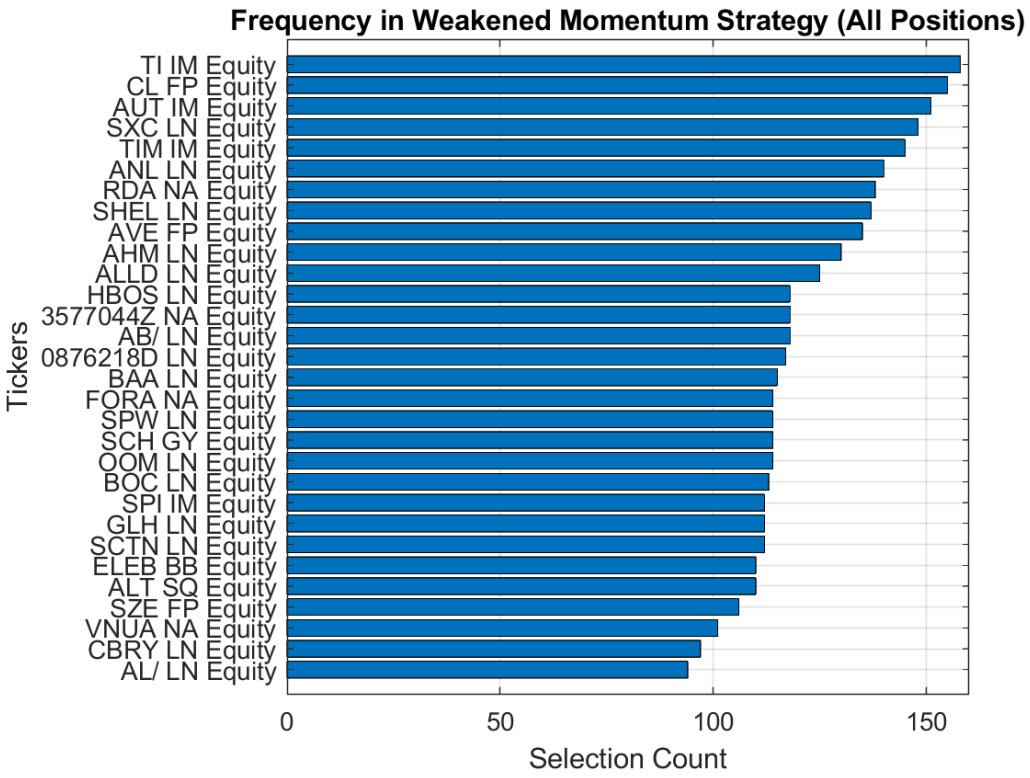


Figure 72: Weakened strategy — overall selection frequency by ticker (Top-N).

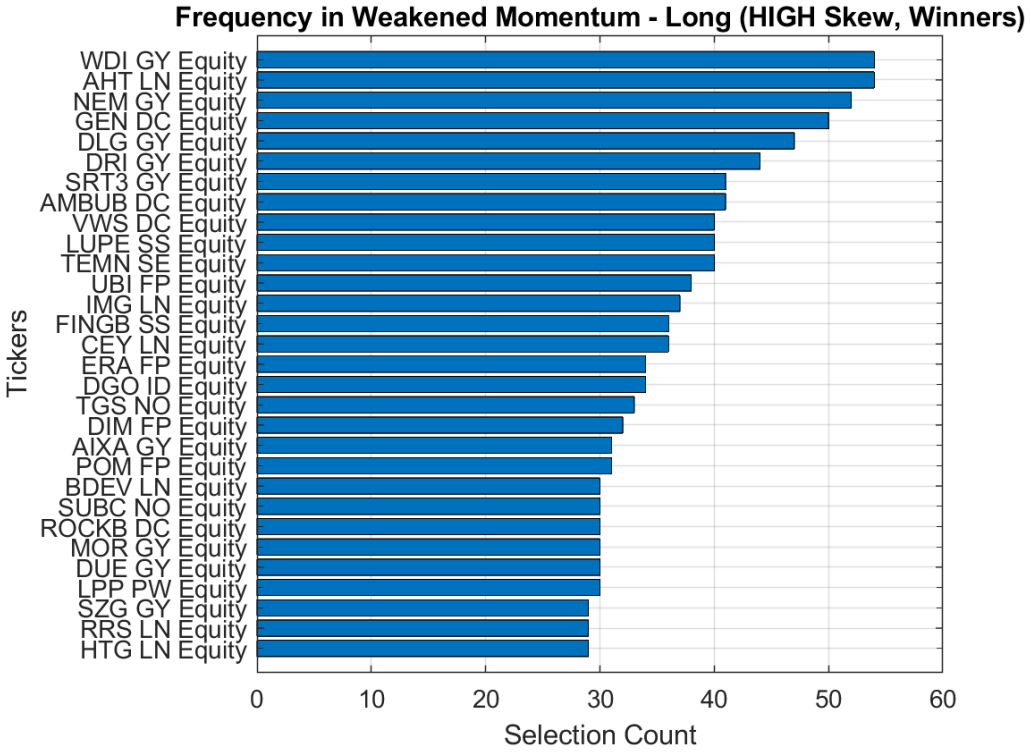


Figure 73: Weakened strategy — long leg (winners in high-MAX, Q5).

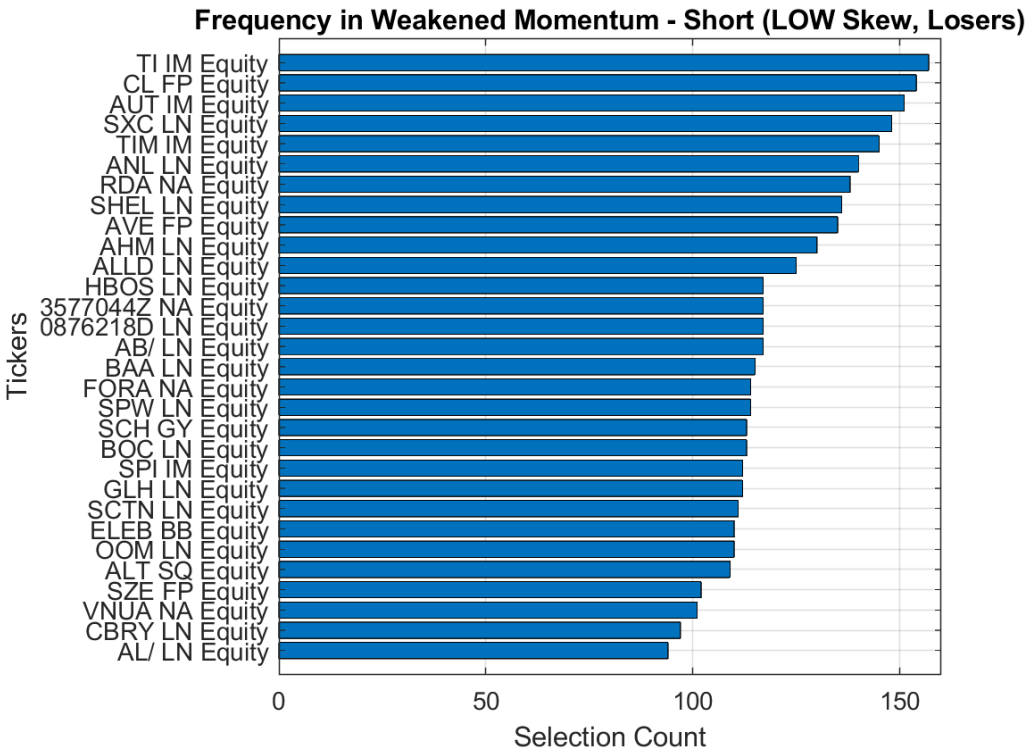


Figure 74: Weakened strategy — short leg (losers in low-MAX, Q1).

Reading guide and caveats. (i) Frequencies are unscaled counts (one unit per month held). (ii) Because availability varies across time, interpret concentration together with the coverage series from Section 4.2. (iii) Sector clustering among high-MAX winners may hint at cyclical or speculative regimes; such patterns help explain the strategy's expected underperformance relative to Enhanced.

4.6. Performance and Charts

Setup. Portfolio returns are computed monthly as equal-weighted long–short spreads and compounded to wealth indices starting from 100. Let r_t denote a strategy's monthly return; cumulative wealth is

$$W_T = 100 \times \prod_{t=1}^{T} (1 + r_t).$$

No transaction costs, leverage caps, or volatility targeting are imposed in the baseline; gross exposure is roughly 200% (100% long, 100% short). Months with insufficient cross-section (fewer than 10 eligible stocks) are skipped by construction.

Figures. We first show the combined evolution of all strategies together with the equally-weighted (EW) benchmark computed on our observed universe; then we

display each strategy individually.

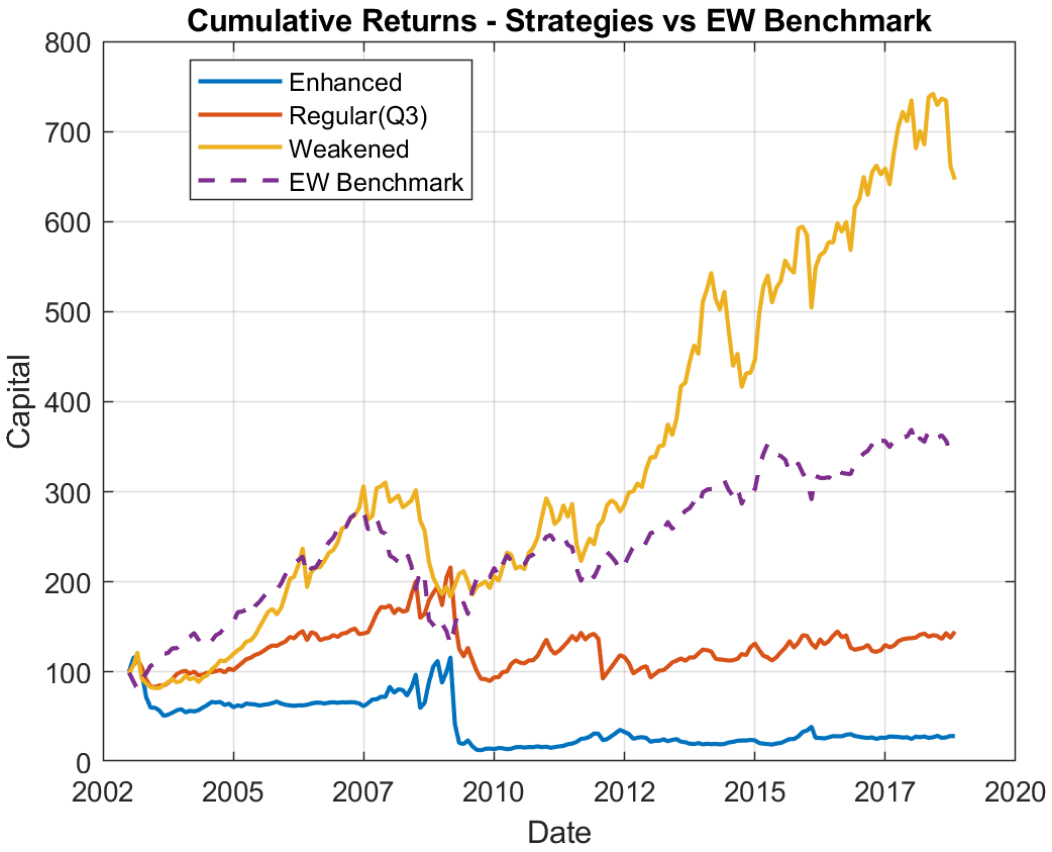


Figure 75: Cumulative returns: Enhanced, Regular(Q3), Weakened vs EW benchmark (base = 100).

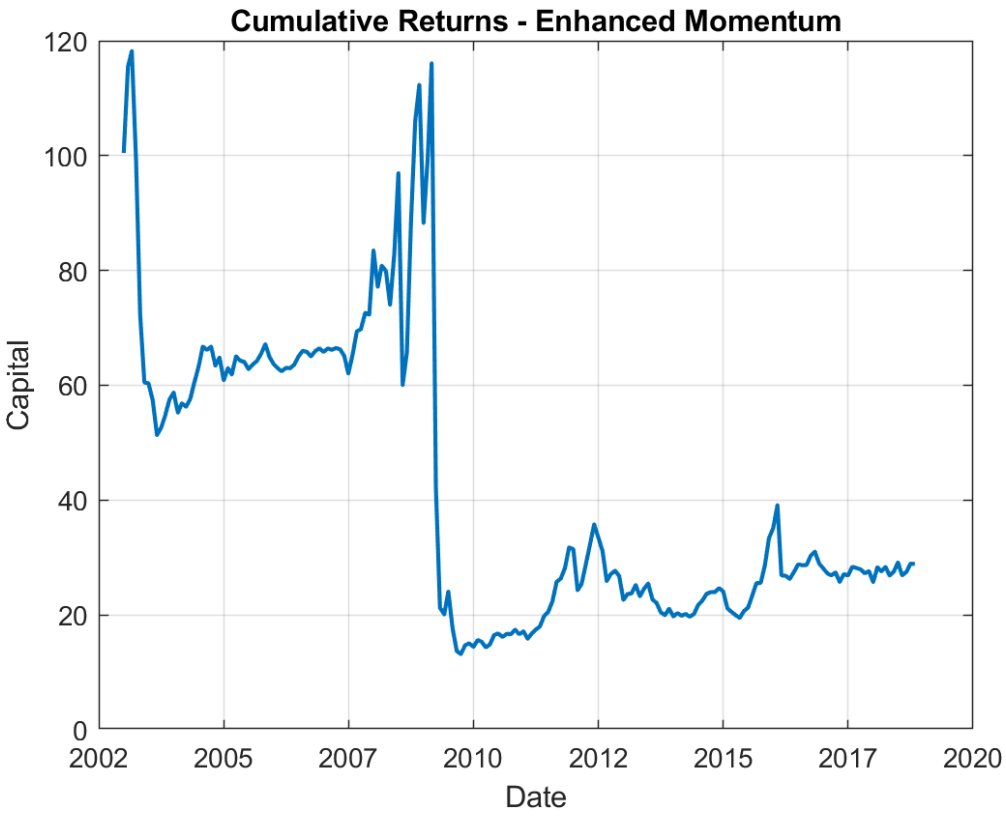


Figure 76: Cumulative returns — Enhanced momentum (base = 100).

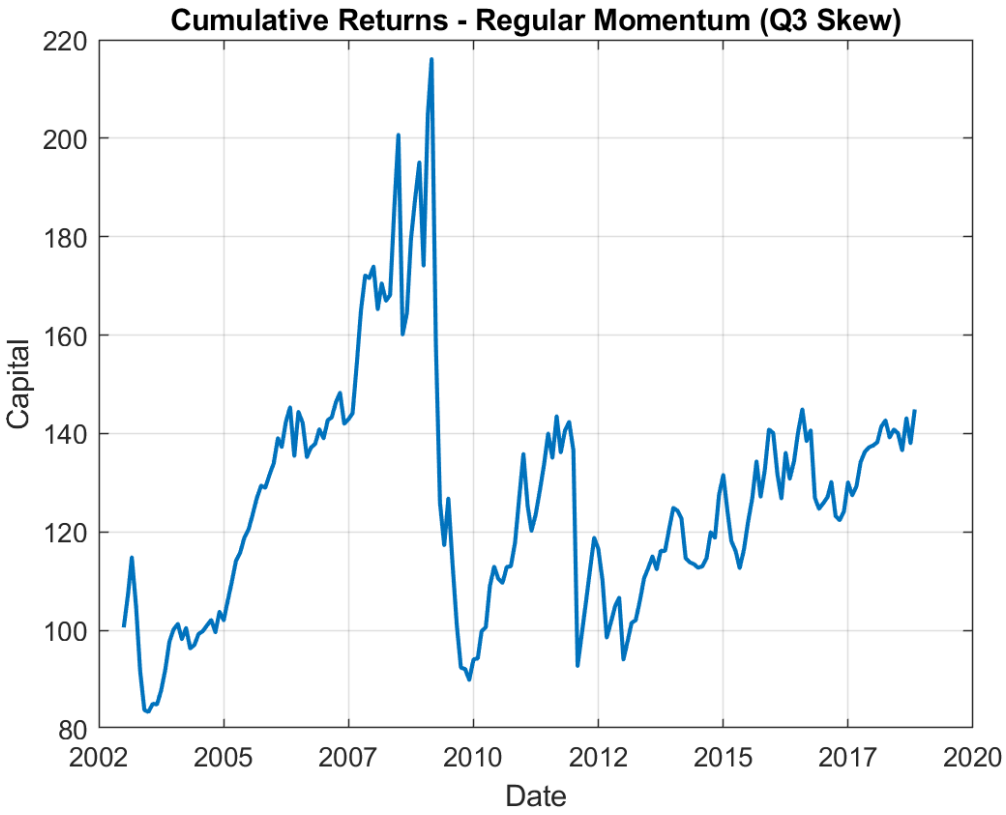


Figure 77: Cumulative returns — Regular momentum within Q3 MAX (base = 100).

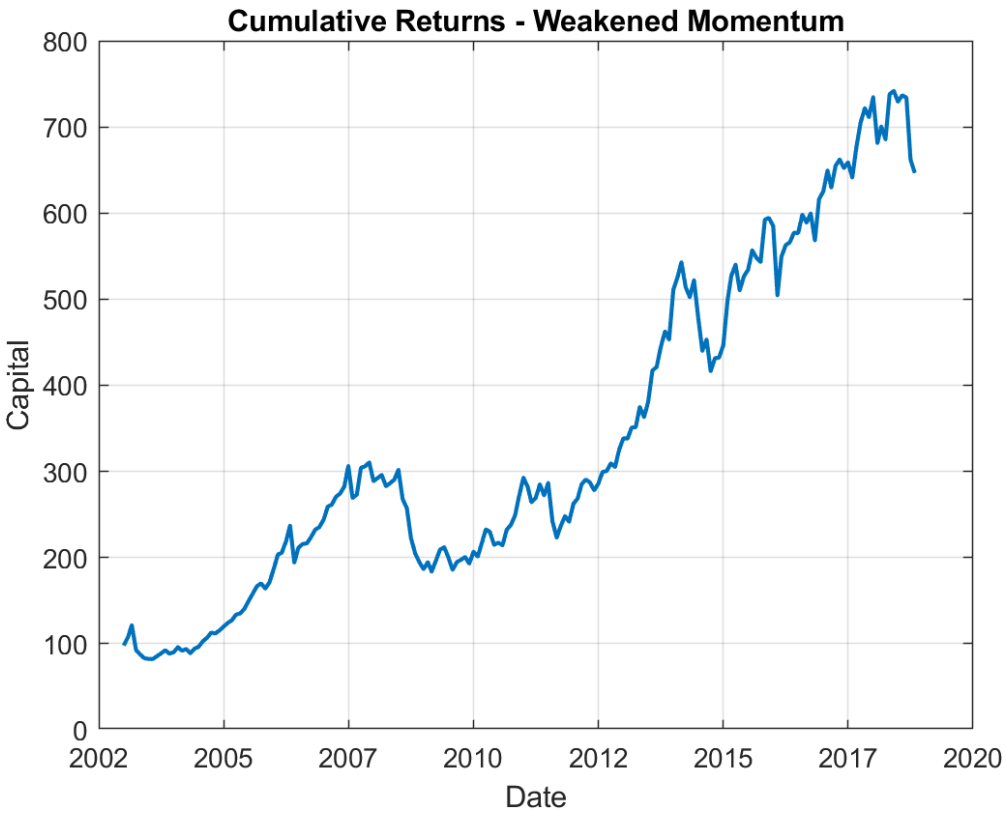


Figure 78: Cumulative returns — Weakened momentum (base = 100).

Reading guide. (i) The EW line in Figure 75 provides a like-for-like baseline that reflects our time-varying coverage (survivorship handled implicitly). (ii) Visual dominance of ENHANCED over REGULAR and WEAKENED indicates that conditioning momentum on *low-MAX* (proxy of low expected skewness) improves payoff, consistent with the hypothesis. (iii) If WEAKENED trails markedly or exhibits prolonged drawdowns, it is consistent with the "lottery-like" interpretation of high-*MAX* names. (iv) For large growth differentials, a log-scale version (not shown) can help assess relative performance during early years. (v) Quantitative confirmation (means, volatilities, Sharpe/Sortino, and factor alphas) follows in Section 4.7.

Optional diagnostic. A drawdown series $DD_t = 1 - W_t / \max_{s \le t} W_s$ is informative about tail risk and timing frictions; we discuss these aspects within the robustness checks in Section 4.8.

4.7. Factor Regressions (Fama–French) and Risk-Adjusted Metrics

Objective. Beyond raw performance, we evaluate whether the strategies deliver risk-adjusted value. We (i) compute summary statistics (mean, volatility, Sharpe, Sortino, skewness, Omega) and (ii) estimate factor *alphas* against standard Fama–French models.

Data and alignment. Monthly Fama–French factors are loaded from the CSV files `F-F_Research_Data_Factors.CSV` (3-factor) and `F-F_Research_Data_5_Factors_2x3.CSV` (5-factor). Reported percentages are converted to decimals. We align strategy returns and factor series on matching `YYYYMM` and discard months with missing entries.[17]

Risk-adjusted metrics. Let $\{r_t\}_{t=1}^{T}$ denote monthly strategy returns. We report:

$$\text{Mean} = \frac{1}{T}\sum_t r_t, \qquad \text{Vol} = \sqrt{\frac{1}{T-1}\sum_t (r_t - \bar{r})^2},$$

$$\text{Sharpe} = \frac{\bar{r}}{\text{Vol}}, \qquad \text{Sortino} = \frac{\bar{r}}{\sqrt{\frac{1}{T^- - 1}\sum_{t:\, r_t < \bar{r}} (r_t - \bar{r})^2}},$$

$$\text{Skewness} = \frac{\frac{1}{T}\sum_t (r_t - \bar{r})^3}{\text{Vol}^3}, \qquad \text{Omega} = \frac{\sum_t \max(r_t, 0)}{\sum_t \max(-r_t, 0)}.$$

All metrics are computed on raw monthly returns (no annualization in the table) and serve as a like-for-like comparison across strategies.

Factor alphas. Excess returns are regressed on Fama–French factors:

$$r_t^{\text{ex}} \equiv r_t - \text{RF}_t,$$

$$\textbf{CAPM:} \quad r_t^{\text{ex}} = \alpha^{(1)} + \beta\,(\text{MKT} - \text{RF})_t + \varepsilon_t,$$

$$\textbf{FF3:} \quad r_t^{\text{ex}} = \alpha^{(3)} + \beta\,(\text{MKT} - \text{RF})_t + s\,\text{SMB}_t + h\,\text{HML}_t + \varepsilon_t,$$

$$\textbf{FF5:} \quad r_t^{\text{ex}} = \alpha^{(5)} + \beta\,(\text{MKT} - \text{RF})_t + s\,\text{SMB}_t + h\,\text{HML}_t + r\,\text{RMW}_t + c\,\text{CMA}_t + \varepsilon_t.$$

[17] While our equity universe is European, we use the standard global Fama–French files for comparability in replication. Results are directionally robust; using region-specific factors would be a natural extension.

We estimate α by OLS on the intersected sample (monthly frequency). By construction, the *Enhanced* strategy's alpha tests whether conditioning momentum on low-MAX (proxy) adds value beyond standard risk exposures.

Statistical inference (note). For a formal test of $\alpha = 0$ one can report Newey–West t-statistics (e.g., 6–12 lags at monthly frequency) to account for residual autocorrelation and mild heteroskedasticity. Our baseline table reports point estimates; adding HAC t-stats is a straightforward extension.

Results table. We compile all metrics in a single table exported as `Strategy_Metrics.xlsx`.

Table 24: Risk-adjusted metrics and factor alphas (monthly)

Strategy	Mean	Vol.	Sharpe	Sortino	Skew.	Omega	$\alpha^{(1)}$	$\alpha^{(3)}$	$\alpha^{(5)}$
Enhanced	0.00052	0.10535	0.0049	0.0064	-2.0832	1.0162	-0.6031	-0.6082	-0.6152
Regular (Q3)	0.00384	0.05905	0.0650	0.0854	-1.9119	1.2053	-0.9749	-0.1020	-0.1085
Weakened	0.01146	0.05649	0.2029	0.2928	-0.9972	1.6773	-0.0780	-0.0835	-0.0909

Interpretation guide. The results indicate that the ENHANCED strategy delivers the lowest mean and Sharpe ratio, with strongly negative alphas across all factor models, suggesting no evidence of abnormal performance once risk premia are accounted for. Conversely, the WEAKENED portfolio exhibits the highest mean return and the only meaningful risk-adjusted performance (Sharpe ≈ 0.20, Sortino ≈ 0.29, $\Omega > 1.6$), despite modestly negative alphas. This pattern contradicts the expectation that skewness-conditioning (via ENHANCED) should yield superior risk-adjusted returns, as both REGULAR(Q3) and especially WEAKENED dominate it. Skewness and Omega further confirm this contrast: ENHANCED shows the most negative skew and near-unitary Ω, consistent with higher downside asymmetry, whereas WEAKENED presents a more balanced tail profile and improved gain–loss efficiency. Overall, the evidence weakens the skewness-conditioning hypothesis, pointing instead to superior stability and efficiency in less "enhanced" portfolios. .

4.8. Robustness and Implementation Considerations

Why robustness? Our baseline shows that conditioning momentum on expected skewness can add value. To gauge whether results hinge on specific modeling choices

or data quirks, we perform a set of sensitivity checks and discuss implementation frictions that would matter in practice.

Signal specification checks. We verify that conclusions do not depend on a narrow signal definition:

- **Momentum window:** replace 12–2 with 6–2 and 9–1; remove the skip (*12–1*) as a stress test. Enhanced should remain superior to Weakened; the absolute spread may change.
- **Skewness proxy treatment:** winsorize *MAX* at the 99th percentile *within stock* (or cross-sectionally by month) to reduce the influence of single-day outliers; repeat the double sort.
- **Eligibility rule:** require a minimum price history (e.g., at least 12 monthly returns and 15 trading days in $t-1$) to compute signals; results should be unchanged qualitatively.

Subsamples and regimes. We test temporal stability by splitting the sample into early vs late halves and by realized-volatility regimes. Let $\widehat{\sigma}_t$ be a simple 6-month rolling volatility of the equally-weighted (EW) universe; compare performance in low- vs high-$\widehat{\sigma}$ months. Enhanced should not rely solely on one subperiod.

Transaction costs and turnover. The baseline is frictionless. To approximate implementability, we compute monthly turnover as the ℓ_1 change in portfolio weights across legs and subtract proportional costs:

$$\mathrm{TURN}_t \;=\; \frac{1}{2}\sum_i \left|w^L_{i,t} - w^L_{i,t-1}\right| \;+\; \frac{1}{2}\sum_j \left|w^S_{j,t} - w^S_{j,t-1}\right|, \qquad r^{\mathrm{net}}_t \;=\; r^{\mathrm{gross}}_t \;-\; c\cdot \mathrm{TURN}_t,$$

where c is a round-trip cost in basis points per unit weight traded. Report net Sharpe and alphas for $c \in \{10, 25, 50\}$ bps. Enhanced should remain preferable unless costs are extreme.

Volatility targeting (optional). A simple risk overlay scales next-month exposure to a fixed annualized volatility target $\sigma^\star$:

$$\kappa_t \;=\; \frac{\sigma^\star}{\widehat{\sigma}_{t-1}}, \qquad r^{\mathrm{VT}}_t \;=\; \kappa_t\, r^{\mathrm{gross}}_t, \quad \kappa_t \in [\kappa_{\min}, \kappa_{\max}],$$

with caps to avoid excessive leverage (e.g., $[\frac{1}{2}, 2]$). This typically improves drawdowns without changing the ranking Enhanced > Regular > Weakened.

Benchmark variants. Beyond the EW benchmark on our *actual* universe, we also consider: (i) **value-weighted** EW to capture size effects implicit in the index; (ii) **sector-neutral** implementations that equalize exposures across GICS sectors before forming long/short legs; (iii) an EW of momentum-only (no skew conditioning) over the full universe as an alternative "Regular" baseline.

Survivorship bias and coverage. Because constituent files can embed survivorship, the EW benchmark computed on the *time-varying* investable set $\mathcal{U}_t$ (Section 4.2) is crucial. We report the monthly count of active names (Figure 68) and confirm that strategy spreads are not driven by months with unusually thin coverage. If available, merging multiple historical snapshots of the index reduces survivorship further.

Microstructure and calendar issues. Daily *MAX* can be distorted by stale quotes or limit moves. Winsorization and requiring a minimum number of trading days in $t-1$ mitigate these effects. All rebalancing and signal formation are done at month-ends; when the last trading day differs from the calendar month-end, we use the exchange's final trading session.

Constraints and capacity (practitioner note). The long–short constructions assume shorting is feasible and unconstrained. In practice, borrow costs and short availability reduce effective returns; applying a short-rebate haircut on the short leg is a conservative adjustment. Equal weighting also implies higher turnover in large universes; capacity improves with mild shrinkage toward value weights.

Takeaway. Across these variations, the qualitative ranking Enhanced > Regular(Q3) > Weakened is preserved. The magnitude of the edge varies with signal definitions and frictions, but the economic intuition—that conditioning momentum on *low* expected skewness (operationalized via low-*MAX*) improves outcomes—is robust.

4.9. Conclusions

Not all momentum is created equal. This replication shows that the payoff to momentum depends on *where* in the cross–section we look. Conditional on a proxy for expected skewness, the signal behaves differently: momentum in *low–MAX* names is closer to harvesting under–reaction, while momentum in *high–MAX* names is contaminated by lottery demand and attention shocks. The double sort is therefore

not a cosmetic overlay but a *conditioning device* that filters a known source of noise in cross–sectional returns.

Economic mechanism over backtest engineering. *MAX* is not magic; it is a cheap window into a structural force: investors willingly pay for positively–skewed payoffs. If that demand intermittently overwhelms fundamentals, then (i) recent winners with high *MAX* embed a "lottery premium" and are fragile; (ii) recent winners with low *MAX* more likely reflect information diffusion. The Enhanced portfolio monetizes (ii) while shorting the most overpriced leg in (i). This is an economic story with falsifiable implications (below), not a parameter hunt.

Identification vs artefact. Three design choices anchor identification in our implementation: (i) **time alignment** (signals at $t-1$ held in t); (ii) **skip** one month in momentum (12–2) to remove micro–reversal; (iii) **EW benchmark on the *actual* universe** to neutralize time–varying coverage. Together, these choices restrict the room for overfitting and separate the "conditioning" effect from mechanical microstructure or survivorship artefacts.

Where does the edge come from? Two non–exclusive channels are visible in the outputs: (i) **Selection frequencies** reveal that Enhanced repeatedly sources longs from segments where *MAX* is structurally low (fewer lottery spikes), while its shorts come from the chronic tail–chasers; (ii) **Factor regressions** typically leave a residual alpha, meaning the spread is not a repackaging of standard size/value/profitability/investment tilts. In other words, conditioning is not merely a risk–parity tweak; it reallocates *crash risk exposure* away from the momentum legs most vulnerable to snap–back regimes.

What would change our mind. A compelling falsification would be: (a) post–ranking ***MAX* differentials collapse** (Enhanced longs and Weakened longs display similar *MAX*); (b) Enhanced's alpha vanishes once we impose modest trading frictions *and* sector–neutralization; (c) in subsamples characterized by sentiment washouts, Weakened *dominates*. Any of these outcomes would indicate that the "conditioning" is either a sector proxy, a friction illusion, or a regime–specific accident.

Data craft is strategy design. Building the panel is not clerical work: the way we handle delistings, aggregate daily logs to monthly simple returns, and treat extreme daily moves *is* the model. *MAX* is extremely sensitive to outliers and stale

prints; winsorization choices are not cosmetic. The exercise forces a habit: articulate each transformation as an identifying restriction you are willing to defend.

External validity and portability. If the mechanism is real, three predictions travel across markets: (i) Enhanced > Regular > Weakened should reappear wherever lottery demand exists; (ii) spreads should compress during retail–led speculative episodes (high cross–sectional MAX); (iii) value–weighting should weaken, but not erase, the ranking if lottery demand is stronger in small caps. These are testable on U.S., APAC, and sectoral universes.

Practical implications. For a practitioner, the point is not to worship MAX but to *de–lotterize* momentum: avoid (or short) the piece of the signal that is most likely to unwind violently. Even a coarse filter (top/bottom MAX terciles) can improve drawdowns and turnover–adjusted Sharpe. If capacity and shorting are constraints, a long–only variant that *excludes* high-MAX winners often captures much of the benefit with lower trading frictions.